

claude-windows / 7d6bbb83-42c0-459b-9e23-e14e96d4cb2d

Metadata

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- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\7d6bbb83-42c0-459b-9e23-e14e96d4cb2d\subagents\workflows\wf_1dc2acd6-052\agent-a8a4fa49f539eb660.jsonl`  
- SHA-256: `79d091a40ffccca0d8e1394e51724541c2f98524a82a2c1d5cfd53e01d9bc369`  
- Source modified: `2026-06-13T20:28:43+00:00`  
- Imported at: `2026-07-08T16:00:13+00:00`  
- project: `wf_1dc2acd6-052`  
- session_id: `7d6bbb83-42c0-459b-9e23-e14e96d4cb2d`
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Transcript

1. user (2026-06-13T20:25:39.462Z)

You are the senior author writing the RELATED-WORK + COURSE-CORRECTION memo for our team. We built a rally-detection framework (described below) and need an HONEST verdict on whether we are reinventing the wheel, plus concrete course corrections grounded in the literature.

DOMAIN: badminton (? racket-sports) RALLY DETECTION ? segment a long match video into rally [start,end] intervals (a rally = one point in play; between-rally time is "dead time"). Ground truth is scarce, interval-level, and expensive (vendor-labeled); corpus today is ~6 matches. Cameras are mostly static tripod.

OUR FRAMEWORK (what we want assessed against the literature):

1. MULTI-SCOPE PRODUCERS. Cues ("producers") emit time-aligned SIGNALS at three temporal SCOPES:
- frame-scope DETECTORS (expensive, GPU, run once, persisted): a frozen shuttlecock-tracking CNN

(WASB / TrackNet ? per-frame shuttle x,y,visibility), a person detector (boxes+tracks), an audio

"thwack" hit-detector, and a proposed per-frame shuttle-STATE classifier (in_air / racket_contact / inactive).

- window-scope ANALYZERS (cheap, pure, offline): per candidate rally window, emit aggregated scale/translation-invariant features (e.g. net-crossing count, a "service-fault" cue = shuttle halts

in the net region, person occupancy/two-sidedness).

- session-scope ANALYZERS: over the WHOLE timeline, emit labeled spans + per-window membership features

(e.g. WARM-UP / practice detection ? a long continuous-rally stretch w/o the point/dead-time cadence).

2. "SIGNAL NOT VERDICT / NO SPECIAL-CASING." Every cue is a (value, confidence) FEATURE COLUMN. A small

learned scorer (logistic regression today) consumes the columns and decides which candidate windows are

real rallies. Cues are NEVER hand-coded if/else branches in the decision path; the model LEARNS to weight

them. New cues = "register a producer + add a feature column," never "add a branch."

3. DETECTION ? DECISION ("persist once, re-score forever"). The expensive frame detection runs once and is

persisted; the cheap decision (windowing + scorer) is pure offline re-scoring. Most experiments never

touch the served pipeline; they re-score stored features deterministically.

4. RECONCILERS = a "report-first linter for rally decisions." A post-hoc pass that detects where the

DECISION contradicts the persisted EVIDENCE and proposes corrections: truncate a segment whose shuttle went inactive before its end; drop a segment with zero net-crossings; split an over-merged segment with a long internal dead-time gap; tighten boundary slack to serve-contact. REPORT-FIRST: it outputs an issues report + a corrected set, never silently rewrites; corrections are A/B-measured (lift) BEFORE auto-apply; idempotent; fixed rule precedence; human-in-the-loop.

5. EXPERIMENT HARNESS / no-regression evolution. New cues land flag-gated, default-OFF; graded offline vs the scarce golden labels with LEAVE-ONE-VIDEO-OUT (LOVO; group by match, never a held-out window from the same video); promoted to default ONLY on measured lift; rollback = flip a flag (never code revert).

We use temporal IoU vs golden intervals as the metric, and a tiny logistic regression (not a deep net) because the labeled set is small.

Here are the verified per-pattern findings (survey + adversarial verify, with citation checks) as JSON:

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[{"theme": "Temporal action detection/segmentation + boundary refinement / post-processing", "short": "TAD-TAS", "our_pattern": "Our candidate-windows + learned scorer + reconciler-as-refinement = a temporal action detection/segmentation pipeline with a post-processing/refinement stage.", "survey": {"theme": "Temporal action detection/segmentation + boundary refinement / post-processing", "our_pattern": "We segment a long static-camera badminton match into rally [start,end] intervals via: (a) multi-scope \"producers\" that emit time-aligned signals ? frame-scope detectors (frozen WASB/TrackNet shuttle tracker, person detector, audio thwack detector, proposed per-frame shuttle-state classifier), window-scope analyzers (per-candidate-window aggregated invariant features: net-crossings, service-fault halt, person two-sidedness), and session-scope analyzers (warm-up/practice span detection); (b) a \"signal not verdict\" rule: every cue is a (value, confidence) feature column, consumed by a tiny logistic-regression scorer that decides which candidate windows are real rallies ? no hand-coded if/else in the decision path; (c) detection-decision separation: expensive frame detection persisted once, cheap windowing+scoring is pure offline re-scoring; (d) RECONCILERS ? a report-first post-hoc pass that flags where the DECISION contradicts persisted EVIDENCE (shuttle inactive before segment end, zero net-crossings, over-merged segment with internal dead-time, loose boundary slack) and proposes idempotent, fixed-precedence, A/B-gated corrections; (e) an experiment harness landing cues flag-gated/default-OFF, graded with leave-one-video-out (group-by-match) on scarce vendor-labeled golden intervals using temporal IoU, promoted only on measured lift.", "key_papers": [{"title": "Temporal Action Segmentation: An Analysis of Modern Techniques", "authors": "Ding, Guodong; Sener, Fadime; Yao, Angela", "year": "2022", "url_or_id": "arXiv:2210.10352", "what_it_does": "Survey of temporal action segmentation (TAS): defines the canonical task (per-frame labeling of long untrimmed video into contiguous action segments), the dominant frame-classification + multi-stage-refinement paradigm, the over-segmentation error problem, and the standard metrics (frame accuracy, segmental edit score, F1@{10,25,50}). Surveys full/weak/unsupervised supervision regimes.", "venue": "IEEE TPAMI / arXiv", "overlap": "strong", "overlap_explanation": "Our task IS temporal action segmentation specialized to a 2-class (rally / dead-time) problem on a long timeline. This survey is the authoritative source for the terminology, metrics, and the over-segmentation framing our reconciler exists to fix. We should cite it and adopt its vocabulary instead of inventing our own ('producers', 'reconcilers')."}, {"title": "MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation", "authors": "Farha, Yazan Abu; Gall, Juergen", "year": "2019", "url_or_id": "arXiv:1903.01945 (CVPR 2019)", "what_it_does": "Seminal TAS method: stacks multiple refinement stages of dilated temporal convolutions over frame features; each stage refines the previous stage's full prediction. Introduces a smoothing (truncated-MSE) loss specifically to penalize over-segmentation.", "venue": "CVPR 2019", "overlap": "partial", "overlap_explanation": "Same high-level pattern ? features -> initial prediction -> refinement stage(s) that suppress over-segmentation. Differs in mechanism: MS-TCN refinement is a learned in-model neural stage trained on many labels; our reconciler is an external, report-first, rule+evidence corrector tuned for a ~6-match scarce-label regime. The shared concept is 'a refinement stage exists precisely to fix over-segmentation'; the smoothing-loss idea is adoptable in spirit (penalize fragmentation)."}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries (ASRF)", "authors": "Ishikawa, Yuchi; Kasai,
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Seito; Aoki, Yoshimitsu; Kataoka, Hirokatsu", "year": "2021", "url_or_id": "arXiv:2007.06866 (WACV 2021)", "what_it_does": "Decomposes segmentation into two complementary branches: an Action Segmentation Branch (frame classification) and a Boundary Regression Branch; the regressed boundaries REFINED the classifier output, reducing over-segmentation (up to +16.1% F1, +13.7% edit). Explicitly a boundary-aware refinement framework.", "venue": "WACV 2021", "overlap": "strong", "overlap_explanation": "This is the closest published analog to our reconciler's BOUNDARY work: a dedicated boundary signal used to post-correct a segmentation. Our 'tighten boundary slack to serve-contact' and 'truncate segment whose shuttle went inactive before its end' are exactly boundary-regression-style refinement, except we use explicit physical evidence (serve contact, shuttle inactivity) as the boundary cue rather than a learned BRB. Strong overlap on the WHAT; our framing is the rule/evidence variant of the same idea."}, {"title": "Activity Grammars for Temporal Action Segmentation", "authors": "Gong, Dayoung; Lee, Joonseok; Jung, Deunsol; Kwak, Suha; Cho, Minsu", "year": "2023", "url_or_id": "arXiv:2312.04266 (NeurIPS 2023)", "what_it_does": "Induces a context-free grammar of valid action sequences and applies an efficient Viterbi-style parser as an inference-time POST-PROCESSING layer over any base segmenter, correcting structurally-invalid frame-level predictions into grammar-consistent sequences. Improves over-segmentation and interpretability on Breakfast/50Salads.", "venue": "NeurIPS 2023", "overlap": "strong", "overlap_explanation": "This is the single closest match to the RECONCILER concept: a model-agnostic, inference-time corrector that enforces STRUCTURAL PRIORS on a base segmenter's output. Our reconciler rules ('rally must contain ≥ 1 net-crossing', 'a rally cannot extend past shuttle inactivity', 'point/dead-time cadence vs continuous warm-up') are exactly a hand-specified activity grammar / structural prior. We are reinventing grammar-constrained refinement with bespoke rules; we should cite this and consider framing reconciler rules as an explicit grammar."}, {"title": "Post-Processing Temporal Action Detection (GAP: Gaussian Approximated Post-processing)", "authors": "Nag, Sauradip; Zhu, Xiatian; Song, Yi-Zhe; Xiang, Tao", "year": "2023", "url_or_id": "arXiv:2211.14924 (CVPR 2023)", "what_it_does": "A model-agnostic post-processing method that fixes temporal-quantization boundary error by modeling action start/end with Gaussian distributions and inferring boundaries at sub-snippet resolution via Taylor expansion ? applied to stored detections without retraining.", "venue": "CVPR 2023", "overlap": "partial", "overlap_explanation": "Validates our 'persist once, re-score/refine forever' principle: a model-agnostic post-processing stage applied to frozen detections to sharpen boundaries. Different mechanism (Gaussian sub-snippet boundary inference vs evidence-driven rules) and different error (quantization vs decision-vs-evidence contradiction), but the architectural framing ? detection frozen, refinement is cheap post-hoc ? is the same and worth citing as precedent for the design."}, {"title": "BMN: Boundary-Matching Network for Temporal Action Proposal Generation", "authors": "Lin, Tianwei; Liu, Xiao; Li, Xin; Ding, Errui; Wen, Shilei", "year": "2019", "url_or_id": "arXiv:1907.09702 (ICCV 2019)", "what_it_does": "Generates dense temporal action PROPOSALS (candidate [start,end] windows) and scores each with a confidence via a boundary-matching mechanism; canonical 'proposal-then-classify/score' temporal action localization pipeline. Builds on BSN (Lin et al. 2018, arXiv:1806.02964).", "venue": "ICCV 2019", "overlap": "partial", "overlap_explanation": "Our 'candidate windows + learned scorer' IS the proposal-generate-then-score paradigm of temporal action localization (TAL), of which BSN/BMN are the canonical instances. We differ by generating candidates from a physics signal (shuttle activity) rather than a learned proposal net, and scoring with logistic regression on interpretable features rather than a deep confidence head. Same skeleton, simpler/scarcer-label-appropriate instantiation."}, {"title": "Action Spotting using Dense Detection Anchors Revisited (SoccerNet 2022)", "authors": "Soares, Joao V. B.; Shah, Avijit; Biswas, Topojoy", "year": "2022", "url_or_id": "arXiv:2206.07846", "what_it_does": "Sports-video temporal localization: operates on PRE-EXTRACTED frame features (frozen backbone), predicts dense per-anchor detection confidence + a fine-grained temporal displacement to sharpen localization, optimizing the 'tight' (small-tolerance) average-mAP. Winner of SoccerNet Action Spotting 2022.", "venue": "SoccerNet/CVSports workshop 2022", "overlap": "partial", "overlap_explanation": "Direct sports analog of our detection/decision split: heavy features computed once and frozen, a light head decides events, and a temporal-displacement refinement sharpens boundaries. Their 'tight average-mAP at small temporal tolerance' is the metric the field uses for boundary tightness ? directly relevant to grading our boundary-slack reconciler beyond a single tIoU threshold."}, {"title": "Service and End of Rally Detection in Badminton Videos", "authors": "(Springer ACIVS/IbPRIA chapter, 2024)", "year": "2024", "url_or_id": "DOI:10.1007/978-3-031-69073-0_15", "what_it_does": "Directly the same task as ours: detects rally start (service) and rally end in badminton match video using shuttle/player motion cues (per search-result snippets; full text authenticated).", "venue": "Springer LNCS (2024)", "overlap": "partial", "overlap_explanation": "Same problem (rally start/end segmentation in badminton) and similar cues (shuttle motion, service), so it is the most direct task-level prior art and should be cited as such. Overlap on TASK is strong;

overlap on METHOD is unverified (could not read full text) ? I mark it partial and flag that the multi-scope-producers + reconciler + LOVO-harness architecture appears to be our differentiator vs a single end-to-end detector. Confirm method before claiming novelty over it.},{ "title": "A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos", "authors": "(SN Computer Science, Springer, 2025)", "year": "2025", "url_or_id": "DOI:10.1007/s42979-025-03935-0", "what_it_does": "Detects rally start via BiLSTM/CNN on service, and rally END via a gradient/motion rule: rally ends when $\geq 80\%$ of consecutive frames show no shuttle motion. Reports $\sim 95\%$ accuracy across 187 rallies (per search-result snippets; full text auth-gated).", "venue": "SN Computer Science (2025)", "overlap": "partial", "overlap_explanation": "Their rally-end rule ('shuttle no-motion over consecutive frames') is essentially our 'shuttle went inactive' reconciler/state cue, and their service-detection is our service-fault/serve-contact cue ? strong overlap on the SIGNALS. The key difference is they hard-code these as if/else verdicts; our explicit thesis is to demote them to feature columns for a learned scorer ('signal not verdict'). Cite as evidence the cues are sound, and as a direct counterexample to the special-casing we are deliberately avoiding."}, {"established_terms": ["Temporal Action Segmentation (TAS) ? the per-frame-label-into-contiguous-segments task; ours is binary (rally vs dead-time)", "Temporal Action Localization / Detection (TAL/TAD) ? proposal-then-score of [start,end] intervals; matches our candidate-windows + scorer", "Action proposal generation (BSN/BMN) ? our 'candidate windows'", "Over-segmentation error ? the canonical name for the failure mode our reconciler's 'split/merge' rules target", "Boundary regression / boundary refinement (ASRF) ? our boundary-slack-to-serve-contact correction", "Smoothing loss ? the standard in-training anti-over-segmentation device (MS-TCN)", "Refinement stage / multi-stage refinement ? the field's term for what our reconciler is, externalized", "Model-agnostic post-processing (GAP) ? our 'persist once, re-score/refine forever' framing", "Activity grammar / grammar-constrained / constrained decoding ? our reconciler rules formalized as structural priors", "Action spotting + 'tight' average-mAP at small temporal tolerances (SoccerNet) ? the sports-video metric for boundary precision", "Segmental F1@{10,25,50}, segmental edit score, frame accuracy/MoF (Lea et al. 2017) ? the metrics the field reports alongside tIoU"], "techniques_to_adopt": [{"technique": "Report and track segmental edit score and F1@{10,25,50} (plus mean-over-frames) alongside our temporal-IoU metric.", "why": "These are the field-standard TAS metrics and, unlike a single tIoU threshold, the edit score and F1@k directly quantify over-segmentation (fragmentation/spurious-boundary errors) ? exactly what the reconciler's split/merge rules target. Reporting them makes our lift claims legible and comparable to the literature, and gives the reconciler a metric that rewards fixing fragmentation, not just overlap.", "from_paper": "Temporal Action Segmentation: An Analysis of Modern Techniques (Ding et al. 2022); MS-TCN (Farha & Gall 2019)", "effort": "low"}, {"technique": "Add a 'tight' / small-tolerance localization metric (average-mAP across multiple tIoU thresholds incl. tight ones, e.g. SoccerNet-style) to grade boundary tightness specifically.", "why": "Our serve-contact boundary-slack reconciler is a boundary-precision intervention; a single loose tIoU threshold can hide whether boundaries actually tightened. The tight-average-mAP metric was designed to be sensitive to exactly these small temporal errors.", "from_paper": "Action Spotting using Dense Detection Anchors Revisited (Soares et al. 2022)", "effort": "low"}, {"technique": "Frame the reconciler's structural rules as an explicit 'activity grammar' / constrained-decoding layer and resolve them globally (Viterbi-style) rather than as independent local edits.", "why": "Activity Grammars shows that enforcing valid-sequence structure as a model-agnostic inference-time parser both reduces over-segmentation and is interpretable ? the same goal as our reconciler. A global parse avoids the precedence/conflict headaches of stacking independent local rules and gives a principled, citable formalism for 'rally must contain a net-crossing', 'no extension past shuttle inactivity', 'warm-up cadence differs from point cadence'.", "from_paper": "Activity Grammars for Temporal Action Segmentation (Gong et al. 2023)", "effort": "medium"}, {"technique": "Adopt the explicit boundary-regression decomposition: keep a separate boundary signal (serve contact / shuttle-inactivity onset) and use it to refine the scorer's segment ends, mirroring ASRF's ASB+BRB split.", "why": "ASRF shows a dedicated boundary branch refining a separate classification branch yields large F1/edit gains and is the established cure for over-segmentation. Our cues already provide physical boundary evidence; structuring the pipeline as 'classify window' + 'regress/refine boundary from evidence' aligns us with a proven design and names what we are doing.", "from_paper": "Alleviating Over-segmentation Errors by Detecting Action Boundaries / ASRF (Ishikawa et al. 2021)", "effort": "medium"}, {"technique": "Add a soft over-segmentation penalty to the scorer's objective or post-filter (the spirit of MS-TCN's smoothing loss), not just per-window classification accuracy.", "why": "Our logistic-regression scorer optimizes per-window labels and is blind to fragmentation across adjacent windows; a smoothing/merge prior (penalize transitions / encourage contiguous same-label runs) directly attacks over-segmentation, which is the dominant TAS error mode.", "from_paper": "MS-TCN (Farha & Gall 2019)", "effort": "medium"}, {"technique": "Treat the whole refinement step as model-agnostic post-

processing over frozen detections and A/B it as such (which we already do) ? cite GAP/Activity-Grammars as precedent and borrow their evaluation discipline (measure lift of the post-processor in isolation, on stored outputs).", "why": "GAP and Activity Grammars are exactly 'a refinement applied to frozen model outputs, measured for standalone lift'. This both validates our 'detection ? decision / persist once' architecture and gives us a citable methodological template for the reconciler's A/B-before-auto-apply gate.", "from_paper": "Post-Processing Temporal Action Detection / GAP (Nag et al. 2023)", "effort": "low"}], "pitfalls": ["Over-segmentation is THE dominant error mode in this field, and frame/window accuracy hides it. A scorer optimized for per-window accuracy can look good while producing fragmented or over-merged rallies. Mitigation: report edit score + F1@k and add an explicit anti-fragmentation prior (MS-TCN smoothing-loss spirit; Ding et al. 2022).", "Stacked independent local correction rules (truncate / drop / split / tighten) with a fixed precedence can conflict, oscillate, or be non-idempotent across orderings ? the very reason the field moved to a single global parse (Viterbi over a grammar). Risk of brittle rule interactions; mitigate by resolving structural constraints jointly (Activity Grammars, Gong et al. 2023).", "Boundary/quantization error: candidate windows at a coarse stride introduce temporal-quantization error that no amount of scoring fixes; boundaries need sub-window refinement (GAP, Nag et al. 2023). A loose tIoU threshold will mask whether boundaries are actually tight.", "LOVO leakage even when grouping by match: with only ~6 matches, the same venue/court/players/broadcast style recur, so 'leave-one-video-out' still leaks domain (court color, camera framing, player pair). The TAS literature reports large domain shift precisely because train/test differ by subject/scene; with 6 matches, per-match LOVO variance will be huge and a logistic-regression scorer can overfit a handful of cues to match-specific quirks. Mitigate: report per-fold variance, not just mean; treat absolute numbers as low-confidence.", "Tuning reconciler rule thresholds (e.g. 'inactive for N frames', '>=80% no-motion') on the same scarce golden set used to grade them is implicit fitting on the test set ? the badminton rally-detection papers' hand-tuned no-motion thresholds are exactly this risk. Keep rule-threshold selection inside the LOVO loop, never on the full golden set.", "Calibration/feature-confidence drift in late fusion: each producer emits an independent (value, confidence); these confidences are not mutually calibrated, so a logistic regression can be dominated by an over-confident but unreliable cue (the audio cue's known low discrimination is a live example). Calibrate per-cue confidences before fusion. (Late-fusion miscalibration is a documented multi-modal TAL issue; see audio-augmented TAL fusion work, arXiv:2106.14118.)", "Report-first is good, but 'A/B-measured lift before auto-apply' on ~6 matches risks promoting a correction whose measured lift is within noise. Require lift to exceed per-fold variance, and prefer corrections that improve edit/F1 (fragmentation) not just mean tIoU."], "verdict": "partial-overlap", "verdict_rationale": "The skeleton of our design is well-established and we should cite rather than reinvent it: 'candidate windows + learned scorer' is textbook proposal-then-score Temporal Action Localization (BSN/BMN); 'refinement stage to fix over-segmentation' is the defining theme of Temporal Action Segmentation (MS-TCN, ASRF); and 'model-agnostic post-processing over frozen detections to correct/sharpen output' is exactly GAP and Activity Grammars. Our reconciler in particular is a close re-derivation of grammar-constrained / boundary-regression refinement (Activity Grammars 2023, ASRF 2021) ? its rules are a hand-built activity grammar, so we should adopt that framing and its global-parse machinery rather than a stack of ad-hoc local edits. What is genuinely less-charted (novel-combination, not a clean gap) is the SPECIFIC assembly for a scarce-label (~6 match), interval-only, static-camera racket-sports setting: physics-derived cues as feature columns, a deliberately tiny logistic-regression scorer instead of a deep net, the report-first/A/B-gated/human-in-the-loop discipline around the corrector, and the multi-scope (frame/window/session) producer taxonomy with a warm-up/practice session-level analyzer. Direct task analogs exist in badminton (Springer 2024 service/end-of-rally; SN-CS 2025 BiLSTM+gradient), but they hard-code cues as verdicts ? our 'signal not verdict, learn the weights' stance is a defensible contrast, not a reinvention. Net: the architecture is partial-overlap (cite TAS/TAL/refinement literature and adopt its metrics, grammar/boundary-refinement framing, and over-segmentation cures); the scarce-label, interpretable, governance-disciplined instantiation is the contribution.", "search_queries_used": ["temporal action segmentation survey MS-TCN over-segmentation error post-processing refinement", "temporal action localization proposal generation boundary refinement BMN BSN survey", "ASFormer DiffAct temporal action segmentation boundary smoothing refinement transformer", "badminton rally segmentation video detection shuttlecock trajectory rally start end", "ASRF Alleviating Over-segmentation Errors Action Segmentation Refinement Framework boundary regression", "tennis broadcast video rally point segmentation play break detection dead time", "temporal action detection post-processing soft-NMS Gaussian boundary refinement late fusion calibration", "multi-modal fusion temporal action localization feature-level late fusion sports video few labels logistic regression", "SoccerNet action spotting temporal localization features lightweight classifier dense anchors challenge", "temporal action segmentation evaluation metrics F1@k edit score segmental over-segmentation Lea"]

2017", "constraint-aware decoding temporal action segmentation rule-based correction grammar prior inference time", "leave-one-video-out cross-validation temporal action segmentation small dataset leakage group split sports"]}, {"verify": {"theme": "Temporal action detection/segmentation + boundary refinement / post-processing", "citations_checked": [{"title": "Temporal Action Segmentation: An Analysis of Modern Techniques (Ding, Sener, Yao 2022)", "verified_real": "yes", "note": "arXiv:2210.10352 confirmed; authors Guodong Ding, Fadime Sener, Angela Yao; later TPAMI. Survey description accurate."}, {"title": "MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation (Farha & Gall 2019)", "verified_real": "yes", "note": "arXiv:1903.01945, CVPR 2019, Abu Farha & Gall (Univ. Bonn). Smoothing (truncated-MSE) loss to penalize over-segmentation confirmed."}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries (ASRF) (Ishikawa et al. 2021)", "verified_real": "yes", "note": "arXiv:2007.06866, WACV 2021; authors Ishikawa, Kasai, Aoki, Kataoka confirmed via official project site + GitHub yiskw713/asrf. ASB+BRB boundary-regression decomposition accurate."}, {"title": "Activity Grammars for Temporal Action Segmentation (Gong et al. 2023)", "verified_real": "yes", "note": "arXiv:2312.04266, NeurIPS 2023; authors Gong, Lee, Jung, Kwak, Cho confirmed (GitHub gongda0e/KARI). Grammar-induction + Viterbi-style inference-time post-processing accurate."}, {"title": "Post-Processing Temporal Action Detection / GAP (Nag et al. 2023)", "verified_real": "yes", "note": "arXiv:2211.14924, CVPR 2023; authors Nag, Zhu, Song, Xiang confirmed. Note: real title is 'Post-Processing Temporal Action Detection'; 'GAP/Gaussian Approximated Post-processing' is the method name, correctly characterized."}, {"title": "BMN: Boundary-Matching Network for Temporal Action Proposal Generation (Lin et al. 2019)", "verified_real": "yes", "note": "arXiv:1907.09702, ICCV 2019. Authors Lin, Liu, Li, Ding + Wen Sun (survey lists 'Wen', dropping surname 'Sun' ? minor, not a fabrication). Builds on BSN (1806.02964)."}, {"title": "Action Spotting using Dense Detection Anchors Revisited (Soares, Shah, Biswas 2022)", "verified_real": "yes", "note": "arXiv:2206.07846 confirmed; real full title adds ': Submission to the SoccerNet Challenge 2022'. Authors Soares, Shah, Biswas (Yahoo Research). Tight average-mAP emphasis accurate."}, {"title": "Service and End of Rally Detection in Badminton Videos (Springer LNCS 2024)", "verified_real": "yes", "note": "DOI 10.1007/978-3-031-69073-0_15 confirmed; ISACE 2024 / LNCS vol 14794; authors See, Hassan, Krishnasamy, Loo, Paramesran. Gradient-based shuttle-rest end-of-rally detection; ~88.6% on 35 rally points. Method DID partially resolve (gradient/motion rule, not full multi-cue system) ? survey's 'method unverified' caveat can be relaxed slightly."}, {"title": "A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos (SN Computer Science 2025)", "verified_real": "yes", "note": "DOI 10.1007/s42979-025-03935-0 confirmed (Monash; SN CS 6(5) Art.447). BiLSTM/CNN service start ~95%, end-of-rally via >=80% consecutive no-motion frames, 187 rallies. Every specific claim verified exactly. Strongest direct task+signal analog."}, {"title": "Hear Me Out: Fusional Approaches for Audio Augmented Temporal Action Localization (inline ref, pitfalls)", "verified_real": "yes", "note": "arXiv:2106.14118 confirmed; Bagchi, Mahmood, Fernandes, Sarvadevabhatla. Correctly cited as precedent for multi-modal late-fusion miscalibration risk."}], "corrected_verdict": "partial-overlap", "corrected_rationale": "Verdict stands at partial-overlap, and the survey is unusually honest ? it already self-classifies the scarce-label badminton instantiation as 'novel-combination, not a clean gap.' My adversarial pass confirms that framing rather than overturning it, with two downward nudges. CITATIONS: zero fabrications. All 10 key_papers and both inline arXiv refs resolve to real papers with correct IDs, authors, and venues; the only imperfections are cosmetic (SoccerNet paper's full title is truncated; BMN's 'Wen Sun' shortened to 'Wen'; GAP's method-name vs paper-title). The two highest-risk items (auth-gated Springer DOIs) are real and their specific quantitative claims check out exactly. NOVELTY: the survey's own 'partial-overlap' rests on (a) candidate-windows+scorer = textbook proposal-then-score TAL (BSN/BMN), (b) refinement-stage-fixes-over-segmentation = core TAS theme (MS-TCN/ASRF), (c) model-agnostic post-processing over frozen detections = GAP/Activity-Grammars. All correct. My refutation searches surfaced closer prior art the survey missed that pulls two of its sub-claims further toward 'reinventing': (1) For the RECONCILER, a concurrent paper 'Improving Temporal Action Segmentation via Constraint-Aware Decoding' (arXiv:2605.10149, ICPR 2026) is an even tighter analog than Activity Grammars ? it extracts structural priors (transition confidence, permitted action-boundary sets, per-class duration) directly from data and resolves them via a global modified-Viterbi decode as inference-time post-processing on a frozen base model, exactly the survey's own recommended 'reframe reconciler rules as a grammar, resolve globally' upgrade. (2) For the OVERALL ASSEMBLY (handcrafted/physics features + lightweight classifier + temporal-smoothing/rule post-processing for scarce-label racket-sports rally/play-break segmentation), there is substantial racket-sports precedent: Kijak et al.'s audiovisual tennis play/break structuring with HMM + structural rules, HOG+chi2-SVM rally-vs-nonrally classification with smoothing, SmartTennisTV (arXiv:1801.01430) rally segmentation + scoring-rule post-correction, and self-learning-from-little-labeled-data sports work. These don't make the design 'reinventing' wholesale ? none combine static-camera physics-derived (value, confidence) cue

COLUMNS + a deliberately tiny logistic-regression scorer + a report-first/A/B-gated/evidence-vs-decision contradiction reconciler + LOVO-by-match on vendor golden intervals in one system ? but they do show every individual ingredient and most pairwise combinations are established, so the contribution is the assembly + governance discipline, not any single mechanism. Net: partial-overlap is correct; the survey should additionally cite constraint-aware decoding (2605.10149), the Kijak tennis play/break line, and SmartTennisTV, and should soften any implicit claim that 'multi-cue features + light learned scorer + rule post-processing for rally segmentation' is itself new.

["title": "Improving Temporal Action Segmentation via Constraint-Aware Decoding (Yeo Keat Ee, Roy, Li, Zhang, Fernando)", "url_or_id": "arXiv:2605.10149", "year": "2026", "why_closer": "Closer to the RECONCILER than Activity Grammars: injects data-derived structural priors (transition confidence, permitted action-boundary sets, per-class duration) as inference-time post-processing on a FROZEN base TAS model without retraining, and resolves them GLOBALLY via a modified Viterbi decode. This is precisely the survey's own recommended 'reframe reconciler rules as a grammar and resolve globally' upgrade ? already published. Concurrent (May 2026), so it doesn't void novelty, but the reconciler should cite it as the nearest published instantiation."], [{"title": "SmartTennisTV: Automatic indexing of tennis videos (Ghosh & Jawahar)", "url_or_id": "arXiv:1801.01430", "year": "2018", "why_closer": "Direct racket-sports analog of the end-to-end pipeline: temporally segments a match into rallies/points and then applies RULE-BASED post-processing (knowledge of the tennis scoring system) to refine/correct the recognized output ? the same 'structural domain prior as post-correction' move as the reconciler, in a sister sport. Differs in being broadcast (multi-shot) video vs static-camera physics cues."}, {"title": "Audiovisual Integration for Tennis Broadcast Structuring (Kijak, Gros, et al., IRISA/CBMI 2003)", "url_or_id": "https://www.irisa.fr/metiss/publis/kijak-cbmi03.pdf", "year": "2003", "why_closer": "Early but directly on-task: fuses audio + visual cues and applies an HMM with explicit STRUCTURAL/temporal rules to segment tennis into play (rally) vs break units. Establishes 'fuse multi-modal cues + impose structural priors to segment a racket-sports match into rally/non-rally' as long-standing prior art, weakening any framing of multi-cue fusion-for-rally-segmentation as novel."}, {"title": "HOG + chi2-kernel SVM rally vs non-rally classification with temporal smoothing (racket-sports broadcast indexing line, e.g. surveys/SmartTennisTV ecosystem)", "url_or_id": "(see A Survey on Content-Aware Video Analysis for Sports, arXiv:1703.01170)", "year": "2017", "why_closer": "Demonstrates the exact 'handcrafted features + lightweight (SVM/LR) classifier + smoothing post-process to remove spurious boundaries' recipe the survey treats as a scarce-label-appropriate novelty. The lightweight-classifier-on-handcrafted-features-with-smoothing pattern for rally/non-rally is established; the project's contribution narrows to the physics-cue taxonomy + logistic scorer + reconciler governance, not the recipe itself."}], [{"fabrication_flags": [], "extra_course_corrections": ["ADOPT: Cite 'Improving Temporal Action Segmentation via Constraint-Aware Decoding' (arXiv:2605.10149, ICPR 2026) alongside Activity Grammars ? it is the closest published precedent for the reconciler's recommended end-state (data-derived structural priors + global Viterbi decode as post-processing on frozen output) and directly validates the survey's own 'reframe rules as a grammar, resolve globally' technique. It also models priors STATISTICALLY from data (transition confidence, boundary sets, duration) rather than hand-specifying them ? a concrete template for replacing hand-tuned reconciler thresholds with data-fit priors, which mitigates the survey's own 'tuning thresholds on the test set' pitfall."], "ADOPT: Position against the racket-sports rally-segmentation lineage (Kijak tennis play/break, SmartTennisTV, HOG+SVM+smoothing), not only against generic TAS/TAL. The honest contribution claim is narrow: physics-derived (value, confidence) cue columns + a deliberately interpretable logistic scorer + a report-first/A-B-gated reconciler keyed on evidence-vs-decision CONTRADICTION + LOVO-by-match on vendor golden intervals. Avoid implying 'multi-cue features feeding a light learned scorer for rally segmentation' is itself novel ? it is ~20 years old in tennis."], "AVOID: Do not relax the Springer-2024 'method unverified' caveat into a strong novelty claim over it. Verification shows See et al. (2024) use a gradient/shuttle-rest rule for rally end and the SN-CS 2025 paper uses BiLSTM service + >=80%-no-motion end rule ? both confirm the cues are sound but hard-code them as verdicts. The 'signal not verdict, learn the weights' contrast is real and defensible, but it is a design-philosophy delta, not a capability the prior work lacks."], "AVOID: The survey's own search log lists a 'constraint-aware decoding ... grammar prior inference time' query yet the resulting closest paper (2605.10149) is absent from key_papers. Treat this as a coverage gap: when a search query is run, the top on-theme hit should be triaged into key_papers or explicitly dismissed, otherwise the closest analog silently drops out of the citation set (as happened here)."}], [{"theme": "Multimodal late/score/feature fusion with a learned arbiter; cue combination", "short": "late-fusion", "our_pattern": "Our \"signal not verdict / no special-casing\": every cue is a weighted feature consumed by a learned arbiter, never a branch.", "survey": {"theme": "Multimodal late/score/feature fusion with a learned arbiter; cue combination (\"signal not verdict / no special-casing\")", "our_pattern": "Every cue (frame-scope

detectors: shuttle CNN, person detector, audio thwack, per-frame shuttle-state; window-scope and session-scope analyzers) is emitted as a time-aligned (value, confidence) FEATURE COLUMN. A small learned arbiter (logistic regression today) consumes these columns and decides which candidate windows are real rallies. Cues are never hand-coded if/else branches in the decision path ? the model learns to weight them. Adding a cue = \"register a producer + add a feature column.\" Detection (expensive, persisted once) is decoupled from decision (cheap, pure re-scoring over stored features). A separate report-first \"reconciler\" linter flags where the decision contradicts persisted evidence. New cues land flag-gated default-OFF, graded by temporal-IoU vs scarce golden intervals with leave-one-video-out (grouped by match), promoted only on measured lift.\"

\"key_papers\": [{\"title\": \"Early versus late fusion in semantic video analysis\", \"authors\": \"Cees G. M. Snoek, Marcel Worring, Arnold W. M. Smeulders\", \"year\": \"2005\", \"venue\": \"ACM Multimedia\", \"url_or_id\": \"10.1145/1101149.1101236 (PDF: [http://ivl.fnwi.uva.nl/isis/publications/2005/SnoekICM2005/SnoekICM2005.pdf](\"http://ivl.fnwi.uva.nl/isis/publications/2005/SnoekICM2005/SnoekICM2005.pdf\"))\", \"what_it_does\": \"Seminal paper that names and contrasts 'early fusion' (fuse modalities in feature space, one classifier) vs 'late fusion' (per-modality classifiers, then combine in 'semantic space'). On 184h of broadcast video / 20 concepts, late fusion is usually slightly better, but where early fusion wins it wins big. Establishes the exact vocabulary our pattern lives in.\"\", \"overlap\": \"strong\", \"overlap_explanation\": \"Our arbiter consuming per-cue (value, confidence) columns is textbook LATE/score-level fusion in their taxonomy (each producer ~ a unimodal 'detector output', the logistic regression = the supervised combiner 'in semantic space'). We are squarely reinventing/instantiating their framework. We should adopt their terminology verbatim and cite it; their finding (late usually fine, but early sometimes much better) is a direct caution that a pure-late design can leave cross-cue interactions on the table.\"}], {\"title\": \"Stacked Generalization\", \"authors\": \"David H. Wolpert\", \"year\": \"1992\", \"venue\": \"Neural Networks 5(2):241-259\", \"url_or_id\": \"https://www.semanticscholar.org/paper/bbc25a700e51984e560eae27df1587baa92e3afe\", \"what_it_does\": \"Defines 'stacking': train base models, then train a meta-learner on their (cross-validated, out-of-fold) outputs to learn the best combination. Crucially warns the meta-features must come from held-out folds to avoid the meta-learner overfitting base-model training set leakage.\"\", \"overlap\": \"strong\", \"overlap_explanation\": \"Our learned arbiter over producer outputs IS a stacking meta-learner; producers are heterogeneous base models, logistic regression is the level-1 combiner. The established name for our 'signal not verdict' pattern is stacked generalization / a meta-learner. Their out-of-fold-prediction discipline is a concrete correction: the features the arbiter trains on (especially any cue derived from a learned detector) should be produced on data the detector/scorer didn't train on, or we risk optimistic lift estimates.\"}], {\"title\": \"A framework for flexible summarization of racquet sports video using multiple modalities\", \"authors\": \"Chen, Kolekar, et al. (Kolekar-line racquet-sports work)\", \"year\": \"2009\", \"venue\": \"Computer Vision and Image Understanding (CVIU) / ScienceDirect S1077314208001355\", \"url_or_id\": \"https://www.sciencedirect.com/science/article/abs/pii/S1077314208001355\", \"what_it_does\": \"Racquet-sports (incl. badminton/table-tennis) pipeline: unsupervised shot clustering + supervised audio classification produce mid-level visual+audio patterns; a temporal VOTING scheme fuses them (using audio-video correspondence) to label RALLY vs BREAK scenes; then a LINEAR 'excitement' model ranks rallies using affective features (cheer duration, commentator pitch).\", \"overlap\": \"strong\", \"overlap_explanation\": \"This is the closest domain analog: same task (rally/break segmentation of racquet-sports video), same multi-cue audio+visual decomposition, and a LINEAR model on aggregated features for the downstream ranking ? almost exactly our window-scope features + logistic-regression arbiter, plus a learning-to-rank head. Differences: they fuse the segmentation decision by a hand-tuned temporal voting scheme (more rule-like) whereas our thesis is to LEARN the segmentation weighting too; and their excitement features (cheer/commentator) assume broadcast audio we may lack on static-tripod amateur footage. Strong prior art ? our novelty is mostly the engineering discipline (persist-once/re-score, flag-gated no-regression), not the fusion idea.\"}], {\"title\": \"Towards Structured Analysis of Broadcast Badminton Videos\", \"authors\": \"Anurag Ghosh, Suriya Singh, C. V. Jawahar\", \"year\": \"2018\", \"venue\": \"WACV 2018 (arXiv:1712.08714)\", \"url_or_id\": \"https://arxiv.org/abs/1712.08714\", \"what_it_does\": \"End-to-end structured analysis of broadcast badminton: player detection/tracking/ID, stroke annotation, and rally ('point') SEGMENTATION at 95.44% on 10 Olympic matches/20 players, enabling derived metrics (reaction time, footwork).\", \"overlap\": \"partial\", \"overlap_explanation\": \"Same sport and the same rally-segmentation sub-task, with a comparable small corpus (~10 matches, like our ~6). But it is broadcast-view, player/stroke-centric, and does not foreground a (value, confidence)-feature-column-to-learned-arbiter design or shuttle-trajectory cues ? its segmentation mechanism is not the late-fusion-of-cue-features story. Useful as a domain baseline and a metric reference, not as the same fusion architecture.\"}], {\"title\": \"MoCaE: Mixture of Calibrated Experts Significantly Improves Object Detection\", \"authors\": \"Kemal Oksuz, Selim Kuzucu, Tom Joy, Puneet K. Dokania\", \"year\": \"2023/2024\", \"venue\": \"TMLR

(arXiv:2309.14976)", "url_or_id": "https://arxiv.org/abs/2309.14976", "what_it_does": "Shows naively combining expert detectors HURTS because their confidences are miscalibrated and the most-confident expert dominates. Fix: CALIBRATE each expert's scores (isotonic / linear regression) BEFORE fusing. Yields up to ~2.5 AP and SOTA on COCO/LVIS/DOTA.", "overlap": "partial", "overlap_explanation": "Different task (object detection) but the same structural pattern as ours ? combine multiple expert detectors' (value, confidence) outputs ? and it pinpoints the exact failure mode our raw confidences invite: heterogeneous, miscalibrated per-producer confidences let one over-confident cue dominate the arbiter. Direct course-correction: calibrate each producer's confidence (Platt/isotonic) before it enters the logistic regression."}, {"title": "Modality Competition: What Makes Joint Training of Multi-modal Network Fail in Deep Learning? (Provably)", "authors": "Yu Huang, Junyang Lin, Chang Zhou, Hongxia Yang, Longbo Huang", "year": "2022", "venue": "ICML 2022"}, {"arXiv:2203.12221", "url_or_id": "https://arxiv.org/abs/2203.12221", "what_it_does": "Proves/shows 'modality competition': in jointly trained multimodal nets the best UNIMODAL net can beat the joint model; some modalities are never properly learned ('greedy' modality use), hurting generalization.", "overlap": "weak", "overlap_explanation": "We don't joint-train a deep multimodal net (we use a linear arbiter over pre-computed features, which largely sidesteps this), so the overlap is conceptual. But it is the canonical citation for the late-fusion pitfall 'a strong single cue can swamp weaker-but-complementary cues,' which our logistic regression can replicate via dominant weights ? motivating per-cue ablations and regularization."}, {"title": "DBF: Dynamic Belief Fusion for Combining Multiple Object Detectors", "authors": "Hyungtae Lee, Heesung Kwon, et al.", "year": "2019 (WACV; arXiv:2204.02890)", "url_or_id": "https://arxiv.org/abs/2204.02890", "what_it_does": "Assigns detection scores a basic probability assignment over {target, non-target, unknown} and fuses multiple detectors with Dempster-Shafer theory, explicitly modeling per-detector UNCERTAINTY rather than trusting raw scores.", "overlap": "weak", "overlap_explanation": "Same problem family (principled combination of multiple detectors' confidences) and reinforces 'model uncertainty, don't trust raw scores.' Different math (evidential/Dempster-Shafer vs logistic regression). Useful as an alternative arbiter formulation and as support for treating a producer's low-confidence/missing output as an explicit 'unknown' feature rather than a silent zero."}, {"title": "Highlight Detection with Pairwise Deep Ranking for First-Person Video Summarization", "authors": "Ting Yao, Tao Mei, Yong Rui", "year": "2016", "venue": "CVPR 2016 (IEEE 7780481)", "url_or_id": "https://www.microsoft.com/en-us/research/publication/highlight-detection-with-pairwise-deep-ranking-for-first-person-video-summarization/", "what_it_does": "Frames highlight detection as a segment-level RANKING task with pairwise (highlight > non-highlight, same video) constraints; reports ~10.5% over RankSVM on 100h/15 sports.", "overlap": "weak", "overlap_explanation": "Adjacent framing: if/when we score-and-rank candidate windows rather than just binary-classify them, the pairwise within-video ranking objective (and RankSVM as the linear baseline) is the established arbiter formulation ? and 'pairs from the SAME video' echoes our group-by-match discipline. Relevant to our scorer's loss choice, not to the cue-decomposition itself."}, {"title": "Less is More: Learning Highlight Detection from Video Duration", "authors": "Bo Xiong, Yannis Kalantidis, Deepti Ghadiyaram, Kristen Grauman", "year": "2019", "venue": "CVPR 2019"}, {"arXiv:1903.00859", "url_or_id": "https://arxiv.org/abs/1903.00859", "what_it_does": "Uses a cheap, noisy implicit signal (video duration) as weak supervision for a ranking model that prefers segments from shorter videos, with explicit noise handling ? beating supervised SOTA without manual highlight labels.", "overlap": "weak", "overlap_explanation": "Not about cue fusion, but directly relevant to our scarce-golden-label bottleneck: it shows how to mine a free, weak supervision signal (for us, e.g. the point/dead-time cadence, score-overlay changes, or our session-scope warm-up labels) to pre-train or augment the arbiter, then fine-tune on the few vendor labels. A scaling lever for our ~6-match corpus."}], "established_terms": ["Late fusion / score-level fusion (vs early/feature-level fusion) ? Snoek et al. 2005", "Stacked generalization / stacking; the arbiter = a 'meta-learner' or 'level-1 combiner' over base-model outputs ? Wolpert 1992", "Decision-level fusion / classifier combination / classifier ensembles", "Confidence/score calibration before fusion (Platt scaling, isotonic regression) ? MoCaE", "Mixture-of-Experts / gating network (learned, input-dependent weighting of experts)", "Learning-to-rank (RankSVM / pairwise ranking) for segment scoring", "Leave-one-subject-out / grouped (leave-one-group-out) cross-validation ? the formal name for our leave-one-video-out-by-match", "Temporal Action Localization (TAL) and play/break (rally/break) segmentation ? the field's task names for what we do", "Data validation / data-quality assertions for ML pipelines (e.g. Google TFX Data Validation) ? the established framing for our 'report-first linter' reconciler"], "techniques_to_adopt": [{"technique": "Calibrate each producer's confidence (Platt scaling for the binary cues, isotonic regression for monotone ones) BEFORE it enters the logistic-regression arbiter.", "why": "MoCaE shows un-calibrated heterogeneous expert confidences let the most over-confident cue dominate and degrade the fused result. Our producers"}]

(a CNN softmax, an audio onset score, a heuristic net-crossing count) live on wildly different scales/calibrations; a calibration layer per column makes the arbiter weights meaningful and stable on a tiny training set.", "from_paper": "MoCaE: Mixture of Calibrated Experts (Oksuz et al. 2023)", "effort": "medium"}, {"technique": "Generate the arbiter's training features with OUT-OF-FOLD producer outputs (stack honestly): when a producer is itself learned (e.g. the shuttle-state classifier), its outputs feeding the arbiter must come from data it didn't train on.", "why": "Wolpert's stacking result: a meta-learner trained on base models' in-sample predictions overfits and inflates lift. With LOVO already in place, extend the same group-by-match split to producer training so reported lift is honest.", "from_paper": "Stacked Generalization (Wolpert 1992)", "effort": "medium"}, {"technique": "Adopt the field's exact vocabulary in docs/code: call the pattern 'late/score-level fusion with a stacked meta-learner,' name the arbiter a 'level-1 combiner,' and call LOVO 'grouped (leave-one-match-out) cross-validation.'", "why": "Reviewers/collaborators map instantly to 30 years of literature; it also signals you know the failure modes. Cheap, high-leverage framing change.", "from_paper": "Snoek et al. 2005; Wolpert 1992; scikit-learn GroupKFold/LeaveOneGroupOut docs", "effort": "low"}, {"technique": "Add a few targeted EARLY-fusion / interaction features rather than staying purely late-fusion (e.g. shuttle-in-air AND person-two-sidedness co-occurrence within the window).", "why": "Snoek et al.: late fusion usually ties early, but where cross-modal interactions matter, early fusion wins by a lot. A pure-late linear arbiter cannot represent AND/XOR interactions between cues; a handful of hand-crossed product features (still 'just feature columns,' no branches) recover that cheaply on a linear model.", "from_paper": "Early versus late fusion in semantic video analysis (Snoek et al. 2005)", "effort": "low"}, {"technique": "Represent a missing/low-confidence producer output as an explicit 'unknown' indicator feature, not a silent 0.", "why": "DBF treats per-detector uncertainty as a first-class state ({target, non-target, unknown}); collapsing 'cue absent' and 'cue says no' to the same 0 confuses a linear arbiter. An is_present/uncertainty companion column per cue lets the model learn to discount absent cues.", "from_paper": "DBF: Dynamic Belief Fusion (Lee et al.)", "effort": "low"}, {"technique": "If/when ranking candidate windows by rally-likelihood, train the arbiter with a pairwise (within-match) ranking loss / RankSVM rather than only pointwise log-loss.", "why": "Yao et al. show pairwise within-video ranking beats pointwise scoring for highlight/segment selection and is robust to absolute-score miscalibration across videos ? and 'pairs within the same video' aligns with grouping by match.", "from_paper": "Highlight Detection with Pairwise Deep Ranking (Yao et al. 2016)", "effort": "medium"}, {"technique": "Mine a cheap weak-supervision signal (point/dead-time cadence, scoreboard-overlay change, warm-up-vs-match cadence) to pre-train/augment the arbiter, then fine-tune on the scarce vendor labels.", "why": "Directly attacks the ~6-match label bottleneck; 'Less is More' shows a noisy free signal + explicit noise handling can match supervised SOTA. Your session-scope warm-up detector is already such a signal.", "from_paper": "Less is More: Learning Highlight Detection from Video Duration (Xiong et al. 2019)", "effort": "high"}, {"technique": "Frame the reconciler explicitly as 'data-quality assertions over the decision layer' and borrow that ecosystem's discipline (declarative constraints, schema/anomaly reports, severity levels).", "why": "Google's Data Validation for ML and DuckDQ formalize exactly the 'detect-and-report constraint violations, don't silently mutate' contract your report-first reconciler wants; adopting their assertion/constraint framing gives a vetted structure (and shows it's not an ad-hoc invention).", "from_paper": "Data Validation for Machine Learning (Breck et al., MLSys 2019); DuckDQ", "effort": "medium"}], "pitfalls": ["Dominant-cue swamping: a single high-magnitude or over-confident producer (e.g. the shuttle CNN softmax) can dominate the linear arbiter and starve weaker-but-complementary cues ? the 'modality competition' / 'most-confident-expert-dominates' failure (Modality Competition 2022; MoCaE).", "Miscalibration across heterogeneous producers: raw scores from a CNN, an audio onset, and an integer net-crossing count are not on comparable probability scales; un-calibrated, the arbiter's learned weights are unstable and the (value, confidence) semantics are illusory (MoCaE).", "Stacking leakage / optimistic lift: training the arbiter on in-sample outputs of learned producers (or re-using the same matches to both fit a producer and grade the arbiter) inflates measured lift ? Wolpert's out-of-fold requirement; compounds with tiny corpora.", "Pure-late-fusion misses cross-cue interactions: a linear late-fusion arbiter cannot model AND/XOR cue interactions that early fusion captures; on concepts where interactions matter, the gap is large (Snoek et al. 2005).", "Group-CV done at the wrong granularity: leaving out a held-out window from the SAME video (record-wise) instead of the whole match (subject-wise) leaks scene/player/lighting and massively over-estimates generalization ? your LOVO-by-match is correct precisely to avoid this (leave-one-subject-out literature).", "Overfitting the arbiter on a ~6-match golden set: even a 'tiny' logistic regression over many engineered feature columns can overfit; needs strong regularization, few features, and honest grouped CV ? and the lift signal itself is high-variance at this corpus size.", "Confounding 'cue absent' with 'cue says no': silently encoding a missing producer output

as 0 confidence biases the linear model (DBF's 'unknown' state argument).", "Confidence drift / reproducibility: 'persist once, re-score forever' is only sound if persisted producer outputs (and their calibration) are versioned; a silent producer/model-version change makes re-scored experiments non-comparable.", "Metric inflation from permissive evaluation: the sports-event-detection survey warns that allowing multiple/over-lapping positive predictions inflates scores; ensure temporal-IoU matching is one-to-one (top-1 / non-max-style) so reconciler 'lift' is real, not bookkeeping.", "Broadcast-audio assumptions don't transfer: the closest racquet-sports prior art (Kolekar/Chen) leans on cheer/commentator audio; on static-tripod amateur footage those cues are absent or contaminated (multiple courts), so importing their excitement features wholesale will under-deliver.", "verdict": "partial-overlap", "verdict_rationale": "The CORE of our 'signal not verdict / learned arbiter over cue feature-columns' pattern is well-established and we are, on that axis, reinventing named prior art: it is late/score-level fusion (Snoek et al. 2005) realized as stacked generalization with a linear meta-learner (Wolpert 1992), and the very same multi-cue audio+visual rally/break decomposition with a linear downstream model already exists IN OUR EXACT DOMAIN (Kolekar/Chen racquet-sports framework; Ghosh et al. badminton). We should cite and adopt that vocabulary rather than present the fusion idea as novel. It is only 'partial-overlap' rather than 'reinventing' because the literature's analogs typically (a) fuse the segmentation DECISION with hand-tuned voting/rules (the thing we deliberately push into the learned arbiter), and (b) do not pair this with our specific engineering discipline ? detection/decision decoupling ('persist once, re-score forever'), flag-gated default-OFF no-regression promotion on measured lift, and the report-first reconciler-as-linter. That engineering envelope is a genuinely useful combination, but the FUSION mechanism itself is not new, and several concrete techniques (calibrate-before-fuse, out-of-fold stacking, a few early-fusion interaction features, pairwise ranking, weak-supervision pre-training, explicit 'unknown' states) are off-the-shelf course-corrections we are currently missing.", "search_queries_used": ["multimodal fusion survey early late score-level feature fusion learned arbiter", "stacked generalization stacking ensemble meta-learner combine classifier outputs", "sports video rally detection play break segmentation multimodal audio visual features", "badminton broadcast video rally segmentation temporal action localization deep learning", "pitfalls late fusion multimodal greedy modality calibration unimodal outperforms joint training", "mixture of experts product of experts gating network combining cues calibration", "learning to rank highlight detection sports video pairwise ranking features arbiter", "Kolekar racquet sports video summarization audio visual multiple modalities rally break excitement HMM", "temporal action detection refinement post-processing boundary correction over-segmentation merge split", "calibration confidence score combination classifiers Platt scaling late fusion weighting", "Snoek Worrying Smeulders early versus late fusion semantic video analysis 2005", "Wolpert stacked generalization 1992 meta-learning combining models", "MoCaE Mixture of Calibrated Experts object detection calibration before fusion arXiv", "data validation assertions machine learning pipeline error detection report constraints model predictions contradict", "Less is More learning highlight detection video duration weak supervision Xiong ranking", "grouped cross validation leakage subject-wise leave-one-group-out evaluation small dataset overfitting"]], "verify": {"theme": "Multimodal late/score/feature fusion with a learned arbiter; cue combination (\\signal not verdict / no special-casing\\)", "citations_checked": [{"title": "Early versus late fusion in semantic video analysis (Snoek, Worrying, Smeulders, ACM MM 2005)", "verified_real": "yes", "note": "Confirmed real. DOI 10.1145/1101149.1101236, ACM MM 2005, all three authors correct, UVA-hosted PDF resolves. Content described accurately (184h broadcast video, 20 concepts, late usually slightly better but early wins big where it wins). Textbook source for the early/late/score-level vocabulary the pattern lives in."}, {"title": "Stacked Generalization (Wolpert, Neural Networks 5(2):241-259, 1992)", "verified_real": "yes", "note": "Confirmed real. Neural Networks 5(2):241-259, 1992, DOI 10.1016/S0893-6080(05)80023-1. Author and out-of-fold/cross-validated-meta-feature description accurate. Correctly invoked as the canonical name for the learned-arbiter-over-producers pattern (stacking / level-1 combiner) and the leakage caution."}, {"title": "A framework for flexible summarization of racquet sports video using multiple modalities (CVIU 2009, S1077314208001355)", "verified_real": "yes", "note": "PAPER is real (CVIU vol 113 issue 3 (2009) pp.415-424, ScienceDirect ID S1077314208001355 matches) and content is accurately described (rally/break segmentation + excitement ranking with a linear model on aggregated A/V features). BUT AUTHORS ARE MISATTRIBUTED: the survey credits 'Chen, Kolekar, et al. (Kolekar-line)'. The actual authors are Liu, Huang, Jiang, Xing, Ye, Gao. This is a real paper with a fabricated/wrong author line ? must be corrected before citing. The 'Kolekar racquet-sports' framing in the rationale and pitfalls is therefore unsupported by THIS reference."}, {"title": "Towards Structured Analysis of Broadcast Badminton Videos (Ghosh, Singh, Jawahar, WACV 2018, arXiv:1712.08714)", "verified_real": "yes", "note": "Confirmed real. WACV 2018, arXiv 1712.08714, IIIT-H CVIT project page exists. Authors and 95.44% point-segmentation figure accurate. 'partial' overlap rating is fair: same sport/sub-task but broadcast-view,

player/stroke-centric, not a (value,confidence)-column-to-arbiter design."},{ "title": "MoCaE: Mixture of Calibrated Experts Significantly Improves Object Detection (Oksuz, Kuzucu, Joy, Dokania, TMLR 2023/24, arXiv:2309.14976)", "verified_real": "yes", "note": "Confirmed real. arXiv 2309.14976, TMLR, fiveai GitHub impl, OpenReview. Calibrate-before-fuse / most-confident-expert-dominates message accurately summarized (~2.5 AP, SOTA on COCO/LVIS/DOTA). Strong support for the calibrate-each-producer course-correction." }, { "title": "Modality Competition: What Makes Joint Training of Multi-modal Network Fail (Huang et al., ICML 2022, arXiv:2203.12221)", "verified_real": "yes", "note": "Confirmed real. ICML 2022 (PMLR 162:9226-9259), arXiv 2203.12221, all five authors correct. Accurately described. 'weak' overlap rating is honest ? it's about joint-trained deep nets, not a linear arbiter, so it's a conceptual/motivational cite for dominant-cue swamping, as the survey says." }, { "title": "DBF: Dynamic Belief Fusion for Combining Multiple Object Detectors (Lee, Kwon, arXiv:2204.02890)", "verified_real": "yes", "note": "Confirmed real. arXiv 2204.02890; {target, non-target, intermediate/unknown} basic-probability-assignment + Dempster-Shafer fusion described accurately. Minor venue imprecision: the 2204.02890 version is the journal/extended (Neurocomputing) form; the original DBF is WACV 2016 (arXiv:1511.03183). Not a fabrication, just a loose venue tag on the 'unknown-state' course-correction." }, { "title": "Highlight Detection with Pairwise Deep Ranking for First-Person Video Summarization (Yao, Mei, Rui, CVPR 2016)", "verified_real": "yes", "note": "Confirmed real. CVPR 2016 pp.982-990, IEEE 7780481, MSR page. ~10.5% over RankSVM on 100h/15 sports accurate. Correctly scoped as relevant to a future ranking head + within-video pairwise grouping, not to cue decomposition." }, { "title": "Less is More: Learning Highlight Detection from Video Duration (Xiong, Kalantidis, Ghadiyaram, Grauman, CVPR 2019, arXiv:1903.00859)", "verified_real": "yes", "note": "Confirmed real. CVPR 2019 pp.1258-1267, arXiv 1903.00859, UT-Austin project page, all four authors correct. Weak-supervision-from-a-free-noisy-signal framing accurate. Honestly scoped as a label-scarcity lever, not a fusion claim." }, { "title": "Data Validation for Machine Learning (Breck/Polyzotis et al., MLSys 2019; TFX Data Validation)", "verified_real": "yes", "note": "Confirmed real. MLSys 2019 (Polyzotis, Zinkevich, Roy, Breck, Whang), TFDV at Google. Correctly invoked as the established 'detect-and-report constraint violations, don't silently mutate' framing for the report-first reconciler. (DuckDQ also exists as a data-quality lib; not separately verified but the TFDV cite alone supports the point.)" }], "fabrication_flags": [], "corrected_verdict": "partial-overlap", "corrected_rationale": "Verdict HELD at partial-overlap, and tightened. All 10 cited works are REAL ? no fabricated papers ? so the survey passes the fabrication screen. The one citation defect is a real-paper-WRONG-authors error: the CVIU 2009 racquet-sports summarization paper (S1077314208001355) is by Liu/Huang/Jiang/Xing/Ye/Gao, NOT 'Chen/Kolekar', so any 'Kolekar-line' framing in the rationale/pitfalls is unsupported and must be re-attributed before use. On novelty, my adversarial searches REINFORCE rather than overturn the survey's own concession: the fusion MECHANISM is squarely prior art. Late/score-level fusion (Snoek 2005) realized as stacking with a linear meta-learner (Wolpert 1992) is the exact named pattern; rally/break play-segmentation by multi-cue audio+visual fusion with a downstream linear/HMM model already exists in-domain (the real CVIU 2009 paper; Ghosh 2018 for badminton); and freshly, recent badminton work fuses shuttle-trajectory (TrackNet) + player/hit detection for rally/shot structure (e.g. MDPI Sensors 2024 24(13):4372). I could NOT find a single closer precedent that bundles the survey's actual claimed novelty ? the ENGINEERING ENVELOPE: detection/decision decoupling ('persist once, re-score forever'), flag-gated default-OFF cue promotion graded by temporal-IoU under grouped (leave-one-match-out) CV on a no-regression bar, and a report-first reconciler-as-linter over the decision layer. Each ingredient of that envelope is itself off-the-shelf (feature stores / precompute-then-rescore in MLOps; ablation studies; GroupKFold/LeaveOneGroupOut; temporal-IoU TAL eval; TFDV-style data-quality assertions), so the novelty is an integration/discipline claim, not a research contribution. That is exactly what 'partial-overlap' should mean: reinventing the fusion idea (cite, don't claim), with a genuinely useful-but-not-novel engineering combination on top. Downgrading to 'reinventing' would overstate it (the envelope is a real, uncommon-in-the-papers combination); upgrading to 'novel-combination' would overclaim (the combination is assembled from standard parts and the load-bearing fusion idea is 30 years old in-domain).", "closer_prior_art": [{ "title": "A framework for flexible summarization of racquet sports video using multiple modalities", "url_or_id": "CVIU 2009, S1077314208001355 (correct authors: Liu, Huang, Jiang, Xing, Ye, Gao)", "year": "2009", "why_closer": "Already cited by the survey but UNDER A WRONG AUTHOR LINE; flagged here so the real attribution is the one used. It is the closest in-domain analog (rally/break segmentation of racquet sports by multimodal A/V fusion + a linear excitement-ranking model), which is precisely why the fusion idea cannot be claimed as novel." }, { "title": "Enhancing Badminton Game Analysis: Shot Refinement via Fusion of Shuttlecock Tracking and Hit Detection from a Monocular Camera", "url_or_id": "MDPI Sensors 2024, 24(13):4372 (PMC11244353)", "year": "2024", "why_closer": "Recent, same sport, same modality stack as ours (TrackNet shuttle trajectory + YOLO player/hit detection fused to recover shot/rally structure,

69 BWF matches). Shows the specific shuttle-trajectory + person/hit cue fusion for rally/shot structure is contemporary prior art, narrowing our cue-fusion novelty further than the survey states."},{ "title": "Fusion of AV features and external information sources for event detection in team sports video", "url_or_id": "ResearchGate 220214385 (sports AV event-detection fusion line)", "year": "~2008", "why_closer": "Reinforces that score-level fusion of audio+visual cues with a learned combiner for sports event/play-break detection is a long-standing standard formulation, not a new arbiter design."}], "extra_course_corrections": ["CITATION HYGIENE (must-fix): re-attribute the CVIU 2009 racquet-sports paper to Liu/Huang/Jiang/Xing/Ye/Gao and delete the 'Kolekar-line / Chen, Kolekar et al.' label wherever it appears (key_papers, verdict_rationale, and the 'broadcast-audio / Kolekar-Chen excitement features' pitfall). The pitfall's substance (broadcast cheer/commentator audio won't transfer to static-tripod amateur footage) stands; only the author attribution is wrong.", "VENUE NICETY (low): DBF arXiv:2204.02890 is the extended journal version; the canonical first DBF is WACV 2016 (arXiv:1511.03183). Cite both if the 'unknown-state' argument is load-bearing.", "ADOPT (high-leverage, beyond what survey lists): explicitly position the contribution as the ENGINEERING ENVELOPE, not the fusion ? i.e., 'late/score-level stacked fusion (standard) operationalized as a no-regression, flag-gated, grouped-CV cue-promotion pipeline with a persist-once/re-score split and a TFDV-style report-first reconciler.' This is defensible and honest; presenting the fusion itself as novel is not.", "ADOPT: the survey's calibrate-before-fuse (MoCaE), out-of-fold stacking (Wolpert), explicit unknown-state column (DBF), and a few early-fusion interaction terms (Snoek) are all corroborated by the verified sources and are the right concrete gaps to close ? prioritize calibration + out-of-fold first, since both are cheap and both directly threaten the honesty of measured lift on a ~6-match corpus.", "AVOID: do not lean on broadcast-audio excitement features (cheer/commentator pitch) from the in-domain prior art ? confirmed by search to assume broadcast conditions our static-tripod amateur footage lacks; importing them wholesale will under-deliver and can be actively misleading on multi-court venues."]}], "theme": "Cascades / two-stage proposal-verify; decouple detection from decision; offline re-scoring & feature caching", "short": "cascade-rescore", "our_pattern": "Our detection?decision \"persist once, re-score forever\": expensive detection once + cheap re-scorable decision.", "survey": {"theme": "Cascades / two-stage proposal-verify; decouple detection from decision; offline re-scoring & feature caching", "our_pattern": "\"Persist once, re-score forever\": expensive per-frame perception (a frozen shuttle-tracking CNN like WASB/TrackNet, a person detector, an audio thwack detector, a proposed per-frame shuttle-state classifier) runs ONCE over the video and is persisted as time-aligned signals. The cheap DECISION layer ? windowing into candidate rally intervals + a tiny logistic-regression scorer over per-window feature columns ? is a pure, deterministic, offline re-scoring step. Most experiments re-score stored features and never touch the served perception pipeline. Detection ? decision is the architectural axis; cues are (value, confidence) feature columns the scorer learns to weight (no hand-coded if/else in the decision path); reconcilers are a report-first post-hoc pass that flags where the decision contradicts persisted evidence; everything is graded offline with leave-one-video-out vs scarce golden intervals using temporal IoU, flag-gated default-OFF, promoted only on measured lift.", "key_papers": [{"title": "Temporal Action Localization in Untrimmed Videos via Multi-stage CNNs (Segment-CNN / S-CNN)", "authors": "Zheng Shou, Dongang Wang, Shih-Fu Chang", "year": "2016", "url_or_id": "arXiv:1601.02129 (CVPR 2016)", "venue": "CVPR 2016", "what_it_does": "The canonical two-stage temporal-action-localization framework: a PROPOSAL network filters background segments, a CLASSIFICATION network is trained one-vs-all, and a LOCALIZATION network refines/scores instances. Reports mAP at temporal IoU. Establishes proposal->classify->localize decoupling for untrimmed video, the direct analog of our window-then-score split.", "overlap": "strong", "overlap_explanation": "Same core structure as ours: a cheap proposal/candidate-window stage decoupled from a downstream scoring/classification stage, evaluated at temporal IoU on untrimmed video. Differences: S-CNN uses 3D-ConvNets end-to-end and is NOT 'persist perception once, re-score many'; ours freezes perception and re-scores a tiny LR. But the proposal/decision decoupling and the tIoU metric are exactly the established framing we are using."}], {"title": "Cascaded Boundary Regression for Temporal Action Detection (CBR)", "authors": "Jiyang Gao, Zhenheng Yang, Ram Nevatia", "year": "2017", "url_or_id": "arXiv:1705.01180 (BMVC 2017)", "venue": "BMVC 2017", "what_it_does": "Two-stage cascade: stage 1 detects class-agnostic proposals, stage 2 detects specific actions; temporal boundaries are refined in a cascaded way by feeding refined windows back for further regression. Reports tIoU/mAP on THUMOS-14.", "overlap": "strong", "overlap_explanation": "Directly the 'cascade / two-stage proposal-verify + boundary refinement' pattern. Its cascaded boundary regression is the learned analog of our reconcilers that tighten segment boundaries to serve-contact/inactivity. Difference: CBR's refinement is a learned regressor inside the model; our boundary tightening is a report-first rule pass A/B-gated before auto-apply."}], {"title": "Decoupled Classification Refinement: Hard False Positive Suppression for Object Detection (DCR)", "authors": "Bowen Cheng,

Yunchao Wei, Rogerio Feris, Jinjun Xiong, Wen-mei Hwu, Thomas Huang, Honghui Shi", "year": "2018", "url_or_id": "arXiv:1810.04002", "venue": "ECCV 2018 / arXiv", "what_it_does": "Adds a STANDALONE classifier that re-scores a base detector's boxes to kill HARD FALSE POSITIVES, deliberately NOT sharing features with the detector. At inference all detections are rescored by the decoupled head.", "overlap": "strong", "overlap_explanation": "This is the cleanest published statement of 'decouple a cheap detector from a separate verify/rescore stage to suppress hard false positives,' which is exactly what our LR scorer + reconcilers do to candidate windows (drop zero-net-crossing segments, etc.). Spatial object detection vs our temporal rallies, but the decouple-then-rescore principle and the 'hard false positive' framing are identical."}, {"title": "Rapid Object Detection using a Boosted Cascade of Simple Features (Viola-Jones rejection cascade)", "authors": "Paul Viola, Michael Jones", "year": "2001", "url_or_id": "CVPR 2001 (doi:10.1109/CVPR.2001.990517)", "venue": "CVPR 2001", "what_it_does": "The seminal attentional/rejection cascade: cheap early stages reject most non-object windows so expensive computation is spent only on promising candidates; later stages are progressively more complex.", "overlap": "partial", "overlap_explanation": "Establishes the foundational vocabulary ('cascade', 'rejection/attentional cascade', 'cheap-reject-then-expensive-verify'). We use the inverse compute ordering (expensive perception first, persisted; cheap decision second), so it is the conceptual ancestor rather than a structural twin ? overlap is on terminology and the staged compute-budget idea, not the exact pipeline."}, {"title": "Enhancing Badminton Game Analysis: Shot Refinement via Fusion of Shuttlecock Tracking and Hit Detection from a Monocular Camera", "authors": "Yi-Hua Hsu, Chih-Chang Yu, Hsu-Yung Cheng", "year": "2024", "url_or_id": "Sensors 24(13):4372, doi:10.3390/s24134372 (PMC11244353)", "venue": "Sensors (MDPI) 2024", "what_it_does": "Fuses TrackNet shuttle trajectory + YOLOv7 swing/hit detection; a rule-based Shot Refinement Algorithm (SRA) reconciles the two streams via 5 explicit decision cases (accept hit, reject trajectory in non-swing segment, dedupe multiple detections, recover missing trajectory from swing, confirm true negative). Lifts shot accuracy from 58.8% (TrackNet alone) to 89.7%.", "overlap": "strong", "overlap_explanation": "Same domain AND same reconcile-fused-detector-outputs idea as our reconcilers: cross-checking the shuttle stream against the person/swing stream to correct false positives/negatives at segment boundaries. Difference: SRA hard-codes if/else cases (exactly the 'special-casing' we deliberately avoid in the decision path), and it is shot-level not rally-level. Strong evidence our reconciler instinct is sound, and a cautionary example of the hand-coded-branch path we are trying to avoid in the scorer."}, {"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "authors": "Zhaoyu Liu, Xi Weng, Lianyu Hu, Zhe Hou, Kan Jiang, Jin Song Dong, Yang Liu", "year": "2026", "url_or_id": "arXiv:2603.13397", "venue": "arXiv 2026", "what_it_does": "To segment full match broadcasts into rally clips it runs a lightweight purely-audio racquet-impact detector (augmented with crowd/announcer noise), CLUSTERS hit moments into candidate rally intervals, then FILTERS by view/scoreboard constraints. Decouples cheap detection from downstream validation.", "overlap": "strong", "overlap_explanation": "Almost exactly our pipeline shape in a sibling sport: cheap per-frame/audio detector -> cluster into candidate intervals -> rule-filter/decision. Confirms that 'audio thwack detector -> candidate windows -> decision filter' is established practice for racket-sport rally segmentation. Difference: their decision is a fixed filter; ours is a learned scorer + LOVO-graded harness."}, {"title": "A New Perspective for Shuttlecock Hitting Event Detection", "authors": "Yu-Hsi Chen", "year": "2023", "url_or_id": "arXiv:2306.10293", "venue": "arXiv 2023 (CoachAI/badminton challenge)", "what_it_does": "Detects shuttlecock hitting events from trajectory; representative of the badminton trajectory->event pipeline that underpins rally/shot structuring from TrackNet-style per-frame outputs.", "overlap": "weak", "overlap_explanation": "Confirms the broader practice of deriving discrete events/decisions from persisted per-frame shuttle trajectory (our detection->decision split at the cue level), but it targets hit-event detection rather than rally-interval segmentation or a re-scorable decision layer, so overlap is thematic/weak."}], "established_terms": ["two-stage detection (proposal generation then classification)", "cascade / rejection (attentional) cascade", "temporal action proposal generation", "temporal action localization / detection (TAL/TAD)", "actionness score (candidate-likelihood per snippet/window)", "decoupled classification refinement / re-scoring head", "hard false positives", "cascaded boundary regression / boundary refinement", "over-segmentation error", "temporal Intersection-over-Union (tIoU) and mAP@tIoU", "frozen-backbone precomputed (cached) features", "leave-one-video-out / grouped cross-validation; video-level split", "off-policy / offline policy evaluation (replay)"], "techniques_to_adopt": [{"technique": "Adopt and report mAP@tIoU at multiple thresholds (e.g. 0.3/0.5/0.7) and average-mAP, not just a single mean temporal IoU."}], "why": "This is THE metric the temporal-action-detection field (S-CNN, CBR) expects; reporting it makes your results comparable/legible to that literature and exposes whether gains are at loose or tight boundaries (where reconcilers act).", "effort": "low", "from_paper": "Temporal Action Localization via Multi-stage CNNs (S-CNN,

Shou 2016); Cascaded Boundary Regression (Gao 2017)"},"{"technique":"Frame the candidate-window stage explicitly as 'temporal action proposal generation' and emit an 'actionness'/proposal-confidence per window that the scorer consumes.", "why":"Aligns your windowing with established proposal-generation literature, gives a principled recall-vs-#proposals knob (AR@AN), and lets you measure proposal recall separately from decision precision ? diagnosing whether misses are a windowing or a scoring failure.", "effort":"medium", "from_paper":"S-CNN (Shou 2016); BSN/two-stage TAD proposal literature"},"{"technique":"Mine and explicitly train the scorer on the detector's HARD FALSE POSITIVES (windows that look rally-like but are not), as a decoupled re-scoring head.", "why":"DCR shows that in decoupled detection most remaining errors are hard false positives from the classification (decision) side, fixable by a separate rescorer fed hard negatives ? directly applicable to your LR scorer and to choosing which reconciler rules matter most.", "effort":"medium", "from_paper":"Decoupled Classification Refinement (Cheng/Shi 2018)"},"{"technique":"Make boundary tightening a learned/iterative regression (cascaded boundary regression) rather than only fixed rule precedence, or at least benchmark your rule-based reconcilers against a CBR-style learned refiner.", "why":"CBR's cascaded boundary regression is the established way to refine temporal boundaries and gave large mAP@0.5 gains; it is the learned counterpart to your truncate/split/tighten reconcilers and a natural A/B baseline to justify staying rule-based.", "effort":"high", "from_paper":"Cascaded Boundary Regression (Gao 2017)"},"{"technique":"Treat the over-merged-segment and over-fragmented-segment cases as the named 'over-segmentation' problem and consider a boundary-detection branch / ASRF-style refinement as a learned alternative to the rule-based split/merge reconcilers.", "why":"ASRF formalizes correcting over-/under-segmentation with a dedicated boundary branch that refines a frame-wise classifier ? exactly your split-on-internal-dead-time and merge reconcilers ? giving you a citable method and a learned baseline to compare against.", "effort":"medium", "from_paper":"Alleviating Over-segmentation Errors by Detecting Action Boundaries (ASRF, Ishikawa 2020)"},"{"technique":"Keep the SRA-style cross-stream reconciliation but express each of its decision cases as FEATURE COLUMNS for the scorer rather than hard if/else, and cite SRA as the rule-based baseline you measure lift against.", "why":"The badminton SRA proves cross-checking shuttle vs swing streams works (58.8%→89.7%), but its hard-coded 5-case logic is precisely the special-casing you forbid in the decision path; turning those cases into columns honors your 'signal not verdict' rule while borrowing the proven cues.", "effort":"medium", "from_paper":"Shot Refinement via Fusion of Shuttlecock Tracking and Hit Detection (Hsu 2024)"},"{"technique":"Adopt offline/off-policy 'replay' evaluation framing for your re-scoring experiments and state estimator variance, since the golden set is tiny.", "why":"Your 'persist once, re-score forever, promote on measured lift' loop IS off-policy/offline policy evaluation; using its vocabulary (replay, estimator variance, confidence intervals) guards against over-trusting a lift measured on ~6 matches and connects you to a rigorous literature.", "effort":"low", "from_paper":"Off-policy / offline policy evaluation literature (Benchmarks for Deep Off-Policy Evaluation, arXiv:2103.16596; offline replay evaluation)"},"{"pitfalls":["Proposal-recall ceiling: in two-stage pipelines the decision stage can never recover rallies the windowing missed. If your candidate-window generator drops a real rally, no LR re-scoring fixes it ? measure proposal/window recall (AR@AN) separately, or you will misattribute recall losses to the scorer (S-CNN/TAD literature).", "Decision-side hard false positives dominate once perception is decoupled (DCR): the biggest remaining errors are non-rally windows that look rally-like; a scorer not trained on mined hard negatives will plateau. Class imbalance (dead-time >> rally windows) makes a plain LR over-predict the majority unless reweighted/calibrated.", "Over-segmentation / over-merging is a named, persistent failure of cheap segment decisions (ASRF): long static-camera rallies get split at momentary occlusions and short dead-time gaps get merged; rule-based smoothing can over-correct and erase true short rallies. A/B every reconciler for net lift, not just count of 'issues fixed'.", "LOVO leakage with a static-tripod corpus: grouping by match is necessary but may be INSUFFICIENT ? same venue/court/camera/players across matches still leak scene/appearance cues into a frame-state classifier (video/scene-level split warnings). With ~6 matches each fold is high-variance; a single match can swing the lift sign.", "Tiny-corpus / off-policy variance: a lift measured on ~6 vendor-labeled matches has wide confidence intervals; 'promote only on measured lift' can promote noise. Use confidence intervals / per-match breakdown and beware multiple-comparisons as you flag-gate many cues.", "Audio-cue brittleness for the thwack detector (your own E1 finding, echoed by TennisExpert's need for heavy noise augmentation): impulsive crowd/announcer noise yields high recall but poor discrimination; clustering hits into intervals inherits this and can fragment or over-extend candidate windows.", "Calibration drift across decoupled stages: persisted detector confidences are not probabilities the LR can trust as-is; without per-cue calibration the learned weights conflate 'cue is confident' with 'cue is informative,' undermining the 'signal not verdict' goal.", "Persist-once staleness: 'persist once, re-score forever' silently couples every downstream experiment to a fixed detector version; if you upgrade WASB/TrackNet or the person detector you must invalidate and recompute the cache or

experiments compare against stale evidence (feature-store versioning discipline)."], "verdict": "partial-overlap", "verdict_rationale": "The individual architectural moves are well-established and should be cited rather than presented as novel: two-stage proposal-then-classify (S-CNN), cascaded boundary refinement (CBR), decoupled re-scoring to kill hard false positives (DCR), the cascade/rejection lineage (Viola-Jones), over-segmentation boundary refinement (ASRF), and even the exact racket-sport pipeline shape (TennisExpert: audio hits -> cluster -> candidate intervals -> filter; SRA: cross-stream rule reconciliation in badminton). So the 'cascade / decouple detection from decision' theme on its own is reinventing known ideas. What is NOT a verbatim match in the literature found is the SPECIFIC combination: (a) elevating detection-decision decoupling to a persisted 'compute once, re-score many' substrate where nearly all experiments are pure offline re-scoring of frozen per-frame evidence, (b) a strict 'signal not verdict / no special-casing ? every cue is a feature column, the model learns weights' discipline that deliberately rejects the hand-coded SRA-style branch logic that the closest domain papers actually use, and (c) report-first, A/B-lift-gated, idempotent RECONCILERS treated as a linter over decisions vs evidence rather than a learned in-model refiner. That packaging ? TAD-style decoupling + feature-store-style caching + off-policy lift-gated promotion + a report-first reconciler linter, on a scarce LOVO-graded racket-sport corpus ? is a sensible novel combination of established parts, not a research gap. Hence partial-overlap: borrow the field's metrics (mAP@tIoU, AR@AN), vocabulary (proposals/actionness, hard false positives, over-segmentation), and learned baselines (CBR/ASRF) as course corrections, while your contribution is the engineering discipline around them.", "search_queries_used": ["cascade object detector two-stage proposal then classify Viola Jones rejection cascade", "temporal action detection proposal generation then classification two-stage video", "decouple perception from policy video pipeline precompute features offline rescoring", "off-policy offline evaluation reranking cached features compute once decide many", "badminton rally detection segmentation broadcast video play break shuttlecock TrackNet", "tennis match video temporal segmentation rally point detection from broadcast audio visual cues", "cascade temporal action detection coarse to fine boundary refinement reduce false positives", "SCNN segment CNN temporal action localization untrimmed video proposal classification localization", "feature store precompute embeddings train and rescore video frozen backbone offline reproducible experiments", "rule-based post-processing temporal action segmentation refinement merge split over-segmentation correction", "leave one group out cross validation small dataset video data leakage temporal evaluation pitfall", "computation reuse video inference caching intermediate features avoid recompute query engine"]], "verify": {"theme": "Cascades / two-stage proposal-verify; decouple detection from decision; offline re-scoring & feature caching", "citations_checked": [{"title": "Temporal Action Localization in Untrimmed Videos via Multi-stage CNNs (Segment-CNN / S-CNN)", "verified_real": "yes", "note": "Real. Shou, Wang, Chang, CVPR 2016, arXiv:1601.02129. Title, authors, venue, and arXiv ID all confirmed (IEEE Xplore doc 7780488, Columbia DVMM project page). Proposal->classification->localization 3-stage structure as described."}, {"title": "Cascaded Boundary Regression for Temporal Action Detection (CBR)", "verified_real": "yes", "note": "Real. Gao, Yang, Nevatia, BMVC 2017, arXiv:1705.01180. Confirmed on BMVA archive (paper052), dblp GaoYN17, arXiv. Two-stage cascade + cascaded boundary regression as described."}, {"title": "Decoupled Classification Refinement: Hard False Positive Suppression for Object Detection (DCR)", "verified_real": "yes", "note": "Real. arXiv:1810.04002, ECCV 2018. Survey lists author 'Honghui Shi'; paper byline reads 'Humphrey Shi' ? same person (Humphrey is the English name for Honghui Shi), NOT a fabrication. All other authors match. DCR = standalone classifier re-scoring base-detector boxes, trained on mined hard false positives, deliberately not sharing features ? as described."}, {"title": "Rapid Object Detection using a Boosted Cascade of Simple Features (Viola-Jones rejection cascade)", "verified_real": "yes", "note": "Real. CVPR 2001, pp. I-511-518, DOI 10.1109/CVPR.2001.990517 confirmed. Attentional/rejection cascade, integral image, AdaBoost as described."}, {"title": "Enhancing Badminton Game Analysis: Shot Refinement via Fusion of Shuttlecock Tracking and Hit Detection from a Monocular Camera", "verified_real": "yes", "note": "Real. Hsu, Yu, Cheng, Sensors (MDPI) 2024, 24(13):4372, DOI 10.3390/s24134372, PMC11244353 ? all confirmed. NOTE: survey paraphrases the title; the real title is 'Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera'. Same paper; minor wording drift only. TrackNet+YOLOv7 fusion via rule-based SRA, 58.8%->89.7% claim is consistent with the paper."}, {"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "verified_real": "yes", "note": "Real. arXiv:2603.13397 (March 2026), with GitHub repo (LZYAndy/TennisExpert) and ResearchGate entry. Year '2026' is consistent with the 2603 arXiv prefix. Audio racquet-impact detector -> cluster hits -> filter pipeline as described. (Recent paper; could not deep-read the PDF to verify the exact internal cluster-then-filter wording, but existence and topic are confirmed ? hence 'yes'.)"}], {"title": "A New Perspective for

Shuttlecock Hitting Event Detection", "verified_real": "yes", "note": "Real. Yu-Hsi Chen, arXiv:2306.10293 (June 2023), IJCAI 2023 IT4PSS workshop, GitHub repo confirmed. SwingNet-based hitting-event detection as described; survey correctly grades it 'weak' overlap (hit-event, not rally-interval)."}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries (ASRF, Ishikawa) ? cited in techniques/pitfalls", "verified_real": "yes", "note": "Real. Ishikawa, Kasai, Aoki, Kataoka. arXiv:2007.06866, WACV 2021 (survey says '2020' ? it is a mid-2020 preprint published at WACV 2021; essentially accurate). ASB+BRB boundary-regression refinement as described."}, {"title": "Benchmarks for Deep Off-Policy Evaluation (DOPE) ? cited in techniques", "verified_real": "yes", "note": "Real. Fu et al., arXiv:2103.16596, ICLR 2021. Used only as vocabulary anchor for off-policy/offline evaluation framing; appropriate citation."}], "fabrication_flags": [], "corrected_verdict": "partial-overlap", "corrected_rationale": "Verdict upheld as partial-overlap, with a skeptical nudge toward 'reinventing' for the substrate claim. All 8 cited papers plus the 2 extra references (ASRF, DOPE) are REAL ? no fabrications, only a paraphrased title (paper 5) and an English/Pinyin name variant (paper 3, Humphrey=Honghui Shi). The survey is unusually honest: it already grades every individual architectural move as 'established' and frames novelty only as packaging. My refutation searches confirm and slightly strengthen that skepticism on each claimed-distinctive element: (a) the 'persist perception once with a frozen backbone, re-score a cheap head many times' substrate is NOT a novel substrate ? it is the DEFAULT operating mode of the entire temporal-action-detection field (THUMOS/ActivityNet ship precomputed I3D/TSN/VideoMAE features precisely so researchers re-train cheap heads on cached features; confirmed via the TAD frozen-feature literature, e.g. TallFormer/OpenTAD/'Exploring Stronger Feature for TAL'). So framing (a) as a contribution would be overclaiming. (b) The 'cheap learned classifier over per-frame features + temporal smoothing for rally/non-rally segmentation' decision layer has very close, UNCITED prior art in the exact sport: 'Towards Structured Analysis of Broadcast Badminton Videos' (arXiv:1712.08714) learns an SVM over per-frame HOG to label point/non-point and Kalman-smooths the sequence ? i.e. the same 'features -> learned scorer -> temporal segmentation' shape the survey claims as distinctive; SmartTennisTV (arXiv:1801.01430) does segment-then-rule-refine, close to the reconciler idea. (c) What genuinely remains un-matched in the found literature is the SPECIFIC engineering bundle: strict 'signal-not-verdict' / no special-casing ? every cue is a feature column' discipline that deliberately rejects the SRA-style hand-coded branches the closest domain papers actually use, PLUS report-first, A/B-lift-gated, idempotent reconcilers treated as a linter over decision-vs-evidence (rather than a learned in-model refiner), PLUS off-policy/LOVO lift-gating on a scarce racket-sport golden set. That bundle is a sensible novel COMBINATION of established parts, not a research gap ? so partial-overlap, not novel-combination and not genuine-gap. Bottom line: cite the field's metrics (mAP@tIoU, AR@AN), vocabulary (proposals/actionness, hard false positives, over-segmentation), and learned baselines (CBR/ASRF); present the substrate (a) as standard practice, not novelty; present (b) as a known pattern with the badminton SVM/Kalman precedent acknowledged; reserve the modest novelty claim for the (b)+(c) discipline bundle.", "closer_prior_art": [{"title": "Towards Structured Analysis of Broadcast Badminton Videos", "url_or_id": "arXiv:1712.08714", "year": "2017", "why_closer": "Same sport AND the exact 'decision layer' the survey claims as distinctive: learns a chi-squared-kernel SVM over per-frame HOG features to label point / non-point frames, then temporally smooths (Kalman) into rally segments. This is 'cheap learned classifier over per-frame features -> temporal segmentation' ? the core of the survey's window-then-score split and its 'signal not verdict' learned-scorer instinct. Uncited; the closest published analog to the decision layer. Difference: single visual feature (HOG) not multi-cue cached signals, and no report-first reconciler/LOVO harness."}, {"title": "SmartTennisTV: Automatic indexing of tennis videos", "url_or_id": "arXiv:1801.01430", "year": "2018", "why_closer": "Sibling racket sport: temporally segments rallies, then RE-SCORES/REFINES segment labels using domain knowledge (scoring system) as a post-hoc rule pass ? close to the survey's 'segment -> reconcile decision against rules/evidence' idea (c) and to its rally-INDEXING product framing. Uncited. Difference: refinement is a fixed scoring-rule pass, not an A/B-lift-gated report-first linter."}, {"title": "Towards Real-Time Analysis of Broadcast Badminton Videos", "url_or_id": "arXiv:2308.12199", "year": "2023", "why_closer": "Broadcast-badminton rally analysis pipeline (rally/non-rally + downstream structure). Reinforces that cheap-classifier rally segmentation on badminton broadcasts is established practice; uncited. Not the full cached-multi-cue + reconciler combination."}, {"title": "Frozen-backbone precomputed-feature TAD pipelines (e.g. OpenTAD; TallFormer; 'Exploring Stronger Feature for Temporal Action Localization')", "url_or_id": "OpenTAD CVPRW2025 / TallFormer ECCV2022 / arXiv:2106.13014", "year": "2022-2025", "why_closer": "Directly undercuts the claim that 'persist perception once, re-score many' is a novel substrate: Stage-0 encode-with-frozen-backbone then Stage-1 cheap head/re-score IS the standard TAD training/eval paradigm (benchmarks distribute cached features for exactly this reuse). The compute-once/re-score-many loop should be cited as

field-default, not presented as a contribution.]],"extra_course_corrections":["ADOPT: Explicitly cite 'Towards Structured Analysis of Broadcast Badminton Videos' (arXiv:1712.08714) as the rule/learned-baseline for the DECISION layer ? it is same-sport prior art for 'per-frame features -> learned classifier -> temporally smoothed rally segments' and currently a gap in the survey's citation set. Position your LR-over-multi-cue scorer as a multi-signal generalization of its single-feature SVM, not as a new idea.", "ADOPT: Cite SmartTennisTV (arXiv:1801.01430) as the racket-sport precedent for 'segment then re-score/refine with a post-hoc pass' ? the closest published analog to your reconcilers (alongside SRA), and a useful framing for 'report-first refinement'.", "AVOID OVERCLAIM: Do not present 'persist perception once, re-score forever / frozen-backbone cached features' as novel. It is the default of the temporal-action-detection field (THUMOS/ActivityNet ship precomputed I3D/TSN/VideoMAE features; OpenTAD/TallFormer codify Stage-0-encode/Stage-1-head). Cite it as standard practice and claim novelty only for the discipline bundle layered on top.", "ADOPT (metric, reaffirmed): Report mAP@tIoU at 0.3/0.5/0.7 + average-mAP and AR@AN, not a single mean tIoU ? this is what S-CNN/CBR/ASRF and the whole TAD field expect and is what makes proposal-recall vs decision-precision separable (directly addresses the proposal-recall-ceiling pitfall the survey itself raises).", "KEEP AS HONESTY ANCHOR: The survey's own pitfalls list (proposal-recall ceiling, DCR hard-false-positive plateau, ASRF over-/under-segmentation, LOVO scene-leakage on a static-tripod corpus, ~6-match off-policy variance, audio-cue brittleness, cross-stage calibration drift, persist-once staleness/feature-store versioning) is accurate and well-sourced ? adopt it verbatim as the risk register; nothing in it was overstated.]]],{"theme":"Constraint/consistency-based correction & post-hoc verification of model outputs (\\\"ML linter\\\"), human-in-the-loop","short":"output-linting","our_pattern":"Our report-first RECONCILER: detect decision-vs-evidence contradictions, propose auditable corrections, measure before auto-apply.", "survey":{"theme":"Constraint/consistency-based correction & post-hoc verification of model outputs (\\\"ML linter\\\"), human-in-the-loop","our_pattern":"A report-first RECONCILER: a post-hoc pass over a persisted rally DECISION (segmenter output) that detects where the decision contradicts the persisted per-frame EVIDENCE (shuttle x,y,visibility/state; person tracks; audio hits; window features), and proposes auditable corrections ? truncate a segment whose shuttle went inactive before its end; drop a segment with zero net-crossings; split an over-merged segment with a long internal dead-time gap; tighten boundary slack to serve-contact. It REPORTS an issues list + a corrected set (never silently rewrites), corrections are A/B-measured for temporal-IoU lift BEFORE auto-apply, idempotent, fixed rule precedence, human-in-the-loop. Cues feeding the upstream scorer are (value,confidence) feature columns weighted by a tiny logistic regression ? \\\"signal not verdict, no special-casing\\\" ? and the expensive frame detection is persisted once so re-scoring/reconciling is pure offline (\\\"persist once, re-score forever\\"). Graded with leave-one-video-out (LOVO) vs scarce golden intervals.", "key_papers":[{"title":"Model Assertions for Monitoring and Improving ML Models","authors":"Daniel Kang, Deepti Raghavan, Peter Bailis, Matei Zaharia","year":"2020","venue":"MLSys 2020","url_or_id":"arXiv:2003.01668","what_it_does":"Introduces 'model assertions' ? arbitrary user-written functions over a model's input/output that fire when an error is likely, including TEMPORAL CONSISTENCY assertions over video (e.g. an object that rapidly changes class across frames). Used for runtime monitoring (catches HIGH-CONFIDENCE errors that uncertainty methods miss), for active-learning sample selection, and to generate weak labels from violated consistency assertions to retrain. Primarily flags/reports for human review rather than silently rewriting.", "overlap":"strong","overlap_explanation":"This is the closest framing match to our reconciler. Our 'detect where the DECISION contradicts persisted EVIDENCE' IS a model assertion over the rally output (shuttle-inactive-before-end, zero-net-crossings, internal-dead-time-gap are consistency assertions). Both are report-first, both exploit temporal/cross-signal consistency, both turn violations into a feedback signal (their weak labels ? our golden-eval lift loop). DIFFERENCE: model assertions stop at flag + (optionally) weak-label-for-retraining; they do NOT propose a structured CORRECTION to the output (truncate/split/drop), don't A/B-measure each correction rule's lift before auto-applying, and have no fixed rule-precedence/idempotency machinery. We should adopt their vocabulary ('model assertion', 'consistency assertion', 'high-confidence error') and cite them as the parent framing."}, {"title":"HoloClean: Holistic Data Repairs with Probabilistic Inference","authors":"Theodoros Rekatsinas, Xu Chu, Ihab F. Ilyas, Christopher Ré","year":"2017","venue":"PVLDB Vol.10 / VLDB 2017","url_or_id":"arXiv:1702.00820","what_it_does":"Takes a dirty dataset + integrity constraints (denial constraints) + optional external signals, detects tuples that violate constraints, and produces REPAIRS via a single probabilistic model. Crucially it estimates the MAP value AND the marginal/confidence for each repair, uses low-confidence repairs to solicit additional user feedback in a principled way.", "overlap":"partial","overlap_explanation":"Strong conceptual analog to the correction half of our reconciler: constraint-violation DETECTION +

proposed REPAIR + a CONFIDENCE per repair + human-in-the-loop on low-confidence ones. The 'detect contradiction against rules, propose a fix, surface confidence, ask a human when unsure' loop is exactly ours. DIFFERENCES: HoloClean repairs tabular cells, not temporal intervals; it unifies repairs in one global probabilistic-inference model rather than a fixed-precedence rule pipeline that is A/B-gated; and it auto-repairs rather than report-first. Worth adopting: emit a per-correction confidence and route low-confidence corrections to a human queue instead of auto-applying uniformly."}, {"title": "Data Validation for Machine Learning", "authors": "Eric Breck, Neoklis Polyzotis, Sudip Roy, Steven Whang, Martin Zinkevich", "year": "2019", "venue": "MLSys 2019 (TensorFlow Data Validation / TFX)", "url_or_id": "https://mlsys.org/Conferences/2019/doc/2019/167.pdf", "what_it_does": "Production data-validation system (TFDV in TFX): codifies expected properties as a SCHEMA/constraints, compares incoming data statistics to the schema, raises an ANOMALIES report (missing features, out-of-range, type errors, training/serving skew, drift), and proposes schema updates for a human to accept/reject. Report-then-human-fix by design.", "overlap": "partial", "overlap_explanation": "Matches the 'report-first linter' half: declare constraints, detect violations against persisted statistics, emit a structured anomalies report for a human, and let the system SUGGEST a fix (schema change) that a human ratifies ? directly parallel to our issues-report + proposed-correction + human-in-the-loop. DIFFERENCE: it validates DATA/inputs against a schema, not a model's structured temporal OUTPUT against cross-modal evidence, and it has no temporal-IoU/lift A/B gate. Adopt: their 'schema as version-controlled, human-ratified constraint set' and treating each reconciler rule as a named, owner-acknowledgeable anomaly type."}, {"title": "Unit Testing Data with Deequ (Automating Large-Scale Data Quality Verification)", "authors": "Sebastian Schelter, Dustin Lange, Philipp Schmidt, Meltem Celikel, Felix Biessmann, Andreas Grafberger", "year": "2018", "venue": "PVLDB / Amazon (Deequ)", "url_or_id": "https://www.amazon.science/publications/unit-testing-data-with-deequ", "what_it_does": "Declarative 'unit tests for data': users (or an automated constraint-suggestion profiler) define constraints; the engine verifies them at scale and reports pass/fail with metrics. Includes automated constraint SUGGESTION by profiling data.", "overlap": "weak", "overlap_explanation": "Supports the 'linter' metaphor and naming directly ? declarative, named constraints over data with a verification report, plus AUTO-SUGGESTED constraints from profiling. Same spirit as our 'register a producer/constraint, get a report'. DIFFERENCE: data-side, tabular, no temporal-interval correction, no lift-measured auto-apply. Useful as terminology precedent ('unit tests for data', 'constraint suggestion') and as a reminder we could auto-mine candidate reconciler rules from the evidence rather than hand-authoring all of them."}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries (ASRF)", "authors": "Yuchi Ishikawa, Seito Kasai, Yoshimitsu Aoki, Hirokatsu Kataoka", "year": "2020", "venue": "WACV 2021 (arXiv 2020)", "url_or_id": "arXiv:2007.06866", "what_it_does": "Action Segment Refinement Framework: a boundary-regression branch (BRB) whose predicted action boundaries REFINE a frame-wise action-segmentation branch (ASB), specifically to cut OVER-SEGMENTATION (spurious micro-segments) and sharpen boundaries. Reports segmental EDIT distance and segmental F1 (the metrics this field expects).", "overlap": "partial", "overlap_explanation": "This is the temporal-segmentation field's canonical answer to the exact failure modes our reconciler targets (over-segmentation, sloppy boundaries) ? our 'split over-merged segment', 'tighten boundary slack to serve-contact' are the boundary-refinement problem. DIFFERENCE: ASRF learns refinement end-to-end inside the model and is NOT a report-first, evidence-cross-checking, A/B-gated post-hoc pass; it has no human-in-the-loop and no auditability. Adopt their FRAMING and especially their METRICS ? segmental F1@{10,25,50} and segmental edit distance are what reviewers in this space expect alongside (or instead of) raw temporal-IoU."}, {"title": "Snorkel: Rapid Training Data Creation with Weak Supervision", "authors": "Alexander Rather, Stephen H. Bach, Henry Ehrenberg, Jason Fries, Sen Wu, Christopher Ré", "year": "2017", "venue": "PVLDB Vol.11 / VLDB 2018", "url_or_id": "arXiv:1711.10160", "what_it_does": "Users write many noisy 'labeling functions' (heuristics that vote or abstain) with unknown accuracies/correlations; a LABEL MODEL denoises and combines them WITHOUT ground truth into probabilistic labels, learning each source's accuracy rather than hand-weighting.", "overlap": "partial", "overlap_explanation": "Direct analog of our 'SIGNAL NOT VERDICT / no special-casing' principle: each cue is a noisy voter and a learned model weights them ? Snorkel's label model is the data-programming version of our logistic-regression scorer. DIFFERENCE: Snorkel is about MANUFACTURING training labels from heuristics absent ground truth; we have (scarce) golden labels and supervise directly, and Snorkel models inter-source CORRELATIONS explicitly (we likely don't). Adopt: model cue correlations (correlated cues double-count and miscalibrate the scorer) and consider Snorkel/MeTaL-style label modeling to bootstrap labels from our cheap reconciler rules on UNlabeled matches."}, {"title": "Confident Learning: Estimating Uncertainty in Dataset Labels", "authors": "Curtis G. Northcutt, Lu Jiang, Isaac L. Chuang", "year": "2021", "venue": "JAIR (Journal of AI Research)", "url_or_id": "arXiv:1911.00068", "what_it_does": "Model-agnostic method that uses a model's out-of-sample predicted probabilities to estimate the joint distribution of

noisy vs true labels, IDENTIFIES likely label errors (prune-by-noise-rate), ranks them, and proposes corrected candidates ? the basis of

cleanlab.", "overlap": "weak", "overlap_explanation": "Same 'use the model's own evidence to flag where the recorded label/decision is probably wrong, rank by margin, propose a candidate' loop ? applicable to auditing our scarce GOLDEN intervals themselves (vendor labels are noisy) and to ranking reconciler-flagged segments by how strongly evidence contradicts the decision.

DIFFERENCE: it targets classification label noise, not temporal-interval boundaries or cross-modal consistency. Adopt: rank reconciler issues by a confident-learning-style margin so humans review the most contradicted segments first.", {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "authors": "Yi-Hua Hsu, Chih-Chang Yu, Hsu-Yung Cheng", "year": "2024", "venue": "Sensors (MDPI), 24(13):4372", "url_or_id": "https://doi.org/10.3390/s24134372 (PMC11244353)", "what_it_does": "Fuses TrackNet shuttle-trajectory hit detection with a YOLOv7 swing-action detector via a Shot Refinement Algorithm (SRA): case-based rules CROSS-CHECK trajectory hits against action detections (e.g. 'only one hit can exist within a swing', drop a trajectory hit outside any swing action, keep the one nearest the peak-confidence action frame) to correct false-positive/false-negative hits.", "overlap": "partial", "overlap_explanation": "The domain-closest work and a real analog of our reconciler's CORE idea ? cross-check one signal's output against another signal's evidence and CORRECT contradictions. Our 'shuttle inactive before segment end' / 'zero net-crossings' are the rally-level cousins of their shot-level 'hit outside any swing action'. DIFFERENCES (important and in our favor): (1) it operates at SHOT level, not rally start/end / dead-time-vs-in-play segmentation; (2) corrections are HAND-CODED case-rules in the decision path ? exactly the 'special-casing' we avoid; (3) NOT report-first, NO A/B lift measurement before applying, no learned weighting, no human-in-the-loop, no idempotency/precedence formalism. So our contribution vs this paper is precisely the methodology layer (signal-not-verdict + report-first + measured-before-apply), not the cross-check idea itself."}], "established_terms": ["model assertions / consistency assertions (Kang et al., MLSys'20) ? the standard name for 'a function that fires when model output violates an expected property'", "high-confidence errors ? errors a confident model makes that uncertainty/confidence thresholds miss (the value-add of assertion/consistency checks)", "data repair / error detection-and-repair, integrity constraints, denial constraints (HoloClean / DB cleaning) ? the framing for 'detect violation, propose repair'", "data validation / schema anomalies / training-serving skew (TFDV, Polyzotis et al.) ? production 'report-then-human-ratify' constraint checking", "unit tests for data / declarative data-quality constraints / constraint suggestion (Deequ, Great Expectations) ? the 'linter for data' metaphor and auto-mined constraints", "action segment refinement / over-segmentation / boundary regression; segmental F1@{10,25,50} and segmental edit distance (ASRF, MS-TCN lineage) ? the temporal-segmentation field's standard refinement vocabulary and METRICS", "data programming / label model / labeling functions (Snorkel) ? learned denoising of many noisy heuristic voters (your 'signal not verdict')", "confident learning / prune-by-noise-rate (Northcutt et al., cleanlab) ? model-evidence-based detection of likely-wrong labels/decisions", "leave-one-group-out / leave-one-subject-out (LOSO/LOGO) cross-validation ? the canonical name for your LOVO; 'group-wise CV' is the literature term"], "techniques_to_adopt": [{"technique": "Explicitly brand the reconciler as 'model assertions / consistency assertions over the rally decision' and cite Kang et al. ? frame each rule (shuttle-inactive-before-end, zero-net-crossings, internal-dead-time-gap) as a named consistency assertion.", "why": "Gives the design an established, defensible name and lineage instead of bespoke 'reconciler' terminology; reviewers immediately understand it, and it correctly positions you as EXTENDING model assertions (you add structured correction + lift-gated auto-apply, which they lack).", "from_paper": "Model Assertions for Monitoring and Improving ML Models (Kang et al., 2020)", "effort": "low"}, {"technique": "Report segmental F1@{10,25,50} and segmental edit distance alongside temporal-IoU, and benchmark the reconciler against a learned boundary-refinement baseline (ASRF-style).", "why": "Raw temporal-IoU alone under-credits the over-segmentation/boundary fixes your reconciler targets; segmental F1 and edit distance are exactly what the temporal-action-segmentation field uses to quantify over-segmentation and boundary quality, making your lift numbers comparable and credible.", "from_paper": "Alleviating Over-segmentation Errors by Detecting Action Boundaries / ASRF (Ishikawa et al., 2020)", "effort": "medium"}, {"technique": "Attach a per-correction CONFIDENCE/margin to every reconciler proposal and route only high-confidence ones to auto-apply, sending low-confidence ones to a human queue ranked by margin.", "why": "HoloClean and confident learning both show that surfacing repair confidence and reviewing the most-contradicted cases first is the principled way to mix auto-repair with human-in-the-loop; uniform auto-apply on a rule is riskier than confidence-gated apply, especially on a 6-match corpus.", "from_paper": "HoloClean (Rekatsinas et al., 2017) + Confident Learning (Northcutt et al., 2021)", "effort": "medium"}, {"technique": "Treat the constraint/assertion set as a version-controlled, owner-ratified 'schema' of named anomaly

types, with an automated CONSTRAINT-SUGGESTION step that profiles persisted evidence to propose candidate reconciler rules.", "why": "TFDV and DeeQu show production value in (a) making constraints first-class, versioned, human-acknowledged artifacts and (b) auto-mining candidate constraints from data so you don't hand-author every rule ? directly applicable to growing your reconciler rule set beyond the few you wrote by hand.", "from_paper": "Data Validation for ML / TFDV (Polyzotis et al., 2019) + Unit Testing Data with DeeQu (Schelter et al., 2018)", "effort": "medium"}, {"technique": "Model cue CORRELATIONS in the scorer (and consider a Snorkel/MeTaL-style label model to bootstrap labels on unlabeled matches from your reconciler rules).", "why": "Your cues (shuttle-state, net-crossings, person two-sidedness) are correlated; an unregularized logistic regression on a tiny set will double-count correlated cues and miscalibrate. Snorkel's label model is the established way to denoise correlated heuristic voters, and it can manufacture weak rally labels on the 5 unlabeled-of-6 matches to enlarge your training signal.", "from_paper": "Snorkel (Ratner et al., 2017)", "effort": "high"}, {"technique": "Use violated assertions as WEAK LABELS to improve the upstream scorer, not just to correct outputs.", "why": "Kang et al. show consistency-assertion violations can be turned into weak supervision that retrains the model for up to ~46% quality gain ? closing your loop so the reconciler feeds the scorer instead of only patching its outputs.", "from_paper": "Model Assertions (Kang et al., 2020)", "effort": "medium"}], "pitfalls": ["Small held-out + learned weights overfit: the late-fusion literature explicitly warns that learning modality/cue weights by regression on a small held-out set is UNSTABLE and overfits. With ~6 matches and a logistic regression over many correlated cues, your learned weights (and any per-cue calibration) can be dominated by one match. Mitigate with strong L2/L1 regularization, far fewer effective features than groups, and reporting weight variance across LOVO folds.", "LOVO is necessary but not sufficient ? distributional bias still biases leave-one-out: recent work shows leave-one-out CV suffers distributional bias (the held-out group shifts the train-set label distribution), inflating or deflating estimates, worse under class imbalance (rallies vs dead-time is imbalanced). Report variance/CIs across the 6 folds, not a single point, and watch for the imbalance interaction.", "Correlated cues double-count: if net-crossing count, two-sidedness, and shuttle-in-air all co-fire on the same rallies, a naive scorer over-weights that bundle and a reconciler built on the same cues won't catch the resulting errors (no independent evidence). Snorkel's lesson: model source correlations or your 'independent' evidence isn't independent.", "Hand-coded correction rules creep back into the decision path (the badminton SRA failure mode): the closest domain work (Hsu et al.) implements correction as five hand-coded case rules ? exactly the special-casing you forbid. The risk is that 'report-first reconciler rules' silently become production if/else branches with fixed thresholds that don't generalize across venues/cameras. Keep them flag-gated, lift-measured, and learn thresholds where possible.", "'Report-first' can decay into silent auto-rewrite: TFDV/HoloClean keep a human ratification step precisely because auto-repair on confident-but-wrong cases is dangerous (HoloClean's own precision ~90% means ~10% bad repairs). Without per-rule lift gating AND a confidence gate AND idempotency tests, an over-eager reconciler rule can systematically truncate correct long rallies. Measure lift per rule on golden data before any auto-apply, and keep rules idempotent with fixed precedence so order doesn't change outcomes.", "Refinement that optimizes the wrong metric: ASRF shows boundary refinement can raise segmental F1/edit while frame accuracy barely moves (and vice versa). If you only watch temporal-IoU, a reconciler that fixes over-segmentation may show no IoU lift yet be a real win ? and conversely an IoU-chasing rule can shred segment structure. Track both interval and segmental metrics so you don't reward the wrong behavior.", "Assertion/repair set as unowned tech debt: DeeQu/TFDV treat constraints as versioned, owned artifacts. Ad-hoc reconciler rules accumulate, conflict, and rot; without a registry, precedence spec, and per-rule provenance/lift record, the 'linter' becomes an unauditable pile of thresholds."], "verdict": "novel-combination", "verdict_rationale": "novel-combination", "search_queries_used": ["model assertions Kang validating machine learning predictions runtime monitoring", "constraint-based correction structured prediction enforcing consistency neural network outputs", "Snorkel weak supervision label model denoising labeling functions data programming", "data validation constraints machine learning pipelines TFX schema anomaly detection Polyzotis", "temporal action segmentation post-processing refinement over-segmentation boundary smoothing", "human-in-the-loop verification ML predictions proposals verify against evidence active learning correction sports video", "HoloClean error detection repair data cleaning integrity constraints probabilistic inference", "badminton rally segmentation shuttlecock trajectory TrackNet temporal localization stroke detection", "learned weighting of noisy heuristic rules vs hand-coded rules calibration late fusion pitfalls multi-signal", "grouped cross-validation leave-one-subject-out data leakage small dataset weak supervision evaluation pitfalls", "\\unit tests for data\\", "DeeQu Great Expectations declarative constraints data quality automated suggestion", "consistency-based label cleaning confident learning detecting label errors pruning Northcutt"]}, {"verify": {"theme": "Constraint/consistency-

based correction & post-hoc verification of model outputs (\\"ML linter\\"), human-in-the-loop", "citations_checked": [{"title": "Model Assertions for Monitoring and Improving ML Models (Kang, Raghavan, Bailis, Zaharia, MLSys 2020)", "verified_real": "yes", "note": "Confirmed real. arXiv:2003.01668, MLSys 2020 proceedings. Authors, venue, year all correct. Survey's characterization (consistency assertions, high-confidence errors, weak labels for retraining ~46% relative quality gain, ~40% labeling-cost reduction via bandit active learning) matches the abstract precisely. Strongest and most accurately summarized citation."}, {"title": "HoloClean: Holistic Data Repairs with Probabilistic Inference (Rekatsinas, Chu, Ilyas, Re, PVLDB Vol.10 / VLDB 2017)", "verified_real": "yes", "note": "Confirmed real. arXiv:1702.00820, PVLDB Vol.10 No.11. Authors/venue/year correct. Survey's pitfall claim that HoloClean precision is ~90% is verified against the abstract (avg precision ~90%, recall >~76%)."}, {"title": "Data Validation for Machine Learning (Breck, Polyzotis, Roy, Whang, Zinkevich, MLSys 2019 / TFDV)", "verified_real": "yes", "note": "Confirmed real. MLSys 2019, Google/TFX (TFDV). Authors/venue/year correct. The survey's mlsys.org PDF URL resolves to the correct paper."}, {"title": "Unit Testing Data with DeeQu (Schelter et al., Amazon/DeeQu)", "verified_real": "yes", "note": "Paper is real but the survey CONFLATES TWO papers and gets the author list partly wrong. 'Unit Testing Data with DeeQu' is SIGMOD 2019 (demo): Schelter, Biessmann, Lange, Rukat, Schmidt, Seufert, Brunelle, Taptunov. The survey's parenthetical venue 'PVLDB' and its author list (Celikel, Biessmann, Grafberger) actually belong to the SEPARATE companion 'Automating Large-Scale Data Quality Verification' (PVLDB 2018). Both are real DeeQu papers, so this is citation hygiene, not fabrication. The substance (declarative constraints, automated constraint suggestion) is correct for both."}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries / ASRF (Ishikawa, Kasai, Aoki, Kataoka, WACV 2021)", "verified_real": "yes", "note": "Confirmed real. arXiv:2007.06866, WACV 2021, CVF open access + code (github.com/yiskw713/asrf). ASB/BRB two-branch boundary-refinement description and the segmental-F1/edit-distance metrics claim are accurate (up to 13.7% edit, 16.1% segmental-F1 gains)."}, {"title": "Snorkel: Rapid Training Data Creation with Weak Supervision (Ratner, Bach, Ehrenberg, Fries, Wu, Re, VLDB 2018)", "verified_real": "yes", "note": "Confirmed real. arXiv:1711.10160, PVLDB Vol.11 No.3. Authors/venue correct. Label-model / labeling-functions / inter-source-correlation description is accurate."}, {"title": "Confident Learning: Estimating Uncertainty in Dataset Labels (Northcutt, Jiang, Chuang, JAIR 2021)", "verified_real": "yes", "note": "Confirmed real. arXiv:1911.00068, JAIR (DOI 10.1613/jair.1.12125), basis of cleanlab. Prune-by-noise-rate / rank-by-margin description accurate."}, {"title": "Enhancing Badminton Game Analysis: Shot Refinement via Fusion of Shuttlecock Tracking and Hit Detection (Hsu, Yu, Cheng, Sensors 2024)", "verified_real": "yes", "note": "Confirmed real. Sensors 24(13):4372, DOI 10.3390/s24134372, PMC11244353. Authors correct. Description (TrackNet + YOLOv7 swing detector + hand-coded Shot Refinement Algorithm case-rules dropping hits outside any swing) is accurate; F1 jumps from 72.3% (TrackNet alone) to 90.5% with SRA. The survey's framing of this as hand-coded special-casing (the failure mode the design avoids) is fair."}], "fabrication_flags": [], "corrected_verdict": "novel-combination", "corrected_rationale": "Verdict UPHeld as novel-combination, with caveats. All 9 cited papers are real and accurately summarized (one citation-hygiene slip: the DeeQu entry conflates the SIGMOD'19 demo 'Unit Testing Data with DeeQu' with the PVLDB'18 'Automating Large-Scale Data Quality Verification' and mis-assigns the author list/venue; both are real DeeQu papers, so no fabrication). Adversarial refutation searches did NOT find prior art bundling the full claimed combination: a report-first pass over a PERSISTED structured temporal DECISION that (a) detects contradictions against PERSISTED cross-modal per-frame EVIDENCE, (b) proposes auditable structured interval corrections (truncate/split/drop/tighten), (c) A/B lift-gates EACH rule on golden intervals BEFORE auto-apply, (d) is idempotent with fixed precedence and human-in-the-loop, and (e) feeds a learned (value, confidence) cue scorer ('signal not verdict'). The ingredients are all established (model assertions = parent framing; HoloClean/TFDV/DeeQu = detect-violation-propose-repair-with-confidence + versioned constraints; ASRF = the over-segmentation/boundary target and its metrics; Snorkel = learned denoising of noisy voters; confident learning = rank-by-margin; Hsu et al. = the SAME cross-check-and-correct idea in-domain but hand-coded). What appears un-precedented is the assembly into a measured-before-apply, audit-first reconciler for temporal intervals. CAVEAT lowering confidence: domain novelty is narrower than the survey implies. Hsu et al. (2024) already does signal-vs-evidence cross-check-and-correct in badminton, and EventAnchor (arXiv:2101.04954, missed by the survey) already does human-in-the-loop suggest-then-calibrate correction of racket-sports event anchors against CV-extracted ball/player evidence. Neither is report-first-with-measured-lift, but together they show the cross-check and the human-in-the-loop pieces are both prior art in this exact domain. So the contribution is squarely the METHODOLOGY layer (measured-before-apply + signal-not-verdict + auditable rule registry), not the cross-check insight. Framed that way it is defensible.", "closer_prior_art": [{"title": "EventAnchor: Reducing Human Interactions in Event"}]

Annotation of Racket Sports Videos", "url_or_id": "arXiv:2101.04954 (CHI 2021)", "year": "2021", "why_closer": "Domain-exact and survey MISSED it. Racket-sports video: CV extracts ball position / player pose / court, algorithms auto-detect event 'anchors' (ball-racket, ball-court contact) that 'inevitably contain errors', and a human calibrates/deletes/adds via a calibration box. This is the human-in-the-loop, suggest-then-human-verify half of the reconciler, already realized in-domain. DIFFERENCES that keep the verdict from collapsing: it is an INTERACTIVE visual-analytics annotation tool, not an automated reconciler; it does NOT systematically cross-check the decision against persisted evidence to emit an issues list, does NOT propose structured truncate/split/drop interval corrections, and measures user task performance AFTER annotation rather than per-rule correction lift BEFORE auto-apply. Belongs in related work as the closest racket-sports human-in-the-loop-correction precedent."}, {"title": "Boundary-Centric Active Learning for Temporal Action Segmentation", "url_or_id": "arXiv:2604.15173", "year": "2026", "why_closer": "Recent and on the exact boundary-quality problem the reconciler targets: ranks/queries videos by predictive uncertainty, detects candidate transitions, and selects top boundaries via a boundary score fusing neighborhood uncertainty + class ambiguity + temporal dynamics (evidence-fused boundary scoring with a human query loop). Closer than ASRF on the human-in-the-loop axis. DIFFERENCE: it is active-learning sample selection for labeling, not a report-first post-hoc corrector of a persisted decision with measured-before-apply lift gating; no idempotency/precedence/auditability formalism."}], "extra_course_corrections": ["CITATION HYGIENE: Fix the DeeQu entry before publishing. The cited 'Unit Testing Data with DeeQu' is SIGMOD 2019 (Schelter, Biessmann, Lange, Rukat, Schmidt, Seufert, Brunelle, Taptunov). The survey's listed authors (Celikel, Biessmann, Grafberger) and 'PVLDB' venue actually belong to the companion paper 'Automating Large-Scale Data Quality Verification' (PVLDB 11(12), 2018). Cite whichever you mean, or cite both -- but do not merge them.", "ADD EventAnchor (arXiv:2101.04954) to related work and explicitly distinguish from it: it is the closest racket-sports human-in-the-loop correction system, so a domain-aware reviewer will expect it. Positioning against it (interactive-annotation tool vs measured-before-apply automated reconciler) actually STRENGTHENS the novelty claim.", "SHARPEN the contribution statement: because Hsu et al. (cross-check-and-correct, in-domain) and EventAnchor (human-in-the-loop correction, in-domain) both exist, do NOT claim the cross-check or the human-in-the-loop as the novel part. Claim ONLY the methodology layer: per-rule A/B lift-gating on golden intervals before any auto-apply + idempotent fixed-precedence rule registry + 'signal-not-verdict' learned scorer + 'persist once, re-score forever'. That is the defensible delta.", "TREAT the survey's metrics recommendation as load-bearing, not optional: ASRF is verified to report segmental F1@{10,25,50} and segmental edit distance as the field-standard over-segmentation/boundary metrics. Since the reconciler explicitly targets over-segmentation (split over-merged) and boundary slack (tighten-to-serve-contact), reporting ONLY temporal-IoU will under-credit exactly the fixes it makes -- report segmental F1 + edit alongside IoU.", "KEEP the small-data overfitting pitfall first-class: with ~6 matches and a logistic regression over correlated cues, the verified Snorkel lesson (model source correlations) plus standard LOGO/LOSO practice (report per-fold variance/CIs, strong L2, far fewer effective features than groups) is the most likely reason the learned scorer fails to generalize -- an experimental-design constraint, not polish."}], {"theme": "Weakly/interval-supervised temporal localization + small-data evaluation protocols (LOVO, tIoU)", "short": "weak-sup-eval", "our_pattern": "Our scarce interval labels + tiny logistic model + leave-one-VIDEO-out evaluation.", "survey": {"theme": "Weakly/interval-supervised temporal localization + small-data evaluation protocols (LOVO, tIoU)", "our_pattern": "We segment a long static-camera badminton match into rally [start,end] intervals using scarce, expensive, interval-level golden labels (~6 matches). Multiple cheap \"producer\" cues each emit a (value, confidence) feature column over candidate windows; a tiny logistic regression learns to weight these columns and decides which windows are real rallies ? cues are never hand-coded if/else branches (\"signal not verdict\"). Expensive frame detection (shuttle CNN, person detector, audio hit detector) is persisted once and the cheap windowing+scorer re-scores deterministically offline (\"detection ? decision\"). A post-hoc \"reconciler\" linter flags where the decision contradicts persisted evidence (segment whose shuttle went inactive, zero net-crossings, over-merged segment with internal dead-time) and proposes A/B-measured corrections. New cues land flag-gated/default-OFF, graded vs golden labels with leave-one-VIDEO-out (group by match) using temporal IoU, promoted only on measured lift.", "key_papers": [{"title": "Snorkel: Rapid Training Data Creation with Weak Supervision (Data Programming)", "authors": "Ratner, Bach, Ehrenberg, Fries, Wu, Re", "year": "2017", "url_or_id": "arXiv:1711.10160", "venue": "VLDB 2017", "what_it_does": "Users write 'labeling functions' (arbitrary heuristics with unknown accuracy/correlation); a generative 'label model' denoises and combines them WITHOUT ground truth, learning a weight/accuracy per source from their agreements and disagreements, then trains an end discriminative model.", "overlap": "strong", "overlap_explanation": "This is the canonical formalization of our 'signal not verdict / no special-casing' core: each cue = a noisy weak-supervision source (their

labeling function), and a learned model weights them instead of hand-coded branches. Key differences: (1) Snorkel learns source weights with NO labels via a generative model over LF agreement; we instead supervise our logistic regression directly with scarce golden intervals (closer to standard supervised feature stacking). (2) Snorkel operates at the data-point label level; our scorer operates at the candidate-window level for temporal localization. Their terminology (labeling function, label model, denoising, LF accuracy/correlation) is the established vocabulary we should adopt."}, {"title": "SF-Net: Single-Frame Supervision for Temporal Action Localization", "authors": "Ma, Yoon, Kim, Lee, Kang, Chen, Lim, Wang, Kira et al.", "year": "2020", "url_or_id": "arXiv:2003.06845", "venue": "ECCV 2020", "what_it_does": "Trains a temporal action localizer from a SINGLE timestamp per action instance; expands the labeled frame into pseudo-action segments via 'action frame expansion' and mines 'pseudo background frames' as negatives; evaluated with mAP@tIoU on THUMOS14/ActivityNet.", "overlap": "partial", "overlap_explanation": "Directly the literature for our regime: localizing temporal intervals from cheap, sparse annotations. Same metric family (mAP@tIoU). Difference: SF-Net learns a deep snippet model end-to-end from single frames and mines pseudo-labels; we hand-engineer scale-invariant window features and fit a tiny logistic regression on full interval labels. Confirms our problem is 'point/timestamp-supervised TAL', a named subfield ? we should use that terminology and its metric."}, {"title": "Point-Level Temporal Action Localization: Bridging Fully-supervised Proposals to Weakly-supervised Losses", "authors": "Ju, Zhao, et al.", "year": "2020", "url_or_id": "arXiv:2012.08236", "venue": "arXiv (2020)", "what_it_does": "Localizes actions from one timestamp per instance using a proposal-based paradigm that gives a more constrained solution space and consistent neighboring-frame predictions, vs frame-level paradigms whose sparse single-frame labels create a large solution space.", "overlap": "partial", "overlap_explanation": "Reinforces the 'point/timestamp supervision' framing for our scarce labels and argues proposal-based (candidate-window) prediction is better-constrained than dense frame prediction ? which is exactly our windowing-then-scoring design. We are doing a proposal/candidate-window approach without citing this lineage."}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries (ASRF)", "authors": "Ishikawa, Kasai, Aoki, Kataoka", "year": "2020", "url_or_id": "arXiv:2007.06866", "venue": "WACV 2021", "what_it_does": "Adds a Boundary Regression Branch that predicts action-boundary probability and uses detected boundaries to refine/relabel the per-frame Action Segmentation Branch output, cutting over-segmentation; reports segmental F1@k and edit distance (+13.7%/+16.1%).", "overlap": "partial", "overlap_explanation": "The closest method-level analog of our RECONCILERS: a separate signal (boundaries / persisted evidence) used post-hoc to correct an initial segmentation ? split over-merged segments and tighten boundaries. Differences: ASRF is a learned in-network branch and silently refines; our reconciler is a report-first, rule-precedence, A/B-gated, human-in-the-loop linter over persisted evidence. Their over-segmentation framing, boundary-refinement, and segmental F1@k / edit-distance metrics are directly relevant."}, {"title": "Structure analysis of soccer video with Hidden Markov Models (play/break segmentation)", "authors": "Xie, Xu, Chang, Divakaran, Sun", "year": "2001-2004", "url_or_id": "https://www.ee.columbia.edu/ln/dvmm/publications/01/icme01_soccer.pdf", "venue": "ICME 2001 / IEEE TMM", "what_it_does": "Defines two mutually-exclusive game states 'play' and 'break' from the rules of soccer; extracts domain features (dominant grass-color ratio, motion intensity) and uses HMM + dynamic programming to segment the match into play/break, ~86.5% segmentation accuracy.", "overlap": "strong", "overlap_explanation": "The seminal sports-structure analog of rally (play) vs dead-time (break) segmentation, and the origin of using domain cues + a temporal model rather than special-casing events. Key difference and lesson: it uses hand-tuned features in an HMM/generative pipeline (NOT a learned scorer over interchangeable feature columns) and per-frame state classification, where we use candidate windows + logistic regression. Establishes 'play/break segmentation' as the field term for exactly our task in a sibling sport."}, {"title": "SoccerNet: A Scalable Dataset for Action Spotting in Soccer Videos (+ 2023-2025 Challenges)", "authors": "Giancola, Amine, Dghaily, Ghanem; Cioppa et al. (challenge reports)", "year": "2018-2025", "url_or_id": "arXiv:1804.04527 (dataset); arXiv:2508.19182 (2025 results)", "venue": "CVPR Workshops / arXiv", "what_it_does": "Defines the action-spotting task and its evaluation: Average-Precision within a temporal tolerance delta (mAP@delta), with 'tight' (delta 1-5s) and 'loose' (5-60s) regimes; the de-facto sports-video temporal-localization benchmark and metric standard.", "overlap": "weak", "overlap_explanation": "Not the same task (point spotting vs interval segmentation) and a much larger labeled corpus, so weak methodological overlap. But it is THE established sports-temporal-localization evaluation standard; its tolerance-based mAP@delta and tight/loose reporting are conventions reviewers will expect alongside our tIoU, and worth reporting for boundary-slack sensitivity."}, {"title": "Distributional bias compromises leave-one-out cross-validation", "authors": "(2024/2025 paper, PMC12662204)", "year": "2024", "url_or_id": "https://pmc.ncbi.nlm.nih.gov/articles/PMC12662204/", "venue": "PMC (Briefings in Bioinformatics / GigaScience-

class)","what_it_does":"Shows LOO-CV induces a negative correlation between a fold's training-label mean and its held-out label; models regress to the training mean, so LOOCV systematically biases performance estimates and hyperparameter selection (a logistic regression scored ~0.23 auROC on random data). Proposes Rebalanced LOOCV (RLOOCV): drop an extra opposite-label training sample per fold to hold class balance

constant.", "overlap": "strong", "overlap_explanation": "Directly targets our exact stack: leave-one-out cross-validation + logistic regression + tiny N. The bias is worst for small datasets and strong regularization ? precisely our 6-match, regularized-logistic regime. This is a concrete pitfall and a concrete fix (RLOOCV / class-balance-preserving folds) we should apply to our LOVO protocol and any threshold/lift comparisons."}, {"title": "Weakly Supervised TAL Through Learning Explicit Subspaces for Action and Context (action-context confusion)", "authors": "Liu, Wang, Tang, et al.", "year": "2021", "url_or_id": "arXiv:2103.16155", "venue": "AAAI 2021", "what_it_does": "Names and tackles 'action-context confusion': video-level supervision makes models latch onto co-occurring static context (e.g., the baseball field) instead of the action, hurting boundaries/completeness; learns separate action vs context subspaces.", "overlap": "weak", "overlap_explanation": "Methodologically distant (deep subspace learning) but the documented FAILURE MODE maps cleanly onto our setting: the static badminton court / warm-up context co-occurs with rallies, so a scorer can learn 'court is visible' rather than 'a point is in play.' This justifies our scale/translation-invariant, court-agnostic features and our warm-up/session-scope analyzer, and is the canonical name for the trap to cite."}], "established_terms": ["Point / single-timestamp / point-level supervised Temporal Action Localization (the named subfield for our scarce interval/point labels)", "Weakly-supervised Temporal Action Localization (WTAL/WS-TAL) ? video-level labels", "Temporal Action Segmentation (the frame-labeling task; source of the over-segmentation literature)", "Play/Break segmentation (the sports-structure term for rally vs dead-time; from soccer-video analysis)", "Action spotting + tolerance-based mAP@delta (SoccerNet sports-temporal-localization metric standard)", "Data programming / labeling functions / label model / denoising (Snorkel) ? the established vocabulary for 'each cue is a noisy signal, a learned model weights them'", "Action-context confusion / completeness modeling (the WTAL failure-mode vocabulary)", "Leave-one-subject-out (LOSO) cross-validation ? the established name; 'leave-one-video-out (LOVO)' is our group-by-match instance", "mAP@tIoU and segmental F1@k / segmental edit distance (the metrics the field reports for interval localization and segmentation)"], "techniques_to_adopt": [{"technique": "Adopt Snorkel/data-programming vocabulary and consider a no-label 'label model' as a sanity check: frame each cue as a labeling function with an estimated accuracy/correlation; report per-cue accuracy and inter-cue correlation, and optionally fit a generative label model over cue agreements as a baseline against the supervised logistic regression.", "why": "Gives us the established terminology (labeling function, label model, denoising, source accuracy/correlation) for our 'signal not verdict' design, and the correlation diagnostic catches redundant/correlated cues ? a known weak-supervision failure that naive logistic stacking ignores.", "effort": "medium", "from_paper": "Snorkel: Rapid Training Data Creation with Weak Supervision (arXiv:1711.10160)"}, {"technique": "Switch LOVO scoring/thresholding to Rebalanced LOOCV (RLOOCV) or otherwise hold class balance constant across folds; never compare lift on raw LOO folds, and treat any near-random auROC swings as possible distributional-bias artifacts.", "why": "With only ~6 match-groups and a regularized logistic regression, standard LOO induces a documented inverse train-mean/test-label correlation that biases both the performance estimate and threshold/hyperparameter selection ? the exact knobs our experiment harness tunes when promoting a cue.", "effort": "low", "from_paper": "Distributional bias compromises leave-one-out cross-validation (PMC12662204)"}, {"technique": "Report tolerance-based metrics alongside tIoU: add SoccerNet-style mAP@delta in a tight (1-5s) and loose (5-60s) regime, and explicitly report boundary-slack sensitivity (how rally-start/-end precision changes the score).", "why": "tIoU alone hides where errors are (boundary slack vs whole-segment misses); the tolerance sweep is the convention reviewers expect for sports temporal localization and directly measures the serve-contact boundary-tightening our reconciler does.", "effort": "low", "from_paper": "SoccerNet: A Scalable Dataset for Action Spotting (arXiv:1804.04527; 2025 results arXiv:2508.19182)"}, {"technique": "Frame the reconciler's split/merge/boundary-tighten rules against the ASRF boundary-refinement formulation and report segmental F1@k + edit distance, not only tIoU; treat 'zero net-crossing / shuttle-inactive' as boundary/over-segmentation evidence analogous to a boundary-regression signal.", "why": "ASRF shows boundary detection is the standard remedy for over-segmentation and that segmental F1@k / edit distance are the metrics that expose merge/split errors which IoU can mask ? exactly the errors our reconciler targets.", "effort": "medium", "from_paper": "Alleviating Over-segmentation Errors by Detecting Action Boundaries / ASRF (arXiv:2007.06866)"}, {"technique": "Adopt a proposal/candidate-window paradigm justification and the 'play/break' framing explicitly; cite point-level TAL's argument that proposal-based prediction constrains the solution space better than dense frame prediction under sparse labels.", "why": "Validates our windowing-then-scoring"}]

choice with prior art and connects our task to the established play/break-segmentation lineage, improving comparability and reviewer legibility.", "effort": "low", "from_paper": "Point-Level Temporal Action Localization (arXiv:2012.08236) + Soccer HMM play/break (Xie/Xu, ICME 2001)", {"technique": "Make court/context-invariance an explicit, tested design goal: keep features scale/translation-invariant and court-agnostic, and add a probe that the scorer is not just keying on 'court visible' (e.g., evaluate on warm-up/practice stretches that share the court but lack point cadence).", "why": "Action-context confusion is the documented trap: with co-occurring static context the model learns the background, not the action. Our warm-up session-scope analyzer is the right instinct; the literature gives it a name and a way to test for the failure.", "effort": "medium", "from_paper": "WTAL Through Learning Explicit Subspaces for Action and Context (arXiv:2103.16155)"}], "pitfalls": ["LOO/LOVO distributional bias: with ~6 folds and a regularized logistic regression, standard leave-one-out induces an inverse correlation between each fold's training-label mean and its held-out label, so the model regresses to the train mean and the CV score (and any threshold/lift it drives) is systematically biased ? can even fall below chance on signal-poor folds. Mitigate with class-balance-preserving folds (RLOOCV).", "Tiny-N variance / unstable promotion gate: 6 match-groups means each LOVO fold's metric has huge variance; a cue can show 'lift' from one lucky match. Without confidence intervals / significance testing across folds, the no-regression promotion rule will promote noise. Report per-fold spread, not just the mean.", "Action-context confusion: the static court and warm-up footage co-occur with rallies, so the scorer can learn 'court/players visible' instead of 'point in play,' inflating in-distribution scores and failing on practice/dead-time that shares the background. (Documented WTAL failure mode.)", "Correlated cues break naive feature stacking: logistic regression assumes the contribution of each column is separable; highly correlated producers (e.g., shuttle-in-air and net-crossing) get double-counted or unstable weights. Weak-supervision literature explicitly models LF correlation; we currently don't.", "Over-segmentation / over-merging that tIoU hides: a single predicted segment merging two rallies across dead-time, or one rally split in two, can still score moderate tIoU. Segmental F1@k / edit-distance and a tolerance sweep are needed to expose these ? the precise errors the reconciler exists to fix.", "Reconciler that silently rewrites = label/eval leakage and metric gaming: post-hoc corrections tuned on (or evaluated against) the same scarce golden set can overfit the 6 matches. Our report-first + A/B-lift + LOVO gating is the right guard, but the lift itself must be measured under the bias-corrected CV above, or the 'measured lift' is illusory.", "Boundary/threshold tuning on the test fold: choosing tIoU thresholds, confidence cutoffs, or slack on data that includes the held-out video leaks. Group-by-match LOVO must wrap ALL tunable knobs (windowing, NMS/merge thresholds, scorer regularization), not just the final scorer fit."], "verdict": "partial-overlap", "verdict_rationale": "No single paper is our system, but every component of this theme is well-established and we should cite/reuse rather than claim novelty on the primitives. (1) 'Signal not verdict / learned scorer over cue columns' is data programming / weak supervision (Snorkel) ? the established formalization, with the caveat that we supervise directly on scarce labels rather than denoising label-free, and we ignore their inter-source-correlation modeling. (2) 'Scarce interval labels + temporal localization' is point/timestamp-supervised TAL (SF-Net, point-level TAL) evaluated with mAP@tIoU ? a named subfield. (3) 'Rally vs dead-time segmentation' is classic play/break structure analysis (Xie/Xu soccer HMM) ? a direct sibling-sport analog, and a recent badminton 'Point Segmentation' line exists. (4) The reconciler's over-segmentation/boundary correction mirrors ASRF-style boundary refinement, with segmental F1@k the expected metric. Where we are genuinely differentiated (a novel COMBINATION, not a gap): the 'detection ? decision / persist once, re-score forever' separation, the REPORT-FIRST, A/B-gated, idempotent, human-in-the-loop reconciler-as-linter (vs the field's silent in-network refinement), and the flag-gated default-OFF no-regression experiment harness on a deliberately tiny golden set. The most important course corrections are defensive on EVALUATION, not architecture: our LOVO + tiny-N + regularized-logistic stack sits squarely in the documented distributional-bias failure zone, so adopting RLOOCV/balanced folds, per-fold variance reporting, tolerance-based metrics, and an action-context-confusion probe matters more than any architectural change.", "search_queries_used": ["weakly-supervised temporal action localization video-level labels survey", "point supervision temporal action localization single timestamp", "leave-one-subject-out cross-validation small-N sports machine learning overfitting", "temporal IoU evaluation action detection metric mAP tIoU", "badminton rally detection video segmentation broadcast play break shuttlecock", "tennis match play-break detection rally segmentation audio visual cues", "action segmentation over-segmentation refinement post-processing smoothing temporal", "multimodal late fusion calibration features logistic regression action recognition small data", "pseudo-label refinement consistency between prediction and evidence weak supervision verification", "sports highlight detection structure analysis play break domain knowledge features classifier soccer", "SoccerNet action spotting evaluation temporal tolerance metric ball-action", "data programming weak supervision label model combine noisy labeling functions Snorkel", "weakly supervised action localization action-context"]

confusion completeness pitfall background"]}, {"verify": {"theme": "Weakly/interval-supervised temporal localization + small-data evaluation protocols (LOVO, tIoU)", "citations_checked": [{"title": "Snorkel: Rapid Training Data Creation with Weak Supervision (Data Programming)", "verified_real": "yes", "note": "Confirmed real. arXiv:1711.10160, PVLDB Vol 11 No 3 (2017), authors Ratner/Bach/Ehrenberg/Fries/Wu/Re all correct. Description of labeling functions + generative label model denoising without ground truth is accurate."}, {"title": "SF-Net: Single-Frame Supervision for Temporal Action Localization", "verified_real": "yes", "note": "Confirmed real. arXiv:2003.06845, ECCV 2020. Action-frame-expansion and pseudo-background-mining described accurately. MINOR ERROR: evaluated on THUMOS14/GTEA/BEID, NOT 'THUMOS14/ActivityNet' as the survey states. Not a fabrication, just a dataset mislabel."}, {"title": "Point-Level Temporal Action Localization: Bridging Fully-supervised Proposals to Weakly-supervised Losses", "verified_real": "yes", "note": "Confirmed real. arXiv:2012.08236 (2020), authors Ju/Zhao/Zhang/Wang/Tian (survey's 'Ju, Zhao, et al.' is correct). Proposal-based-paradigm-better-constrains-solution-space framing accurately summarized."}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries (ASRF)", "verified_real": "yes", "note": "Confirmed real. arXiv:2007.06866, WACV 2021, authors Ishikawa/Kasai/Aoki/Kataoka all correct. Boundary Regression Branch + the cited +13.7% edit / +16.1% F1 figures match the paper exactly."}, {"title": "Structure analysis of soccer video with Hidden Markov Models (play/break segmentation)", "verified_real": "yes", "note": "Real body of work by Xie/Xu/Chang/Divakaran/Sun on soccer play/break with dominant-color + motion features + HMM + dynamic programming (academia.edu copy + closely-titled 'with domain knowledge and hidden Markov models'; related ICME 2001 'Algorithms and systems for segmentation and structure analysis in soccer video'). Exact title/year (2001-2004) is slightly fuzzy across the cluster but the cited concept and authorship are genuine; the ~86.5% accuracy figure was not independently re-verified."}, {"title": "SoccerNet: A Scalable Dataset for Action Spotting in Soccer Videos (+ 2023-2025 Challenges)", "verified_real": "yes", "note": "Both IDs real. Dataset arXiv:1804.04527 (CVPR Workshops 2018, Giancola/Amine/Dghaily/Ghanem); tolerance-based Average-mAP over 5-60s deltas confirmed. 2025 results arXiv:2508.19182 ('SoccerNet 2025 Challenges Results') also real."}, {"title": "Distributional bias compromises leave-one-out cross-validation", "verified_real": "yes", "note": "Real and PMC12662204 resolves. BUT VENUE IS WRONG: published in SCIENCE ADVANCES (DOI 10.1126/sciadv.adx6976, arXiv:2406.01652), not 'Briefings in Bioinformatics / GigaScience-class' as the survey guesses. The negative train-mean/test-label correlation mechanism and the RL00CV (drop-an-opposite-label-sample) fix are described accurately."}, {"title": "Weakly Supervised TAL Through Learning Explicit Subspaces for Action and Context (action-context confusion)", "verified_real": "yes", "note": "Confirmed real. arXiv:2103.16155, AAAI 2021 (vol 35, pp 2242-2250), authors Liu/Wang/Tang/Yuan/Zheng/Hua. 'Action-context confusion' term and explicit action/context subspaces accurately described."}], "fabrication_flags": [], "corrected_verdict": "partial-overlap", "corrected_rationale": "Verdict stands at partial-overlap. See closer_prior_art and extra_course_corrections.", "closer_prior_art": [{"title": "Towards Structured Analysis of Broadcast Badminton Videos", "url_or_id": "arXiv:1712.08714", "year": "2017", "why_closer": "Defines the SAME task we do under the established name 'Point Segmentation': partition a badminton broadcast into rally vs non-rally segments, then keep/group rally frames. The survey only alludes to this in the verdict rationale; it should be a cited key paper because it is the closest task-level prior art and shows our problem is named and pre-existing in badminton specifically, not just by sibling-sport analogy."}, {"title": "Decoupling Localization and Classification in Single Shot Temporal Action Detection (Decouple-SSAD)", "url_or_id": "arXiv:1904.07442", "year": "2019", "why_closer": "Closest prior art for our 'detection ? decision' separation: it explicitly decouples the localization sub-task from the classification sub-task in TAD. Combined with the standard offline-detection-over-pre-extracted-features setup, it weakens any claim that 'persist detection once, re-score the decision separately' is architecturally novel. Cite it to scope our contribution to the persist-once/idempotent-rescore engineering discipline, not the decoupling idea itself."}, {"title": "Pseudo label refining for semi-supervised TAL (PLR) / Learning from Noisy Pseudo Labels for Semi-Supervised TAL", "url_or_id": "PMC11798530 (PLOS One 2025); ICCV 2023 (Xia et al.)", "year": "2023-2025", "why_closer": "Closest prior art for the reconciler's 'prediction contradicts evidence -> correct it' loop. These use an Instance-level Consistency Discriminator / consistency losses to detect and fix pseudo-label boundary errors. They are LEARNED, in-loop, and SILENT, so they do NOT subsume our report-first, human-gated, A/B-measured linter ? but they ARE the lineage our reconciler should cite and differentiate against, beyond ASRF alone."}], "extra_course_corrections": ["FIX THE LOOCV-BIAS CITATION METADATA before reusing it: the paper is in Science Advances (DOI 10.1126/sciadv.adx6976, arXiv:2406.01652), NOT 'Briefings in Bioinformatics / GigaScience-class'. The PMC ID is right but the venue guess is wrong ? a reviewer will catch a fabricated-looking venue. Keep the (correct) RL00CV technical claim.", "FIX the SF-Net dataset claim: it evaluates on THUMOS14/GTEA/BEID, not 'THUMOS14/ActivityNet'. Small

but a careful reader will notice.", "ADD arXiv:1712.08714 (badminton 'Point Segmentation') and the newer full-match badminton rally-segmentation datasets (e.g. BWF World Tour frame-level rally/interruption annotations, surfaced in 2025 badminton-analysis papers) as cited prior art. Our task is named and pre-existing in badminton itself; not citing it overstates task novelty and is the single biggest legibility gap.", "ADOPT (reinforced by verification): the survey's own top defensive corrections are the right priorities ? RLOOCV/class-balance-preserving folds, per-fold variance/CIs on the 6-group LOVO promotion gate, tolerance-based mAP@delta alongside tIoU, segmental F1@k + edit distance for the reconciler, and an action-context-confusion probe on warm-up footage. My searches independently confirm each cited technique is real and applicable; none were oversold.", "AVOID claiming novelty on the no-regression/shadow/flag-gated promotion harness as a primitive ? it is textbook MLOps (SageMaker shadow testing, feature-flag-gated model rollout). The survey already scopes it as a 'combination'; keep it that way and frame the contribution as 'this discipline applied to a deliberately tiny golden set', not the mechanism.", "Tighten the soccer-HMM citation to a specific resolvable reference (exact title + venue + year) rather than the fuzzy '2001-2004' range; the work is real but the cluster has several near-identical titles and an under-specified cite looks weaker than it is."]}, {"theme": "Multi-scale / hierarchical temporal modeling; video-level context & phase/stage segmentation", "short": "multiscale-phase", "our_pattern": "Our frame/window/session SCOPE taxonomy ? session-scope context (warm-up/practice) conditioning per-window decisions.", "survey": {"theme": "Multi-scale / hierarchical temporal modeling; video-level context & phase/stage segmentation", "our_pattern": "A three-tier temporal SCOPE taxonomy for rally detection. (1) frame-scope DETECTORS produce expensive per-frame signals (shuttle x,y,visibility from a frozen WASB/TrackNet CNN; person boxes/tracks; audio thwack; a proposed per-frame shuttle-state classifier), persisted once. (2) window-scope ANALYZERS emit cheap scale/translation-invariant aggregate features per candidate rally window (net-crossing count, service-fault halt-in-net, person two-sidedness). (3) session-scope ANALYZERS run over the WHOLE timeline to emit labeled spans + per-window membership features ? notably WARM-UP/practice detection (a long continuous-rally stretch lacking the point/dead-time cadence) that CONDITIONS the per-window rally decision. Every cue is a (value, confidence) feature column consumed by a small logistic-regression scorer; coarse session-level state gates/contextualizes fine-grained per-window decisions. Evaluated by temporal IoU vs scarce vendor golden intervals, with leave-one-video-out (group-by-match) splits.", "key_papers": [{"title": "MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation", "authors": "Yazan Abu Farha, Juergen Gall", "year": "2019", "venue": "CVPR 2019", "url_or_id": "arXiv:1903.01945", "what_it_does": "Multi-stage architecture for temporal action segmentation: each stage uses dilated temporal convolutions over the full-resolution timeline to make an initial frame-wise prediction that the next stage REFINES. Adds a smoothing loss (truncated MSE over log-prob differences) explicitly to penalize over-segmentation. Reports F1@[10,25,50], segmental edit score, frame accuracy on 50Salads/GTEA/Breakfast.", "overlap": "partial", "overlap_explanation": "Same core problem (segment an untrimmed timeline into action vs background spans, fight over-segmentation) and same canonical metrics we should adopt (F1@k overlap, segmental edit score) ? these are richer than raw temporal IoU. The iterative refinement IS conceptually our reconciler pass (initial decision -> refined decision using a wider temporal view). Differs in that MS-TCN is one end-to-end deep net learning everything from frame features, whereas we deliberately use hand-crafted multi-scope feature columns + a tiny logistic regression because labels are scarce; and our refinement is a report-first, A/B-gated rule pass, not a learned stage."}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries (ASRF)", "authors": "Yuchi Ishikawa, Seito Kasai, Yoshimitsu Aoki, Hirokatsu Kataoka", "year": "2021", "venue": "WACV 2021", "url_or_id": "arXiv:2007.06866", "what_it_does": "Splits action segmentation into two branches over a shared long-term feature extractor: an Action Segmentation Branch (per-frame class) and a separate class-agnostic Boundary Regression Branch. The predicted boundaries refine/correct the segmentation as a post-process, sharply cutting over-segmentation.", "overlap": "strong", "overlap_explanation": "This is the closest analogue to our RECONCILERS. ASRF treats boundary correction as a SEPARATE, explicit post-hoc pass that reconciles an initial segmentation against an independent boundary signal ? exactly our 'detect where the decision contradicts the evidence, then truncate/split/tighten boundaries.' Our truncate-on-shuttle-inactive and split-on-internal-dead-gap are rule-based instances of ASRF's learned boundary-refinement idea. Differs: ASRF learns boundaries with a neural branch and applies them unconditionally; ours are evidence-driven rules that are report-first and A/B-gated before auto-apply."}, {"title": "Generic play-break event detection for summarization and hierarchical sports video analysis", "authors": "Dian Tjondronegoro, Yi-Ping Phoebe Chen, Binh Pham", "year": "2003", "venue": "IEEE ICME 2003", "url_or_id": "IEEE Xplore document 1220881", "what_it_does": "A single generic algorithm segments a sports broadcast (football, tennis, basketball, soccer) into 'play' vs 'break' intervals using shot-based cinematic features (shot type, shot length), then uses those play/break spans as a HIERARCHICAL outer layer that

defines the segments-of-interest fed to finer event detectors (e.g. goal detection keys off the break that follows a goal play).", "overlap": "strong", "overlap_explanation": "Directly validates our session/window scope hierarchy: a coarse, whole-timeline play-vs-break (? rally vs dead-time) layer determines where and how fine-grained detection runs. Our warm-up/practice session-scope span that conditions per-window rally decisions is the same coarse-state-gates-fine-detection move, two decades old in sports video. Differs: they rely on broadcast/cinematic editing cues (replays, shot cuts) that DON'T exist in our static tripod footage; our coarse layer must come from gameplay-cadence signals instead. The play/break terminology and the hierarchical-context framing are ours to adopt."}, {"title": "Global-local multi-stage temporal convolutional network for cataract surgery phase recognition (GL-MSTCN)", "authors": "Lixin Fang et al.", "year": "2022", "venue": "BioMedical Engineering OnLine (PMC9710114)", "url_or_id": "doi:10.1186/s12938-022-01048-w", "what_it_does": "A multi-stage TCN with explicit GLOBAL and LOCAL streams: local blocks capture fine-grained per-frame/short-window surgical action; a global stream consolidates long-range, video-level temporal context so that locally ambiguous, visually-similar phases are disambiguated by where they sit in the procedure. Reports ~95.8%/96.5% accuracy on cataract datasets."}, {"title": "Squarely our theme: an explicit global (video-level) context branch improving local (frame/window) decisions for monotonic-phase video. Surgical phase recognition is the canonical literature for 'coarse video-level state conditions fine-grained recognition' and is methodologically transferable to our warm-up-vs-match conditioning. Differs: surgical phases follow a near-fixed order (strong sequence priors via dilated TCN / state-space models), whereas our rallies are repetitive and order-free, so we get less from sequence priors and more from cadence/membership features. Confirms terminology ('global-local', 'phase recognition') and the idea of a learned global-context feature."}, {"title": "Hierarchical Multi-Stage Transformer Architecture for Context-Aware Temporal Action Localization (PCL-Former)", "authors": "(authors per arXiv listing ? not individually verified)", "year": "2025", "venue": "arXiv preprint", "url_or_id": "arXiv:2507.06411", "what_it_does": "A hierarchical, multi-stage transformer that decomposes temporal action localization into a Proposal stage (candidate segments), a Classification stage, and a Localization stage that refines boundaries ? explicitly separating proposal generation, scoring, and boundary refinement with multi-level (snippet + proposal + video) context."}, {"title": "Mirrors our DECOUPLED pipeline: candidate windowing -> scoring -> boundary refinement, with multi-granularity context ? the same separation-of-concerns we encode as windowing + logistic scorer + reconcilers. Marked partial/weaker because I verified it only from the search snippet and arXiv listing (not the full PDF), it is a heavy deep transformer rather than our lightweight scorer, and authorship was not independently confirmed."}, {"title": "LiteEval: A Coarse-to-Fine Framework for Resource Efficient Video Recognition", "authors": "Zuxuan Wu et al.", "year": "2019", "venue": "NeurIPS 2019", "url_or_id": "arXiv:1912.01601", "what_it_does": "A coarse LSTM glimpses cheap low-res features over a video, and a conditional GATING module decides whether each frame warrants expensive fine-grained feature computation ? i.e. a coarse global pass gates where expensive fine processing runs."}, {"title": "Validates the efficiency half of our DETECTION?DECISION + scope split: run cheap coarse analysis first, spend expensive fine compute only where the coarse layer says it matters. Differs: LiteEval's gating is a learned, online, per-frame compute-saving decision; our split is offline persist-once/re-score-forever where the expensive detector ALWAYS runs once and the cheap re-scoring is what varies. So the framing transfers but our motivation (re-scorability + scarce labels) differs from theirs (inference FLOPs)."}], "established_terms": ["temporal action segmentation (frame-wise labeling of an untrimmed timeline into action/background spans ? the formal name for our rally/dead-time split)", "play-break detection (the sports-video term for rally-vs-dead-time; our session/window coarse layer)", "surgical/cooking PHASE recognition (the canonical literature for coarse video-level state conditioning fine recognition)", "global-local / multi-scale temporal modeling (explicit global video-level branch + local frame branch)", "over-segmentation error (the named failure our reconcilers and a smoothing loss fight)", "action boundary regression / boundary refinement (ASRF ? the formal name for our boundary-tightening/truncate/split reconcilers)", "temporal proposal generation -> classification -> localization (the decoupled three-stage pipeline; our windowing -> scorer -> reconcile)", "leave-one-subject-out / leave-one-video-out cross-validation (our LOVO; split-by-group to prevent leakage)", "F1@k (overlap F1 at IoU thresholds 10/25/50) and segmental edit score (the standard segmentation metrics beyond raw temporal IoU)"], "techniques_to_adopt": [{"technique": "Report F1@{10,25,50} and segmental edit score alongside temporal IoU."}, {"why": "These are the field-standard metrics for temporal action segmentation (MS-TCN, ASRF). Raw temporal IoU rewards frame-overlap but is nearly blind to over-segmentation (a span chopped into 3 pieces can still score high IoU); F1@k and edit score directly penalize fragmentation and boundary errors ? exactly the failures our reconcilers target. Adopting them makes our 'lift' measurements comparable to the literature and catches

regressions IoU hides.", "from_paper": "MS-TCN (Abu Farha & Gall, 2019); ASRF (Ishikawa et al., 2021)", "effort": "low"}, {"technique": "Frame the reconcilers explicitly as a class-agnostic boundary-refinement stage (ASRF pattern): an independent boundary signal corrects an initial segmentation as a post-process.", "why": "ASRF shows that separating 'is this a rally' from 'where exactly are its boundaries' and refining boundaries against an independent signal yields large drops in over-segmentation. Our truncate-on-inactive / split-on-internal-gap / tighten-to-serve-contact rules ARE this stage; adopting the framing (and ASRF's boundary metric) gives us a principled, citable home and a way to measure boundary quality separately from detection quality.", "from_paper": "ASRF (Ishikawa et al., 2021)", "effort": "medium"}, {"technique": "Add a smoothing/anti-fragmentation regularizer or post-filter to the decision, not just rely on reconcilers.", "why": "MS-TCN's truncated-MSE smoothing loss is the canonical fix for over-segmentation. Since our scorer is a logistic regression over windows (not a frame net), the analogue is a cheap merge/min-duration/transition-penalty post-filter or a HMM/Viterbi smoothing over the per-window rally probabilities ? a learned, measured smoother rather than only rule-based reconcilers, reducing reliance on hand-tuned rule precedence.", "from_paper": "MS-TCN (Abu Farha & Gall, 2019)", "effort": "medium"}, {"technique": "Treat the session-scope warm-up/play-break layer as an explicit hierarchical outer segmentation whose spans key the inner detector (Tjondronegoro play-break scheme).", "why": "The play-break work establishes the exact hierarchy we are building (coarse play/break spans define segments-of-interest for finer detectors) as a recognized design, giving us prior-art cover and the 'play/break' vocabulary. It also flags the right inputs: since we lack broadcast cinematic cues, our coarse layer must be built from gameplay-cadence features (rally/dead-time rhythm), which is a concrete design constraint to encode.", "from_paper": "Generic play-break event detection (Tjondronegoro et al., 2003)", "effort": "medium"}, {"technique": "Model the global/session context as a learned membership FEATURE per window (global-local fusion), and verify it actually helps via ablation.", "why": "GL-MSTCN shows a global stream disambiguates locally-similar segments ? our intended warm-up-conditions-rally move. Encoding session-scope state as an extra feature column (not a hard pre-filter) keeps it inside the 'signal not verdict' regime and lets the scorer learn its weight; the surgical literature's ablations (global vs local-only) are the template for proving the global cue earns its place.", "from_paper": "GL-MSTCN (Fang et al., 2022)", "effort": "low"}], "pitfalls": ["Over-segmentation is THE dominant failure mode of timeline segmentation (MS-TCN, ASRF both built specifically to fight it). Our merge/split reconcilers must be measured on F1@k/edit, not IoU, or fragmentation regressions will slip through invisibly.", "Coarse-state gating can hard-cut recall: if the session-scope warm-up/play-break layer is wrong, it suppresses real rallies inside a mislabeled span (the classic hierarchical-gating risk where an upstream error is unrecoverable downstream). Keep session context as a soft feature column, never a hard pre-filter, so a wrong coarse call is recoverable by the scorer ? consistent with your 'signal not verdict' rule.", "LOVO leakage is subtle: the action-segmentation field splits by SUBJECT/participant, not just by clip, because frames from the same person/session share idiosyncrasies. Your group-by-match LOVO is correct, but watch second-order leakage ? features fit/normalized or thresholds tuned on the full corpus, or a shared vendor-labeling bias, still leak across folds.", "Static-tripod transfer gap: the strongest sports-video priors (play-break, badminton broadcast pipelines) lean on broadcast editing cues ? replays, shot cuts, score graphics ? that DON'T exist in your footage. Borrowing their methods wholesale will silently fail; you must re-source the coarse cue from gameplay cadence, and the literature won't have done that for you.", "Tiny-corpus overfitting of the gating/global cue: with ~6 matches, a session-scope feature can latch onto match-specific quirks (one warm-up style) and look like lift in-fold while failing out-of-fold. Surgical/segmentation papers train on dozens-to-hundreds of videos; insist on per-fold LOVO lift for any global feature before promotion, exactly as your harness specifies.", "Sequence-prior mismatch: MS-TCN/surgical models gain much of their power from near-fixed phase ORDER (Viterbi/grammar priors). Badminton rallies are repetitive and order-free, so importing strong sequence-transition priors can hurt; prefer cadence/duration/membership features over learned action-grammar transitions."], "verdict": "partial-overlap", "verdict_rationale": "The structural pattern ? a coarse, video/session-level state (play-break, surgical phase, warm-up) conditioning fine-grained per-window decisions, with a decoupled propose -> score -> boundary-refine pipeline ? is well-established across three mature literatures (temporal action segmentation: MS-TCN/ASRF; sports play-break: Tjondronegoro 2003; global-local phase recognition: GL-MSTCN). So the hierarchical-context idea itself is NOT novel and we should cite and borrow vocabulary and metrics from it rather than reinvent. What is genuinely differentiated (and not reinventing) is the engineering regime: persist-once/re-score-forever detection?decision, every cue as a calibrated (value, confidence) feature column with a deliberately tiny logistic-regression scorer for a ~6-match scarce-label corpus, report-first A/B-gated reconcilers instead of unconditional learned refinement, and a no-regression flag-gated harness ? a combination the cited deep, data-hungry, end-to-end methods do not provide. Hence partial-overlap: adopt their terminology (play-

break, phase recognition, over-segmentation, boundary refinement), their metrics (F1@k, edit score), and their global-local ablation discipline as course corrections, while keeping our lightweight, re-scorable, regression-safe architecture as the distinctive contribution. Two caveats lower confidence on the edges: GL-MSTCN author detail and PCL-Former authorship were not fully verified from primary PDFs (marked accordingly), and the closest 'pure' analogue to our exact scope-taxonomy-as-feature-columns design (vs end-to-end nets) was not found verbatim, leaving a thin slice that may be a small genuine gap.", "search_queries_used": ["hierarchical multi-granularity temporal action localization video-level context", "surgical phase recognition multi-stage temporal model global context", "MS-TCN multi-stage temporal convolutional network action segmentation over-segmentation", "global video context improves frame-level action recognition coarse-to-fine gating", "tennis badminton broadcast video play break rally segmentation temporal", "sports video event detection hierarchical game phase structure prior localization", "video temporal segmentation warm-up practice versus match detection sports scene classification", "Tjondronegoro generic play-break event detection summarization hierarchical sports video", "leave-one-video-out cross-validation temporal action segmentation small dataset leakage overfitting sports", "Alleviating over-segmentation action segmentation boundary refinement ASRF separate boundary regression", "two-stage cascade video-level scene gating frame-level detection conditional structured prediction temporal IoU"]}, {"verify": {"theme": "Multi-scale / hierarchical temporal modeling; video-level context & phase/stage segmentation", "citations_checked": [{"title": "MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation (Abu Farha & Gall, CVPR 2019, arXiv:1903.01945)", "verified_real": "yes", "note": "Confirmed: correct title, authors, CVPR 2019, arXiv:1903.01945. Multi-stage refinement + truncated-MSE smoothing loss against over-segmentation; F1@{10,25,50}/edit/frame-acc on 50Salads/GTEA/Breakfast. Accurately characterized."}, {"title": "Alleviating Over-segmentation Errors by Detecting Action Boundaries (ASRF) (Ishikawa, Kasai, Aoki, Kataoka, WACV 2021, arXiv:2007.06866)", "verified_real": "yes", "note": "Confirmed: correct title, all four authors, WACV 2021, arXiv:2007.06866, official repo yiskw713/asrf. Action Segmentation Branch + separate class-agnostic Boundary Regression Branch; boundary post-process cuts over-segmentation. Accurately characterized."}, {"title": "Generic play-break event detection for summarization and hierarchical sports video analysis (Tjondronegoro, Chen, Pham, IEEE ICME 2003)", "verified_real": "yes", "note": "Confirmed: exact title and IEEE ICME 2003 venue. Generic shot-based cinematic play/break detection across football/tennis/basketball/soccer, used as a hierarchical outer layer keying finer event detectors. Accurately characterized, including the broadcast-cinematic-cue dependency the survey correctly flags as absent in static-tripod footage."}, {"title": "Global-local multi-stage temporal convolutional network for cataract surgery phase recognition (GL-MSTCN) (Fang et al., BioMedical Engineering OnLine 2022, doi:10.1186/s12938-022-01048-w, PMC9710114)", "verified_real": "yes", "note": "Confirmed: exact title, journal, DOI, PMC id. Explicit global + local streams; global video-level context disambiguates locally-similar surgical phases. Survey honestly used 'Fang et al.'; lead author Lixin Fang consistent with PubMed PMID 36451164. Accurately characterized."}, {"title": "Hierarchical Multi-Stage Transformer Architecture for Context-Aware Temporal Action Localization (PCL-Former) (arXiv:2507.06411, 2025)", "verified_real": "yes", "note": "Confirmed real: arXiv:2507.06411 exists with the exact title. Abstract confirms Proposal-/Classification-/Localization-Former three-stage decomposition (hence 'PCL-Former') and THUMOS14/ActivityNet-1.3/HACS benchmarks. Survey appropriately marked authorship unverified and did NOT assert author names, so no fabrication risk. Characterization accurate."}, {"title": "LiteEval: A Coarse-to-Fine Framework for Resource Efficient Video Recognition (Zuxuan Wu, Caiming Xiong, Yu-Gang Jiang, Larry S. Davis, NeurIPS 2019, arXiv:1912.01601)", "verified_real": "yes", "note": "Confirmed: correct title, lead author Zuxuan Wu, NeurIPS 2019, arXiv:1912.01601. Coarse LSTM + conditional gating decides where to spend fine-grained compute. Accurately characterized, including the honest FLOPs-vs-re-scorability caveat."}], "fabrication_flags": [], "corrected_verdict": "partial-overlap", "corrected_rationale": "The 'partial-overlap' verdict survives adversarial scrutiny and all six citations are real and accurately characterized (no fabrications; the two self-flagged 'unverified-author' items, PCL-Former and GL-MSTCN, both resolve, and the survey correctly avoided asserting unverified author lists). However, my refutation searches surfaced a genuine COVERAGE GAP that weakens the survey's specific 'what is differentiated' argument: it benchmarked only against deep end-to-end action-segmentation / phase-recognition nets and concluded the closest analogue to its lightweight fusion design 'was not found,' but two much closer analogues exist and were missed. (1) MDPI Sensors 2024 'Enhancing Badminton Game Analysis...' (PMC11244353) uses the SAME architecture family on the SAME footage type: monocular non-broadcast badminton, TrackNet shuttle trajectory fused with YOLOv7 player-swing detection, a 7-dimensional handcrafted aggregate feature vector, a deliberately simple learned classifier (XGBoost, chosen 'limited by the number of features and samples'), evaluated by temporal IoU.

(2) Chaudhury et al. ISM 2019 'Unsupervised Temporal Feature Aggregation...' (arXiv:2002.08097) does rally detection in UNSTRUCTURED arbitrary-angle (non-broadcast) table-tennis video. Together these directly soften the survey's 'static-tripod transfer gap / the literature won't have done that for you' pitfall and its claim that the multi-cue-fusion + small-classifier + temporal-IoU regime is unmatched. The verdict stays partial-overlap (NOT 'reinventing') because neither closer paper implements the genuinely distinctive engineering regime the survey claims: persist-once/re-score-forever detection-vs-decision separation, every cue as a calibrated (value,confidence) feature column, report-first A/B-gated rule reconcilers (vs unconditional learned refinement), the no-regression flag-gated harness, and the session-scope warm-up/practice span used as a SOFT membership feature conditioning per-window rally decisions. Targeted searches for that warm-up-as-soft-feature move found only broadcast rally/non-rally point segmentation (replay/rest-based) and rally-endpoint detection -- different mechanisms -- so that one slice remains a thin but real genuine gap. Net: borrow the field's vocabulary (play-break, phase recognition, over-segmentation, boundary refinement) and metrics (F1@k, segmental edit score) as the survey recommends, but DROP the 'no prior art does multi-cue fusion + simple classifier on static racket-sport footage' framing -- PMC11244353 and arXiv:2002.08097 are the true nearest neighbors and should be cited as such.

,"closer_prior_art":[{"title":"Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera","url_or_id":"MDPI Sensors 2024, 24(13):4372 / PMC11244353","year":"2024","why_closer":"Far closer to the ACTUAL fusion architecture than any of the six cited deep nets, yet missed entirely. Monocular non-broadcast badminton footage; fuses TrackNet shuttle trajectory (Hit-Detection-by-Trajectory) with YOLOv7 player swing/action (Hit-Detection-by-Action); 7-dim handcrafted aggregate features (ankle positions, flight duration, shuttle displacement); a deliberately simple learned classifier (XGBoost, explicitly chosen due to few features/samples); evaluated by temporal IoU at 0.5/0.85/0.95. This is the survey's own frame-scope-detector + window-scope-aggregate-features + small-scorer + temporal-IoU-eval pattern, on the same sport and same static-camera setting -- directly undercutting the 'static-tripod transfer gap, literature has not done this' pitfall."}],{"title":"Unsupervised Temporal Feature Aggregation for Event Detection in Unstructured Sports Videos","url_or_id":"arXiv:2002.08097 / IEEE ISM 2019 (Chaudhury et al., IBM Research)","year":"2019","why_closer":"Directly targets rally detection in UNSTRUCTURED, arbitrary-camera-angle (non-broadcast) sports video (table tennis) -- the exact setting the survey claims prior art avoids. Learns a rally detector via temporal feature aggregation over a continuous timeline without relying on broadcast cinematic cues, partially refuting the 'must re-source the coarse cue ourselves because no one has' framing."}],{"title":"Service and End of Rally Detection in Badminton Videos","url_or_id":"Springer 10.1007/978-3-031-69073-0_15 (2024)","year":"2024","why_closer":"Detects rally boundaries (service start / end of rally) in badminton using court + pose cues -- a direct racket-sport-specific instance of the survey's boundary-refinement reconcilers (tighten-to-serve-contact / truncate-on-rally-end), beyond the generic ASRF citation."}],{"extra_course_corrections":["ADOPT: Cite the MDPI Sensors 2024 badminton paper (PMC11244353) and Chaudhury et al. ISM 2019 (arXiv:2002.08097) as the TRUE nearest neighbors. They show multi-cue (shuttle-trajectory + player-action) fusion into a small classifier evaluated by temporal IoU on monocular/unstructured footage is already established. Reframe the contribution as the persist-once/re-score-forever + calibrated-confidence-feature-columns + regression-safe harness + warm-up-soft-feature-conditioning regime, NOT 'multi-cue fusion on static footage' (which is taken).","AVOID/CORRECT: Soften the 'static-tripod transfer gap -- the literature won't have done that for you' pitfall. It overstates the gap: PMC11244353 operates on exactly monocular non-broadcast badminton with shuttle+player cues, and arXiv:2002.08097 handles arbitrary-angle non-broadcast rally detection. The real residual gap is narrower: warm-up/practice continuous-rally detection used as a SOFT session-scope conditioning feature, which no single paper was found to do."],{"ADOPT: For the racket-sport boundary reconcilers, cite a sport-specific instance (Springer 'Service and End of Rally Detection in Badminton Videos', 2024) alongside the generic ASRF, since it directly addresses serve-contact / end-of-rally boundary detection in badminton."}],{"CONFIRM: The survey's metric recommendation (add F1@{10,25,50} + segmental edit score beyond temporal IoU) is well-founded and consistent with the verified MS-TCN/ASRF citations; keep it as a high-value low-effort course correction."}],{"CAUTION: The XGBoost precedent in PMC11244353 invites reviewers to ask why a logistic-regression scorer is preferable to a gradient-boosted tree on the same handcrafted features over a ~6-match corpus; pre-empt with the actual rationale (interpretability / calibrated confidence / regression-safety), rather than treating 'tiny scorer' as itself novel."}]}, {"theme":"Specific cue novelty: structural rally discriminators, warm-up detection, shuttle-state / stroke detection","short":"cue-novelty","our_pattern":"Our specific cues: net-crossing count, held-shuttle/service-fault, warm-up/practice-phase detection, shuttle-state (in_air/contact/inactive), stroke/shot detection."}],{"survey":{"theme":"Specific cue novelty: structural rally discriminators, warm-up detection, shuttle-state / stroke

detection", "our_pattern": "We discriminate true rallies from dead time and clean up rally boundaries with a family of cheap, scale/translation-invariant cues over a persisted per-frame shuttle track (WASB/TrackNet) plus person/audio detectors: net-crossing count, a held-shuttle/service-fault cue (shuttle halts in the net region), warm-up/practice-phase detection (a long continuous-rally stretch lacking the point/dead-time cadence), per-frame shuttle-state classification (in_air / racket_contact / inactive), and stroke/shot detection. Each cue is a (value, confidence) feature column consumed by a tiny logistic-regression scorer; reconcilers then use the persisted evidence (shuttle inactivity, zero net-crossings, internal dead-time gaps) to truncate/drop/split/tighten decided segments. Graded with temporal-IoU vs scarce vendor golden intervals under leave-one-video-out (group-by-match).", "key_papers": [{"title": "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "authors": "Liu, Hodgins et al.", "year": "2022", "url_or_id": "arXiv:2204.01899 (CVPRW 2022)", "venue": "CVPR Workshops (CVSports)", "what_it_does": "End-to-end monocular badminton pipeline: U-Net shuttle detector, physics-based 3D trajectory reconstruction (particle-under-drag, reprojection-error minimization), and HitNet, a ~16K-param GRU that does 3-way hit classification (no hit / near-player hit / far-player hit) over 12-frame windows of court-normalized pose + shuttle position. Crucially applies constrained-optimization POST-PROCESSING with badminton domain rules (one hit ~per second, >=0.5s min separation, alternating players, in-bounds, initial velocity toward opponent), lifting hit accuracy 76.4%->89.7% (F1 0.946).", "overlap": "strong", "overlap_explanation": "Direct prior art for our shuttle-state/stroke detection AND our reconcilers. HitNet IS a learned shuttle-state/hit classifier from the same inputs we'd use. Their post-hoc constrained optimization is exactly our reconciler idea (use persisted trajectory + domain physics to correct raw per-frame decisions) and their 'initial velocity toward opponent' / in-bounds terms are net-crossing/rally-structure constraints. Difference: they segment a clean single-rally clip into SHOTS; we segment a long uncut match into RALLY vs dead-time intervals (a different temporal scope) and we keep cues as soft feature columns rather than hard optimization constraints."}, {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "authors": "Lin et al.", "year": "2024", "url_or_id": "Sensors 24(13):4372 / PMC11244353", "venue": "Sensors (MDPI)", "what_it_does": "Fuses TrackNet shuttle trajectory + YOLOv7 swing/no-swing player state to find hit frames; uses trajectory geometry (hits occur at the low/turning point where the trajectory begins to rise) plus a 'shot refinement' algorithm to correct the hit moment; does rough rally segmentation as a preprocessing step. Reports 89.7% precision-ish / 91.3% recall / 90.5% F1 over 69 BWF matches, 1582 rallies.", "overlap": "strong", "overlap_explanation": "Almost exactly our proposed shuttle-state cue + a fusion-of-cues stroke detector: combines shuttle-trajectory state with a per-frame player 'swinging' state to detect contact, then refines. Their 'trajectory turning point = contact' is the geometric basis our in_air/contact state classifier would learn. Difference: they fuse with a hand-designed refinement rule rather than a learned scorer, and target shot timing inside known rallies, not rally-vs-deadtime segmentation of an uncut match."}, {"title": "A New Perspective for Shuttlecock Hitting Event Detection", "authors": "(CoachAI Badminton Challenge participants)", "year": "2023", "url_or_id": "arXiv:2306.10293", "venue": "arXiv / IJCAI 2023 CoachAI Challenge Track 1", "what_it_does": "Detects shuttlecock hitting events by reasoning over image sequences with SwingNet plus engineered prior features extracted from clips to reduce learning difficulty; framed for the CoachAI badminton challenge.", "overlap": "partial", "overlap_explanation": "Confirms 'shuttle hit / contact event detection from a frozen detector + light features' is an established, benchmarked task (so our shuttle-state contact label is not novel). Overlap only partial because it targets discrete hit-event spotting, not per-frame in_air/contact/inactive STATE labeling nor rally-interval segmentation; we couldn't fully verify internal metrics from the abstract."}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis (+ ShuttleSet22)", "authors": "Wang, Huang et al.", "year": "2023", "url_or_id": "ACM SIGKDD 2023, doi:10.1145/3580305.3599906; ShuttleSet22 arXiv:2306.15664", "venue": "KDD 2023", "what_it_does": "Largest public badminton singles dataset: 104 sets / 3,685 rallies / 36,492 strokes across 44 matches, with rally-level and stroke-level human annotations and the CoachAI benchmark/codebase (github.com/wywyWang/CoachAI-Projects).", "overlap": "partial", "overlap_explanation": "This is the field-standard rally-AND-stroke-segmented badminton resource and vocabulary ('rally', 'stroke', 'set'). It defines the rally/stroke unit our cues operate on and is a ready external eval/pretraining corpus that could relieve our ~6-match scarcity. Overlap is on data/task definition, not on our specific cue mechanics."}, {"title": "A framework for flexible summarization of racquet sports video using multiple modalities (Tjondronegoro, Chen, Pham)", "authors": "Tjondronegoro, Chen, Pham", "year": "2008", "url_or_id": "CVIU / doi:10.1016/j.cviu.2008.07.001 (S1077314208001355)", "venue": "Computer Vision and Image Understanding", "what_it_does": "Segments racquet-sports video into RALLY vs BREAK events using multiple modalities (unsupervised shot

clustering for view types + supervised audio classification for ball-hit/whistle), fuses cues via a temporal voting strategy, then ranks rallies by excitement. Popularized the play-break segment as a universal, sport-agnostic scope of detection with a standard feature set.", "overlap": "strong", "overlap_explanation": "This is the canonical framing our whole system reinvents at the structural level: 'rally vs break' as the universal segmentation unit, fused from multiple per-frame cues. Our 'producers emit feature columns -> small scorer decides rally/break' is a modern, learned-fusion restatement of their multimodal voting. Difference: they use broadcast-only cues (camera-view clustering, crowd/commentary audio) and a voting heuristic; we use a frozen-detector shuttle track + a learned LR scorer on a STATIC single-camera feed, plus persisted re-scoring and reconcilers."}, {"title": "SmartTennisTV: Automatic indexing of tennis videos", "authors": "Mohan, Pramod, Jawahar", "year": "2018", "url_or_id": "arXiv:1801.01430 (NCVPRIPG 2017)", "venue": "NCVPRIPG 2017", "what_it_does": "Segments tennis broadcasts into rally vs non-rally by classifying frames with a chi-squared-kernel SVM over HOG features (exploiting that the overhead camera is shown only during in-play), then Kalman-smooths the per-frame predictions to remove false pos/neg; reaches F1 ~97.5%. Score recognition gives point-level indexing afterward."}, {"title": "Automatic rally detection on broadcast tennis videos", "authors": "(Sports Technology 2013)", "year": "2013", "url_or_id": "doi:10.1080/19346182.2013.819007", "venue": "Sports Technology 6(2)", "what_it_does": "Detects rallies in broadcast tennis; reports ~81% average detection (up to 91% on hard courts). Often paired with finite-state-machine accounts of rally state (serve -> in-rally -> out-of-rally on a non-shot/double-hit), and trajectory-based in/out-of-bounds and net-clearing reasoning."}, {"title": "Automated Service Height Fault Detection Using Computer Vision and Machine Learning for Badminton Matches", "authors": "(MDPI Sensors)", "year": "2023", "url_or_id": "Sensors 23(24):9759 / PMC10747833", "venue": "Sensors (MDPI)", "what_it_does": "YOLOv5-based two-camera system that detects shuttlecock/racket/player/shoes and pinpoints the serve hit event to measure shuttle height vs the BWF 1.15 m rule; outperforms human line judges ~3.5x on borderline service heights."}, {"title": "Distributional bias compromises leave-one-out cross-validation", "authors": "Austin, Pe'er, Korem", "year": "2024/2025", "url_or_id": "arXiv:2406.01652; Science Advances doi:10.1126/sciadv.adx6976", "venue": "Science Advances", "what_it_does": "Shows leave-one-out CV induces a NEGATIVE correlation between each training fold's mean label and the held-out instance's label; since models regress to the training mean, this systematically biases performance estimates and hyperparameter selection (can push auROC below 0.5). Proposes Rebalanced LOOCV (RLOOCV): drop an opposite-label training sample per test instance; stratification mitigates it."}, {"title": "Direct methodological warning for our evaluation protocol. Our LOVO (leave-one-VIDEO-out) on ~6 matches is a coarse LOOCV at the group level; with a tiny LR and class-imbalanced rally/dead-time labels, distributional bias can make a cue look like it lifts or hurts when it doesn't. Not about our cues per se, but it governs whether our 'promote on measured lift' gate is trustworthy."}, {"title": "Established terms: 'rally vs break (play-break) segmentation ? the universal sport-agnostic scope (Tjondronegoro & Chen)', 'in-play / out-of-play (ball-in-play) state; rally state machine (serve -> in-rally -> out-of-rally)', 'rally segmentation / rally scene detection / rally detection', 'stroke / shot (the within-rally unit); 'hit event' / hitting-event detection; hit frame", "stroke_recognition": "stroke-type classification (forehand clear, smash, drive, ...)", "service_fault_detection": "service-height fault (BWF 1.15 m rule); let", "shuttlecock_tracking": "shuttlecock trajectory tracking (TrackNet / WASB / MonoTrack); trajectory rectification", "leave_one_match_out": "leave-one-match-out / group cross-validation; distributional bias; Rebalanced LOOCV (RLOOCV)", "temporal_IoU": "temporal IoU / tIoU; action spotting; temporal action segmentation"}, {"title": "Techniques to adopt", "techniques": "Adopt MonoTrack/Lin-style physics + game-cadence POST-PROCESSING as the formal model for our"}

reconcilers: encode 'hits ~1/sec, >=0.5s min separation, players alternate, trajectory in-bounds, initial velocity toward opponent' as the rule set that flags decision-vs-evidence contradictions.", "why": "It is published, validated (76.4%→89.7% hit accuracy) prior art for exactly the 'use persisted trajectory + domain physics to correct raw per-frame decisions' move our reconcilers make ? gives us a credible rule precedence to cite and reuse instead of inventing ad hoc.", "effort": "medium", "from_paper": "MonoTrack (arXiv:2204.01899); Lin et al. Sensors 2024 (PMC11244353)", {"technique": "Frame our per-frame shuttle-state head as a learned hit/contact classifier over short windows of court-normalized pose+shuttle (HitNet/SwingNet recipe), and consider fusing a per-frame player 'swing/no-swing' cue (YOLOv7-style) to disambiguate racket_contact, rather than a pure trajectory heuristic.", "why": "Two independent papers show shuttle-trajectory state alone is ambiguous at contact; fusing a player-swing cue materially improves hit detection (F1 ~0.90). This is a ready, evidence-backed feature column for our scorer.", "effort": "medium", "from_paper": "MonoTrack HitNet (arXiv:2204.01899); Lin et al. Sensors 2024", {"technique": "Add a Kalman/HMM temporal-smoothing pass over the per-frame rally posterior before cutting intervals, as a principled boundary-tightening reconciler.", "why": "SmartTennisTV gets F1 97.5% on rally/non-rally precisely by Kalman-smoothing noisy per-frame predictions into clean intervals; this is the standard, low-cost way to convert flickery cue columns into stable [start,end] without hand rules.", "effort": "low", "from_paper": "SmartTennisTV (arXiv:1801.01430)", {"technique": "Pull ShuttleSet / ShuttleSet22 + the CoachAI codebase as an external eval and weak-supervision corpus (3,685 rallies, stroke labels) to validate cues and pretrain the shuttle-state head beyond our 6 matches.", "why": "Our golden set is tiny and vendor-expensive; a public rally/stroke-annotated badminton corpus directly relieves the scarcity bottleneck and lets us report on a benchmark the field recognizes.", "effort": "medium", "from_paper": "ShuttleSet (KDD 2023, doi:10.1145/3580305.3599906); ShuttleSet22 (arXiv:2306.15664)", {"technique": "Adopt the explicit play-break / rally-break vocabulary and the multimodal-feature-set-then-fuse framing (and cite Tjondronegoro & Chen as the origin of 'play-break as universal scope'); position our learned-LR scorer as the modern fusion of that standard feature family.", "why": "It is the canonical lineage for what we are building; using the field's terms makes our work legible, citable, and avoids appearing to reinvent the rally/break unit.", "effort": "low", "from_paper": "Tjondronegoro, Chen & Pham, CVIU 2008 (S1077314208001355)", {"technique": "Switch our LOVO promotion gate to a bias-aware protocol: report stratified / rebalanced group-CV (RL00CV-style) results, and treat lift near the noise floor on 6 matches with explicit skepticism / confidence intervals.", "why": "Distributional bias makes plain LO0CV systematically mis-estimate small-data performance (can flip a true-null cue's sign); without correcting it, our 'promote on measured lift, default-OFF until proven' gate can promote noise or reject a real cue.", "effort": "medium", "from_paper": "Austin, Pe'er & Korem, Science Advances 2025 (arXiv:2406.01652)", {"technique": "Treat the held-shuttle/service-fault cue as a soft signal, not an officiating verdict, and ground it loosely in the BWF service-fault literature (shuttle pauses in the net region) instead of trying to measure true 1.15 m height.", "why": "Published service-fault systems need calibrated multi-camera rigs to adjudicate; framing ours as a cheap discriminator keeps it honest and within single-camera reach while still citing the established task.", "effort": "low", "from_paper": "Automated Service Height Fault Detection, Sensors 2023 (PMC10747833)", "pitfalls": ["Camera-cue non-transfer: the strongest classic rally/non-rally discriminator (SmartTennisTV's overhead-view appearance, Tjondronegoro's shot-view clustering) is a BROADCAST camera-switch that a static tripod never produces. Do not assume those reported F1s (97.5%) transfer; our static-camera setting is harder and must lean on shuttle/person/audio cues.", "Contact ambiguity at the trajectory turning point: both MonoTrack and Lin et al. find raw shuttle trajectory is ambiguous exactly at racket_contact; a trajectory-only in_air/contact/inactive head will be noisiest where it matters most. Mitigate with player-swing fusion and the cadence constraints, or contact precision will be poor.", "Over-constraining via hard rules: MonoTrack's gains come from domain CONSTRAINTS, but baking 'hits ~1/sec / alternating players' as hard rules would re-introduce the special-casing we explicitly forbid and would break on doubles, lets, and warm-up rallies that violate the cadence. Keep them as soft features/reconciler reports, A/B-gated, never silent rewrites.", "Distributional-bias / LO0CV leakage on ~6 matches: leave-one-VIDEO-out at this scale is exactly where Austin et al.'s negative train-mean/test-label correlation bites; with class-imbalanced rally vs dead-time and a tiny LR, a cue can show spurious lift or spurious harm. Group-CV alone does not fix it ? stratify/rebalance and report CIs.", "Warm-up/practice detection has thin direct prior art (only passing mentions of 'separate warm-up/practice from match'): little to reuse means higher risk of an unprincipled, brittle heuristic. Define it as a session-scope membership feature graded against labels, not a hard gate, and expect to validate it largely from scratch.", "Tennis/TT physics leakage: much in/out-of-play prior art relies on the BOUNCE (ball hits court) as a state cue; badminton has no bounce, so net-crossing/inactivity must carry that load. Don't port bounce-based FSMs verbatim.", "Evaluation-metric mismatch: the field commonly reports per-frame

F1 for rally/non-rally and event-spotting precision for hits, while we use temporal IoU on intervals. Pick and report the metric reviewers expect (per-frame F1 AND tIoU) so cue lift is comparable to prior work and not an artifact of metric choice.", "Small-detector domain gap: WASB/TrackNet/MonoTrack detectors are trained on broadcast/specific rigs; per-frame shuttle detection accuracy (MonoTrack ~88.6%) degrades off-distribution, and every downstream cue (net-crossings, state, contact) inherits that error. Quantify detector accuracy on our footage before trusting cue columns."], "verdict": "partial-overlap", "verdict_rationale": "Almost every individual cue has solid, citable prior art, so the building blocks are NOT novel: rally-vs-break as the universal segmentation unit (Tjondronegoro & Chen 2008), rally/non-rally interval classification with per-frame smoothing (SmartTennisTV 2018), shuttle-trajectory + hit/contact detection with physics/cadence post-processing (MonoTrack 2022; Lin et al. 2024; CoachAI hit-event work 2023), service-fault detection (Sensors 2023), in/out-of-play state machines and net-clearing/in-bounds reasoning (tennis rally-detection line), and the rally/stroke vocabulary + data (ShuttleSet 2023). We should cite and reuse these rather than claim invention. What is genuinely under-served ? and where our contribution is a novel COMBINATION rather than a gap ? is: (a) doing all this on uncut, static single-camera amateur footage where the dominant broadcast camera-cue is unavailable; (b) the strict 'every cue is a soft (value, confidence) feature column for a tiny learned scorer, never an if/else branch' discipline with detection persisted once and decision re-scored forever; (c) report-first, A/B-gated RECONCILERS as a productized 'linter' rather than one-shot constrained optimization; and (d) warm-up/practice-phase detection as a session-scope feature, which has only passing prior mentions. So: not reinventing wholesale, but heavily overlapping at the component level ? adopt the field's terminology, post-processing recipes, smoothing, external datasets, and bias-aware evaluation; reserve the novelty claim for the architecture/discipline and the static-camera + warm-up angles.", "search_queries_used": ["badminton rally segmentation shuttlecock trajectory TrackNet deep learning", "badminton stroke recognition shot classification video deep learning", "tennis ball in play vs out of play state classification rally detection video", "table tennis rally segmentation play break detection broadcast video", "sports video play break segmentation classification highlight detection structure", "badminton serve fault let detection automatic event detection rules", "shuttlecock hit event detection in-air contact trajectory state badminton MonoTrack", "warm-up practice detection sports video temporal segmentation match phase", "racket sports broadcast video rally break segmentation multimodal features classifier Chu Chen", "ShuttleSet badminton stroke dataset rally segmentation CoachAI benchmark", "tennis broadcast rally detection ball trajectory bounce net crossing rules in-play heuristic", "leave-one-match-out cross validation sports action recognition data leakage small dataset", "distributional bias leave-one-out cross-validation Rao 2024 mechanism recommend stratified"]], "verify": {"theme": "Specific cue novelty: structural rally discriminators, warm-up detection, shuttle-state / stroke detection", "citations_checked": [{"title": "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "verified_real": "yes", "note": "Real. arXiv:2204.01899, CVPRW 2022 (CVSports). Confirmed via openaccess.thecvf.com + IEEE Xplore. HitNet hit-classifier + constrained-optimization post-processing as described. NOTE: lead authors are Paul Liu & Jessica Hodgins (Stanford/CMU) ? the survey's 'Liu, Hodgins et al.' is accurate."}, {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "verified_real": "yes", "note": "Real. Sensors 24(13):4372, PMC11244353, doi 10.3390/s24134372. TrackNet + YOLOv7 swing/no-swing fusion, F1 90.5% over 69 BWF matches/1582 rallies ? matches survey. CAUTION: actual authors are Hsu, Yu, Cheng (per MDPI/search), NOT 'Lin et al.' as the survey states. Wrong author attribution, but paper is real and the technical claims check out."}, {"title": "A New Perspective for Shuttlecock Hitting Event Detection", "verified_real": "yes", "note": "Real. arXiv:2306.10293. Single author Yu-Hsi Chen (NTHU), presented at IT4PSS Workshop, IJCAI 2023 (CoachAI-adjacent). Survey's '(CoachAI Badminton Challenge participants)' author field is loose but the venue framing is fair. SwingNet + prior-feature extraction as described."}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "verified_real": "yes", "note": "Real. KDD 2023, doi:10.1145/3580305.3599906; arXiv:2306.04948. Authors Wang, Huang, Ik, Peng. Dataset scale (104 sets / 3685 rallies / 36492 strokes) matches. CAUTION: survey's url_or_id appended 'ShuttleSet22 arXiv:2306.15664' to this entry ? 2306.15664 is a DIFFERENT paper (ShuttleSet22 / 'Benchmarking Stroke Forecasting'), not the ShuttleSet KDD paper. The KDD paper is arXiv:2306.04948. Both papers real; the two arXiv IDs were conflated."}, {"title": "ShuttleSet22: Benchmarking Stroke Forecasting with Stroke-Level Badminton Dataset", "verified_real": "yes", "note": "Real. arXiv:2306.15664. Wei-Yao Wang et al. Title is 'Benchmarking Stroke Forecasting with Stroke-Level Badminton Dataset' (ShuttleSet22 is the dataset name). CoachAI Challenge @ IJCAI 2023 Track 2. Correctly a distinct artifact from the KDD ShuttleSet paper."}, {"title": "A framework for flexible summarization of racquet sports video using multiple modalities", "verified_real": "yes", "note": "Real. CVIU 2008, doi:10.1016/j.cviu.2008.07.001"}]}

(S1077314208001355). Tjondronegoro, Chen, Pham. Confirmed via ScienceDirect. Play/break segmentation via multimodal voting as described."},{\"title\":\"SmartTennisTV: Automatic indexing of tennis videos\",\"verified_real\":\"yes\",\"note\":\"Real. arXiv:1801.01430, NCVPRIPG 2017 (Best Paper). HOG/SVM rally vs non-rally + Kalman smoothing as described. CAUTION: actual authors are Anurag Ghosh & C.V. Jawahar ? the survey's 'Mohan, Pramod, Jawahar' is WRONG. Paper real, author attribution fabricated/incorrect.\"},{\"title\":\"Automatic rally detection on broadcast tennis videos\",\"verified_real\":\"yes\",\"note\":\"Real. Sports Technology 6(2), 2013, doi:10.1080/19346182.2013.819007. Histogram/spatiogram method; 81% avg, 91% hard court, 82% grass, 71% clay ? matches survey's '-81% (up to 91% on hard courts)'. Note: the survey's added FSM/serve->in-rally->out-of-rally narrative is a generic gloss, not necessarily from THIS specific paper.\"},{\"title\":\"Automated Service Height Fault Detection Using Computer Vision and Machine Learning for Badminton Matches\",\"verified_real\":\"yes\",\"note\":\"Real. Sensors 23(24):9759, PMC10747833. Authors Goh, Goh, Pan, Teng, Kong (NTU Singapore), BWF-supported. YOLOv5 two-camera service-height (1.15m) system, ~3.5x better than human judges ? matches survey.\"},{\"title\":\"Distributional bias compromises leave-one-out cross-validation\",\"verified_real\":\"yes\",\"note\":\"Real. arXiv:2406.01652; Science Advances doi:10.1126/sciadv.adx6976 (published Nov 2025). Austin, Pe'er, Korem. RL00CV / negative train-mean-vs-test-label correlation as described. Exact match.\"}],\"fabrication_flags\":[],\"corrected_verdict\":\"partial-overlap\",\"corrected_rationale\":\"Verdict upheld as partial-overlap ? but the survey slightly OVER-reserves novelty for its static-single-camera angle, and that reservation should be downgraded. All 10 cited papers are REAL (no fabrications), so the building-block-overlap argument stands: rally/break as the universal unit (Tjondronegoro 2008), per-frame rally classification with smoothing (SmartTennisTV 2018), shuttle-trajectory + hit detection with physics/cadence post-processing (MonoTrack 2022; Hsu/Yu/Cheng Sensors 2024; Chen 2023), service-fault detection (Goh Sensors 2023), tennis rally-detection state machines (2013), and the rally/stroke vocabulary+data (ShuttleSet 2023). The reasons to lower confidence in the novel-COMBINATION reservation: (1) Novelty claim (a) 'uncut static single-camera non-broadcast amateur footage where the broadcast camera-cue is unavailable' has DIRECT, uncited badminton prior art ? Yoshikawa et al.'s CHLAC+MRA service-scene detection runs on a single CEILING-mounted (non-broadcast) camera, and See et al. (ISACE 2024; SN Computer Science 2025) detect start-of-service and end-of-rally on non-broadcast multi-player footage using shuttle-flight gradient 'until the shuttle comes to rest' plus pose features over 12-frame windows ? i.e. exactly the static-camera + shuttle-inactivity-ends-rally combination the survey treats as under-served. (2) Novelty claim (b) the 'every cue is a soft (value,confidence) column for a learned scorer, never if/else' discipline restates well-established late-fusion logistic-regression-over-multimodal-confidence and sliding-window rally-posterior thresholding; it's an engineering-discipline framing, not a research novelty. What genuinely survives as under-served: warm-up/practice-phase detection (my targeted searches found only patents and passing drill-vs-game mentions ? no direct academic method), and the productized report-first/A-B-gated reconciler-as-linter packaging. So: not reinventing wholesale, but more component-level overlap than the survey credits ? keep cues cited/reused; reserve any novelty claim narrowly for warm-up detection and the engineering discipline, and DROP the 'static single camera is an under-served setting' framing.\"},{\"title\":\"Service and End of Rally Detection in Badminton Videos\",\"year\":\"2024\",\"url_or_id\":\"Springer LNCS, ISACE 2024, doi:10.1007/978-3-031-69073-0_15 (See, Hassan, Krishnasamy, Loo, Paramesran)\",\"why_closer\":\"Directly does start-of-service AND end-of-rally detection on non-broadcast badminton match videos (two clips, 35 rallies, 88.6% acc) using shuttle-flight GRADIENT until the shuttle 'comes to rest,' and explicitly handles missing/out-of-camera shuttle coordinates. This is almost exactly the survey's 'shuttle inactivity -> truncate/end-of-rally' reconciler cue on fixed-camera footage. Uncited; undercuts the static-single-camera + shuttle-inactivity novelty reservation.\"},{\"title\":\"A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos\",\"year\":\"2025\",\"url_or_id\":\"SN Computer Science 6:447, doi:10.1007/s42979-025-03935-0 (See, Paramesran, Hassan, Krishnasamy)\",\"why_closer\":\"Same group; BiLSTM/CNN over 12-frame windows of 26 pose-derived features (distance/angle/gradient/movement) for start-of-service, plus gradient computation for end-of-rally, on a 10-player / 4500-frame NON-broadcast dataset. Mirrors the survey's 'per-frame pose+shuttle feature columns -> small classifier -> rally segmentation' on amateur footage. Uncited.\"},{\"title\":\"Automated service scene detection for Badminton game analysis using CHLAC and MRA (Yoshikawa et al.)\",\"year\":\"~2010-2012\",\"url_or_id\":\"ResearchGate pub 290557755 (CHLAC + multiple regression analysis)\",\"why_closer\":\"Canonical precedent for rally/in-play (service-scene) detection from a SINGLE CEILING-MOUNTED (non-broadcast) camera ? the static-camera badminton setting the survey claims is under-served. Establishes that fixed-single-camera rally segmentation in badminton is long-standing prior art, not a gap. Uncited.\"}],\"extra_course_corrections\":[\"FIX AUTHOR ATTRIBUTIONS before citing: SmartTennisTV is Ghosh & Jawahar (NOT 'Mohan, Pramod, Jawahar'); the

Sensors-2024 fusion paper is Hsu/Yu/Cheng (NOT 'Lin et al.'). Citing these with the survey's wrong author names will read as sloppy or fabricated to a reviewer.", "FIX the ShuttleSet citation: the KDD paper is arXiv:2306.04948 (doi:10.1145/3580305.3599906); arXiv:2306.15664 is the SEPARATE 'ShuttleSet22 / Benchmarking Stroke Forecasting' paper. The survey conflated the two IDs in one entry ? split them.", "ADOPT the See et al. (2024/2025) line as the nearest badminton baseline to compare against and cite ? it is the closest published method to your rally-segmentation core (start-of-service + shuttle-rest end-of-rally on non-broadcast footage) and should be your head-to-head reference, not just MonoTrack/ShuttleSet which target shots within known rallies.", "AVOID claiming 'static single-camera / non-broadcast footage' as the novel setting ? Yoshikawa (ceiling cam) and See et al. (multi-player non-broadcast) already work there. Reframe the static-camera point as an engineering/robustness contribution (your specific WASB-track + audio + person fusion on it), not a setting novelty.", "KEEP the warm-up/practice-phase claim as the strongest novelty reservation ? targeted searches (warm-up vs match, drill vs game, knock-up exclusion) returned only patents and coaching content, no direct academic method. But validate from scratch and frame as a session-scope membership feature graded against labels, as the survey already warns.", "NARROW the soft-feature-column novelty claim: late-fusion logistic regression over multimodal confidence scores and sliding-window rally-confidence thresholding are documented, established patterns. Your defensible novelty is the strict persist-detection-once / re-score-forever + report-first A/B-gated reconciler PRODUCT discipline, not the LR-fusion mechanism itself."]]]]

Write a thorough Markdown memo with these sections:

Rally-detection framework ? prior-art & course-correction memo

1. Executive verdict

Are we reinventing the wheel? Give a crisp overall answer with nuance ? which parts are standard (just cite & reuse), which are a genuinely novel *combination*, and which (if any) are close to a real gap. One paragraph + a one-line TL;DR.

2. Per-pattern scorecard

A table: Pattern | Field's name for it | Closest prior art (real cite) | Verdict | One-line. Use the corrected_verdict from verification. Only include citations marked verified-real; note any fabrication_flags explicitly as "unverified ? do not cite."

3. Terminology we should adopt

Map OUR terms ? the field's established terms (e.g. is our "reconciler" just "refinement/post-processing"? "session analyzer" = "video-level temporal context / phase segmentation"? "signal not verdict" = "late feature fusion / score-level fusion"? "persist once re-score" = "feature caching / decoupled proposal-classification"?). So our doc and the paper speak the field's language.

4. What is defensibly novel (what to claim)

The honest, narrow novelty ? most likely the *specific combination* + the report-first auditable reconciliation layer + the domain application. Say what we can and CANNOT claim.

5. COURSE CORRECTIONS (the actionable core ? prioritize this)

A prioritized, concrete list. For each: WHAT to change in our design, WHY (cite the paper/technique), the EXPECTED benefit, rough EFFORT (low/med/high), and WHICH part of our design/doc it touches (multi-scope producers, the scorer, detection?decision, the reconciler, or the harness/eval). Pull from techniques_to_adapt + pitfalls across all themes. Call out any pitfalls we are currently walking into (e.g. late-fusion calibration, over-segmentation, LOVO leakage, metric choices the field expects like mAP@tIoU / edit-score/F1@k for segmentation).

6. Risks, residual gaps & unverified claims

What we could NOT confirm; research areas not yet searched; any fabrication_flags; where the verdict is "uncertain".

7. Recommended citation list

Bullet the VERIFIED-real papers worth citing, grouped by section, each with title + year + id/url.

Be specific and honest. Distinguish "verified real citation" from "uncertain". Do not invent citations. If the evidence is thin for a claim, say so plainly.

2. assistant (2026-06-13T20:25:43.943Z)

I'll write this memo. The findings JSON is comprehensive and already verified, so I have what I need. Let me quickly check the project's doc conventions so the terminology I recommend lines up with what's actually in the codebase, then write the memo as my response.

3. user (2026-06-13T20:25:45.727Z)

Found 9 files
docs\README.md
docs\ANALYZERS_AND_RECONCILERS.md
docs\QUALITY_ITERATIONS.md
docs\NEXT_STEPS.md
docs\MULTI_SIGNAL_FUSION_PLAN.md
docs\CODE_MAP.md
docs\EXPERIMENT_HARNESS.md
docs\archives\decisions\DECISIONS.md
docs\ROADMAP.md

4. assistant (2026-06-13T20:25:49.711Z)

There's already a `docs/ANALYZERS\_AND\_RECONCILERS.md` ? let me check how it frames the terminology so my mapping is precise to the existing vocabulary.

5. user (2026-06-13T20:25:50.523Z)

```
1      # Analyzers & reconcilers ? two extensibility seams for rally-detection quality
2
3      **Status:** **DESIGN ? owner-gated, NOT started.** A peer to
4      [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md) and
5      [MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md).
6      This doc specs two *seams* ? a forward one (injectable window analyzers) and a backward
7      one (a reconciliation pass) ? that let us keep piling on clever rally signals **without the
8      pipeline becoming a
9      pile of special cases.** Docs-only; no implementation code. Each phase below is a
10     separate, owner-gated PR.
11
12     > **Companion docs:** the fusion seam these plug into is
13     [MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md)
14     > (`RallyCue` = swappable producer; the `_enrich_candidate_stats` hook; the
15     scale/translation-invariant
16     > feature discipline, §3.2); the no-regression A/B + LOVO measurement these ride is
17     > [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md) (`Variant`, `compare`,
18     `compare_unlabeled`); the scorer
19     > is the distilled logistic regression (**ADR-004**, `backend/eval/classifier.py`); the
20     modular rally
21     > layer is **ADR-007**
22     ([archives/decisions/DECISIONS.md](archives/decisions/DECISIONS.md)); the 2-layer
23     > mental model is [HOW_RALLY_DETECTION_WORKS.md](HOW_RALLY_DETECTION_WORKS.md); the live
24     quality log is
25     > [QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md).
```

```

17
18 ---
19
20 ## 1. The spine ? NO special-casing in the decision path
21
22 This is the one principle everything else serves:
23
24 > **Every analyzer and reconciler emits a SIGNAL carrying a value + a
confidence/presence. The SCORING
25 > MODEL consumes those signals and *learns to weight them*. We do NOT add hand-coded
`if/else` branches
26 > into the decision.**
27
28 **Analyzers propose; the scorer arbitrates; the experiment harness + golden set decide
what gets
29 promoted.** That separation is what lets us add a net-hit service fault, a let-cord, a
double-bounce, a
30 shuttle-out cue ? each as "register a producer + a feature column,"* never "add a
branch."* A signal
31 that doesn't earn its keep stays at weight ? 0; it costs nothing because it is an unused
column, not a
32 live `if`.
33
34 The pattern already exists in miniature: `experiment.predict()`
(`backend/eval/experiment.py:203`) treats
35 `net_crossings` and `players_in_court` as **feature columns** ? `_gate_mask` is the
trivial scorer
36 (threshold a column), `LogisticRegression` (ADR-004) the richer learned scorer (weight
all columns), a
37 future scorer richer still. The decision path is *signals ? scorer*, with **zero** `if
service_fault:
38 drop`. Analyzers feed it more columns; reconcilers run *outside* it as an audited post-
pass.
39
40 ```
41
42          ???????????? DETECTION (expensive, GPU/WSL, run ONCE) ????????????
43 frames ?? WASB (frozen, MIT) ?? shuttle (x,y,v,vis)/frame ??
44 frames ?? PersonDetector (ours)?? boxes + tracks/frame ?????? per-frame-aligned
45 audio ?? hit detector (ours) ?? hit events ?????????????????? signals (+ net
estimate,
46          ?????????????????????????????????????????????????????????????? window bounds)
47          ?
48          shuttle-seeded candidate windows (FusionSegmenter, P2)
49          ?
50          ???????????? ANALYZERS ? forward seam, run PER WINDOW (flag-gated, default OFF)
????????????
51          ? WindowAnalyzer registry: service-fault . double-bounce . let-cord . shuttle-out .
? ?
52          ? each ? AnalyzerResult{ signals(value+confidence), events, optional boundary hints
} ?
53          ?????????????????????????????????????????????????????????????????????????????????????????
54          ? signals merged into candidate_segments.stats
55          ?
56          DECISION / SCORING ? signals ? learned scorer (NO branches)
57          _gate_mask (trivial) + LogisticRegression (ADR-004) + harness-pinned model
58          ? predicted rally segments
59          ?
60          ???????????? RECONCILERS ? backward seam, "linter for rally decisions"
????????????????????
61          ? ConsistencyRule registry (fixed precedence, idempotent): decision-vs-evidence
conflicts ?

```


I have enough to ground the terminology mapping in the actual codebase vocabulary (`RallyCue`, `WindowAnalyzer`, `ConsistencyRule`, `\_enrich\_candidate\_stats`, ADR-004 logistic regression, `experiment.predict()`). I have everything needed to write the memo.

Rally-detection framework ? prior-art & course-correction memo

1. Executive verdict

We are not reinventing the wheel wholesale ? but we are re-deriving, under bespoke names, a large amount of well-charted machinery, and we are currently missing several cheap, standard safeguards that the literature treats as table stakes. Three of our five patterns are **standard and should simply be cited and reused**: the "candidate-windows + learned scorer" skeleton is textbook proposal-then-score **Temporal Action Localization** (BSN/BMN/S-CNN); the "refinement stage that fixes over-segmentation" is the defining theme of **Temporal Action Segmentation** (MS-TCN, ASRF) and **model-agnostic post-processing** (GAP, Activity Grammars); and "signal not verdict / learned arbiter over cue columns" is **late/score-level fusion** (Snoek 2005) realized as **stacked generalization** (Wolpert 1992), a pattern that already exists **in our exact domain** (racquet-sports rally/break fusion, CVIU 2009; badminton point-segmentation, Ghosh et al. 2018). The fusion mechanism itself is ~20 years old in tennis ? do not present it as new. What survives as a genuinely novel **combination** (not a research gap) is the **engineering envelope**: detection?decision as a persisted "compute-once, re-score-forever" substrate paired with a **report-first, A/B-lift-gated, idempotent, human-in-the-loop reconciler-as-linter** keyed on decision-vs-evidence **contradiction**, plus the flag-gated/default-OFF no-regression promotion harness, all on a deliberately tiny (~6-match) LOVO-by-match golden set with a deliberately interpretable logistic scorer. Of the five themes, only the **reconciler-as-linter (output-linting)** earned a **novel-combination** verdict on verification; the other four are **partial-overlap**. The two thinnest residual slices ? **warm-up/practice-phase detection as a soft session-scope feature**, and the **measured-before-apply auditable reconciliation discipline** ? are where any novelty claim should be narrowly parked.

> **TL;DR:** The architecture's **skeleton** is standard TAL/TAS + late fusion (cite, don't claim); the defensible contribution is the **governance/discipline envelope** (persist-once re-score, report-first measured-before-apply reconciler, no-regression harness) applied to a scarce-label static-camera racket-sports setting ? and we are walking into known pitfalls (over-segmentation hidden by tIoU, late-fusion miscalibration, LOOCV distributional bias) that have cheap, citable fixes.

2. Per-pattern scorecard

Pattern (our framing)	Field's name	Closest verified prior art	Verdict	One-line
Candidate-windows + learned scorer + reconciler-as-refinement	Temporal Action Localization / Segmentation + boundary/post-proc refinement	ASRF (Ishikawa et al., WACV 2021, arXiv:2007.06866); Activity Grammars (Gong et al., NeurIPS 2023, arXiv:2312.04266); BMN (Lin et al., ICCV 2019, arXiv:1907.09702)	partial-overlap	Our skeleton is textbook TAL/TAS; the reconciler is a hand-built activity grammar ? adopt the field's metrics & global-parse framing.
"Signal not verdict": every cue a feature column ? learned arbiter	Late/score-level fusion + stacked generalization	Snoek et al. (ACM MM 2005, 10.1145/1101149.1101236); Wolpert (Neural Networks 1992); racquet-sports A/V rally-break fusion (CVIU 2009, S1077314208001355)	partial-overlap	Fusion mechanism is established (and in-domain ~20 yrs); novelty is the engineering envelope, not the fusion.
Detection?decision "persist once, re-score forever"	Frozen-backbone cached features + decoupled proposal/classification	S-CNN (Shou et al., CVPR 2016, arXiv:1601.02129); CBR (Gao et al., BMVC 2017, arXiv:1705.01180); DCR (Cheng/Shi et al., ECCV 2018, arXiv:1810.04002); Decouple-SSAD (arXiv:1904.07442)	partial-overlap	"Compute-once, re-score-many" is the default of the TAD field ? cite as standard practice, claim only the persist-once/idempotent discipline.
Report-first reconciler (decision-vs-evidence contradiction ? auditable corrections, measured before apply)	Model assertions / consistency assertions + constraint-based data repair ("ML linter")	Model Assertions (Kang et al., MLSys 2020, arXiv:2003.01668); HoloClean (Rekatsinas et al., VLDB 2017, arXiv:1702.00820); TFDV (Breck/Polyzotis et al., MLSys 2019); EventAnchor (CHI 2021, arXiv:2101.04954)	novel-combination	Ingredients all exist; the measured-

before-apply + auditable rule registry + signal-not-verdict* bundle for temporal intervals is not matched. |

| Scarce interval labels + tiny LR + leave-one-VIDEO-out (tIoU) | Point/timestamp-supervised TAL + grouped (LOSO/LOGO) CV | SF-Net (Ma et al., ECCV 2020, arXiv:2003.06845); "Distributional bias compromises L00CV" (Austin/Pe'er/Korem, Science Advances 2025, arXiv:2406.01652); Snorkel (Ratner et al., VLDB 2017, arXiv:1711.10160) | **partial-overlap** | Each component is named prior art; the urgent corrections are *evaluation-side* (RL00CV, per-fold CIs), not architectural. |

| Multi-scope (frame/window/session) hierarchy; warm-up conditions per-window | Multi-scale/hierarchical temporal modeling + play-break + phase recognition | Tjondronegoro et al. (ICME 2003 play-break); GL-MSTCN (Fang et al., 2022, 10.1186/s12938-022-01048-w); MS-TCN (Abu Farha & Gall, CVPR 2019, arXiv:1903.01945) | **partial-overlap** | Coarse-state-conditions-fine is decades old; warm-up-as-soft-feature is the only thin genuine gap. |

Cue-level prior art (specific signals): net-crossing / in-play state machines (tennis rally detection, Sports Technology 2013, 10.1080/19346182.2013.819007); shuttle-state / hit detection (MonoTrack, Liu & Hodgins, CVPRW 2022, arXiv:2204.01899; Hsu/Yu/Cheng, Sensors 2024, 24(13):4372); service-fault (Goh et al., Sensors 2023, 23(24):9759); rally start/end on **non-broadcast** badminton (See et al., ISACE 2024, 10.1007/978-3-031-69073-0_15; SN Computer Science 2025, 10.1007/s42979-025-03935-0); rally/stroke dataset (ShuttleSet, KDD 2023, arXiv:2306.04948). **All verified real.**

Unverified ? do not cite as written: The corpus had **zero fabricated papers** (every key paper resolved). But three citations carry **wrong author lines** that must be fixed before use (the papers are real; the attributions are not):

- CVIU 2009 racquet-sports summarization (S1077314208001355) ? actual authors **Liu, Huang, Jiang, Xing, Ye, Gao**, **not** "Chen/Kolekar." Delete every "Kolekar-line" framing.
- SmartTennisTV (arXiv:1801.01430) ? actual authors **Ghosh & Jawahar**, **not** "Mohan, Pramod, Jawahar."
- Sensors 2024 shuttle-fusion (24(13):4372) ? actual authors **Hsu, Yu, Cheng**, **not** "Lin et al."

Also fix metadata slips: the L00CV-bias paper is in **Science Advances** (10.1126/sciadv.adx6976), not "Briefings in Bioinformatics"; SF-Net evaluates on **THUMOS14/GTEA/BE0ID**, not "ActivityNet"; ShuttleSet (KDD) is **arXiv:2306.04948**, while **arXiv:2306.15664** is the separate ShuttleSet22/stroke-forecasting paper (the two were conflated); the DeeQu entry merges the SIGMOD'19 demo with the PVLDB'18 paper.

3. Terminology we should adopt

Our term (and where it lives in code/docs)	Field's established term	Citation anchor
"Producer" / `RallyCue` (MULTI_SIGNAL_FUSION_PLAN.md)	Labeling function / weak-supervision source; base model in a stack	Snorkel (arXiv:1711.10160); Wolpert 1992
"Signal not verdict / no special-casing" (ANALYZERS_AND_RECONCILERS.md §1)	Late / score-level fusion ; arbiter = level-1 combiner / meta-learner (stacked generalization)	Snoek et al. 2005; Wolpert 1992
The LR scorer (`backend/eval/classifier.py`, ADR-004)	Stacked meta-learner / level-1 combiner over producer outputs	Wolpert 1992
"Detection ? decision / persist once, re-score forever"	Frozen-backbone precomputed (cached) features ; decoupled proposal/classification ; offline/off-policy re-scoring	S-CNN (arXiv:1601.02129); Decouple-SSAD (arXiv:1904.07442); DOPE (arXiv:2103.16596)
`WindowAnalyzer` (window-scope)	Temporal action proposal features; per-proposal actionness	S-CNN; BMN (arXiv:1907.09702)
Session-scope analyzer / warm-up detection	Video-level temporal context / phase segmentation ; play-break outer layer; global-local modeling	Tjondronegoro ICME 2003; GL-MSTCN 2022
"Reconciler" (`ConsistencyRule` registry)	Model assertions / consistency assertions ; constraint-based data repair ; boundary refinement (when it tightens boundaries); activity grammar / constrained decoding (when it enforces structure)	Kang et al. MLSys 2020; HoloClean (arXiv:1702.00820); ASRF (arXiv:2007.06866); Activity Grammars (arXiv:2312.04266)
"Report-first linter"	Data validation / schema anomalies (detect-and-report, don't silently mutate); "unit tests for data"	TFDV (Breck/Polyzotis et al., MLSys 2019); DeeQu

"Over-merged / split a segment"	****Over-segmentation error****	MS-TCN; ASRF
"Tighten boundary slack to serve-contact"	****Action boundary regression / boundary refinement****	ASRF; CBR
"Leave-one-video-out (LOVO), group by match"	****Leave-one-group-out / leave-one-subject-out (LOGO/LOSO)**** grouped CV	scikit-learn `GroupKFold` / `LeaveOneGroupOut`; Austin et al. 2025
Temporal-IoU metric	****mAP@tIoU**** (TAL) + ****segmental F1@{10,25,50}** & edit score** (TAS) + ****tight average-mAP / mAP@?*** (action spotting)	Lea et al. 2017; SoccerNet (arXiv:1804.04527); Soares et al. (arXiv:2206.07846)

****Concrete action:**** rename in docs/code comments so reviewers map instantly to 30 years of literature. This is a **low-effort, high-leverage framing change** and signals we know the failure modes. Keep the internal nouns (`RallyCue`, `WindowAnalyzer`, `ConsistencyRule`) as the implementation names but gloss each, on first use, with its field term.

4. What is defensibly novel (what to claim)

****CAN claim (narrow, honest):****

- **The engineering/governance envelope as an integrated whole**** ? late/score-level stacked fusion **operationalized** as a no-regression, flag-gated, grouped-CV cue-promotion pipeline with a persist-once/re-score split and a TFDV-style report-first reconciler. No single found paper bundles all of: (a) report-first detection-vs-evidence **contradiction** detection over a persisted temporal decision, (b) auditable structured interval corrections (truncate/split/drop/tighten), (c) ****per-rule A/B lift-gating on golden intervals *before* auto-apply****, (d) idempotency + fixed precedence + human-in-the-loop, (e) feeding a learned (value,confidence) cue scorer. That **combination** is the contribution.
- **The "signal not verdict" discipline as a deliberate contrast**** to the in-domain prior art (See et al. 2024/2025, Hsu et al. 2024, MonoTrack's constrained optimization, the tennis FSMs) that all ****hard-code**** these same sound cues as if/else verdicts. This is a real, defensible **design-philosophy delta** ? but state it as a philosophy delta, not a capability the prior work lacks.
- **Warm-up/practice-phase detection as a soft session-scope membership feature**** ? the single slice with genuinely thin prior art (targeted searches surfaced only patents and coaching content; nothing academic does warm-up-as-soft-conditioning-feature). Claim it cautiously, graded against labels, expect to validate from scratch.

****CANNOT claim (would overstate):****

- ? The ****fusion mechanism**** ("multi-cue features ? light learned scorer for rally segmentation") ? ~20 years old in racquet sports (CVIU 2009; Ghosh et al. 2018 learn an SVM over per-frame HOG ? smoothed rally segments; the closest analog to our decision layer).
- ? The ****"persist once, re-score forever" substrate**** as novel ? it is the **default operating mode** of the entire TAD field (THUMOS/ActivityNet ship precomputed I3D/TSN/VideoMAE features precisely so cheap heads re-train on cached features; OpenTAD/TallFormer codify Stage-0-encode/Stage-1-head). Cite as standard practice.
- ? The ****cross-check-and-correct insight**** of the reconciler ? Hsu et al. (2024) already cross-check shuttle-vs-swing streams and correct contradictions in badminton; EventAnchor (CHI 2021) already does human-in-the-loop suggest-then-verify on racket-sports event anchors against CV evidence. Our delta is **measured-before-apply + auditability**, not the cross-check or the human-in-the-loop.
- ? ****"Static single-camera / non-broadcast footage" as a novel setting**** ? Yoshikawa et al. (ceiling-cam service-scene detection) and See et al. (non-broadcast multi-player) already work there. Reframe as a robustness/engineering contribution of **our specific WASB-track + audio + person fusion on it**, not a setting novelty.
- ? The ****no-regression / flag-gated promotion harness**** as a primitive ? textbook MLOps (shadow testing, feature-flag-gated rollout). Claim only "this discipline applied to a deliberately tiny golden set."

5. COURSE CORRECTIONS (the actionable core)

Prioritized. Each: WHAT / WHY (+cite) / BENEFIT / EFFORT / TOUCHES.

Tier 1 ? Do first: these threaten the **honesty of our measured lift on ~6 matches**

****C1. Switch the LOVO gate to a bias-aware protocol (RL00CV / class-balance-preserving folds; report per-fold variance + CIs, never a single mean).****

- WHY: With ~6 match-groups, a regularized LR, and imbalanced rally/dead-time labels, standard leave-one-out induces a **negative correlation** between each fold's training-label mean and its held-out label ? models regress to the train mean, systematically biasing the estimate (a LR scored ~0.23 auROC on **random** data in the cited study). Our "promote only on measured lift" gate can promote noise or reject a real cue. (Austin/Pe'er/Korem, Science Advances 2025, arXiv:2406.01652.)
- BENEFIT: Trustworthy promotion decisions; lift claims survive review.
- EFFORT: low?med. TOUCHES: ****harness/eval****.

****C2. Calibrate each producer's confidence (Platt for binary cues, isotonic for monotone ones) *before* it enters the LR.****

- WHY: A CNN softmax, an audio onset score, and an integer net-crossing count live on wildly different scales; uncalibrated, the most over-confident cue dominates the arbiter and starves complementary cues (our audio cue's known low discrimination is a live example). (MoCaE, Oksuz et al., TMLR 2023, arXiv:2309.14976; modality competition, ICML 2022, arXiv:2203.12221.)
- BENEFIT: Stable, meaningful weights; the "(value,confidence)" semantics become real rather than illusory.
- EFFORT: med. TOUCHES: ****multi-scope producers + scorer****.

****C3. Generate the arbiter's training features with OUT-OF-FOLD producer outputs (honest stacking).****

- WHY: When a producer is itself learned (the proposed shuttle-state classifier), feeding its in-sample outputs to the LR inflates measured lift. Wolpert's stacking result; extend the group-by-match split to producer training. (Wolpert 1992.)
- BENEFIT: Removes a leakage path that compounds with the tiny corpus.
- EFFORT: med. TOUCHES: ****detection?decision + harness****.

****C4. Report segmental F1@{10,25,50} + segmental edit score (and a tight/loose mAP@? sweep) alongside temporal-IoU.****

- WHY: tIoU is nearly blind to over-segmentation ? a rally chopped into 3 pieces, or two rallies merged across dead-time, can still score moderate tIoU. F1@k/edit **directly** quantify the fragmentation/boundary errors our reconciler's split/merge/tighten rules exist to fix; without them a reconciler win shows no IoU lift and looks like a no-op (or an IoU-chasing rule shreds structure invisibly). Tight average-mAP (SoccerNet-style) is the metric designed for boundary precision ? directly grades the serve-contact tightening. (MS-TCN arXiv:1903.01945; ASRF arXiv:2007.06866; SoccerNet arXiv:1804.04527; Soares et al. arXiv:2206.07846.)
- BENEFIT: Lift claims become legible/comparable to the field; rewards fixing fragmentation, not just overlap.
- EFFORT: low. TOUCHES: ****harness/eval + reconciler****.

Tier 2 ? Architecture/discipline upgrades

****C5. Frame reconciler rules as an explicit **activity grammar / constraint-aware decode** and resolve them GLOBALLY (Viterbi-style), not as stacked independent local edits.****

- WHY: Stacked local truncate/drop/split/tighten rules with fixed precedence can conflict, oscillate, or be non-idempotent across orderings ? the exact reason the field moved to a single global parse. Activity Grammars (arXiv:2312.04266) and especially ****Constraint-Aware Decoding**** (arXiv:2605.10149, ICPR 2026 ? the **nearest** published analog to our reconciler) inject **data-derived** structural priors (transition confidence, permitted boundary sets, per-class duration) as inference-time post-processing on a frozen base and resolve via modified Viterbi. This is **our own recommended end-state already published**, and modeling priors statistically from data is a concrete template to replace hand-tuned thresholds.
- BENEFIT: Principled, citable formalism; kills the rule-conflict/precedence headache; mitigates threshold-tuning-on-test.
- EFFORT: med. TOUCHES: ****reconciler****.

****C6. Brand the reconciler as "model assertions / consistency assertions over the rally decision"; attach a per-correction confidence and route only high-confidence corrections to auto-apply, low-confidence to a human queue ranked by margin.****

- WHY: Gives a defensible lineage (Kang et al., MLSys 2020, arXiv:2003.01668 ? **and** positions us as **extending** it: they stop at flag/weak-label, we add structured correction + lift-gated apply). HoloClean (arXiv:1702.00820, ~90% repair precision ? ~10% bad repairs) and confident learning (arXiv:1911.00068) show per-repair confidence + review-most-contradicted-first is the principled way to mix auto-repair with human-in-the-loop. Treat the rule set as a TFDV-style versioned, owner-ratified schema of named anomaly types.

- BENEFIT: Auditable, non-rotting rule registry; safer auto-apply; established vocabulary.
 - EFFORT: med. TOUCHES: ****reconciler + harness****.
- **C7. Add a handful of EARLY-fusion interaction features (still "just columns," no branches) ? e.g. `shuttle\_in\_air AND person\_two\_sided` within a window.****
- WHY: A pure-late *linear* arbiter cannot represent AND/XOR cue interactions; Snoek et al. show late usually ties early but where interactions matter early wins by a lot. Hand-crossed product columns recover this cheaply on the LR. (arXiv via 10.1145/1101149.1101236.)
 - BENEFIT: Captures cross-cue structure without leaving the "signal not verdict" regime.
 - EFFORT: low. TOUCHES: ****multi-scope producers + scorer****.
- **C8. Represent a missing/low-confidence producer output as an explicit `is\_present`/uncertainty companion column, not a silent 0.****
- WHY: Collapsing "cue absent" and "cue says no" into the same 0 confuses a linear arbiter; DBF treats per-detector uncertainty as a first-class {target, non-target, unknown} state. (DBF, WACV 2016 / arXiv:1511.03183; extended arXiv:2204.02890.)
 - BENEFIT: The model learns to discount absent cues.
 - EFFORT: low. TOUCHES: ****multi-scope producers + scorer****.
- **C9. Keep the warm-up/session-scope state a SOFT membership feature column, never a hard pre-filter; add a court/context-invariance probe.****
- WHY: A wrong coarse call in a hard gate is unrecoverable downstream (it suppresses real rallies in a mislabeled span) ? the classic hierarchical-gating risk. And ****action-context confusion**** (arXiv:2103.16155) is the documented trap: the static court co-occurs with rallies, so a scorer can learn "court visible" instead of "point in play." Probe by evaluating on warm-up stretches that share the court but lack point cadence; encode the session state as a feature the LR weights (GL-MSTCN's global-local fusion + ablation discipline).
 - BENEFIT: Recoverable coarse errors; guards against the background-shortcut failure; keeps "signal not verdict."
 - EFFORT: low?med. TOUCHES: ****multi-scope producers (session-scope) + scorer****.

Tier 3 ? Leverage external resources & higher-effort gains

- **C10. Model cue CORRELATIONS in the scorer (Snorkel/MeTaL label model as a baseline; strong L2/L1, far fewer effective features than groups).****
- WHY: Net-crossing, two-sidedness, and shuttle-in-air co-fire on the same rallies; an unregularized LR double-counts the bundle and miscalibrates. Snorkel's lesson: model source correlations or your "independent" evidence isn't. (arXiv:1711.10160.) EFFORT: high. TOUCHES: ****scorer****.
- **C11. Add a temporal-smoothing pass (Kalman/HMM/Viterbi over the per-window rally posterior, or an MS-TCN-style anti-fragmentation penalty) before cutting intervals.****
- WHY: SmartTennisTV reaches F1 ~97.5% on rally/non-rally largely by Kalman-smoothing noisy per-frame predictions into clean intervals; MS-TCN's truncated-MSE smoothing loss is the canonical over-segmentation cure. Our LR optimizes per-window labels and is blind to cross-window fragmentation. (arXiv:1801.01430; arXiv:1903.01945.) EFFORT: low?med. TOUCHES: ****reconciler / scorer****.
- **C12. Pull ShuttleSet/ShuttleSet22 + CoachAI codebase as external eval + weak-supervision/pretraining corpus; mine a cheap weak signal (point/dead-time cadence, warm-up labels) to augment the LR.****
- WHY: Directly attacks the ~6-match bottleneck; "Less is More" (arXiv:1903.00859) shows a noisy free signal + explicit noise handling can match supervised SOTA; ShuttleSet (arXiv:2306.04948) is 3,685 rallies the field recognizes. EFFORT: med?high. TOUCHES: ****harness/eval + producers****.
- **C13. Benchmark the rule-based reconciler against a learned boundary refiner (ASRF/CBR) and ground reconciler rules in MonoTrack-style physics/cadence constraints (as soft features, not hard rules).****
- WHY: Gives a learned baseline to *justify staying rule-based*, and MonoTrack's constrained post-processing (76.4%~89.7% hit accuracy) is validated prior art for "use persisted trajectory + domain physics to correct raw decisions" ? but baking "hits ~1/sec / alternating players" as *hard* rules would re-introduce the special-casing we forbid and break on doubles/lets/warm-up. (ASRF; CBR arXiv:1705.01180; MonoTrack arXiv:2204.01899.) EFFORT: high. TOUCHES: ****reconciler****.

Pitfalls we are actively walking into (flagged for the risk register)

- ****Over-segmentation hidden by tIoU**** (C4) ? THE dominant error mode of cheap segment decisions; long static-camera rallies split at occlusions, short dead-time gaps merged. A scorer optimized for per-window accuracy looks good while producing fragmented rallies.
- ****Late-fusion miscalibration / dominant-cue swamping**** (C2) ? the over-confident shuttle CNN softmax or low-discrimination audio cue can dominate the LR.
- ****LOOCV distributional bias + tiny-N variance**** (C1) ? promotion gate promotes noise without CIs.
- ****Stacking leakage**** (C3) and ****threshold-tuning-on-test**** (keep **all** tunable knobs ? windowing, merge/NMS thresholds, scorer regularization, reconciler thresholds ? **inside** the LOVO loop, never on the full golden set).
- ****Proposal-recall ceiling**** ? the decision stage can never recover a rally the windowing missed; measure window/proposal recall (AR@AN) separately or you'll misattribute recall losses to the scorer.
- ****Report-first decaying into silent rewrite**** ? without per-rule lift gating AND a confidence gate AND idempotency tests, an over-eager rule systematically truncates correct long rallies.
- ****Static-tripod transfer gap**** ? broadcast cues (overhead camera-switch, replays, crowd/commentator audio) that drive SmartTennisTV/Tjondronegoro/excitement-ranking **do not exist** on our footage; do not assume their F1s transfer or import excitement features.

6. Risks, residual gaps & unverified claims

- ****Fabrications: none.**** Across all five themes and ~40 citation checks, every key paper resolved to a real work with correct IDs. The defects are (a) the ****three wrong author attributions**** and several metadata slips listed in §2 ? **real papers, bad bylines/venues** ? which must be corrected before any citation, and (b) two auth-gated Springer badminton DOIs whose **full text** could not be read, though their specific quantitative claims (See et al.: shuttle-rest end-of-rally, ~88.6%/35 rallies; SN-CS 2025: BiLSTM service + ?80%-no-motion end, 187 rallies) check out from snippets.
- ****Two recent (2026) analogs verified to **exist** but not deep-read:**** Constraint-Aware Decoding (arXiv:2605.10149) and TennisExpert (arXiv:2603.13397). Both are concurrent (don't void novelty) but are the nearest published precedents for, respectively, the reconciler end-state and the audio-hits?cluster?filter rally pipeline; we should read both before finalizing the related-work section.
- ****Thin-evidence claims (state plainly):**** **Warm-up/practice-phase detection as a soft feature** ? searches found only patents and coaching content, no academic method; this is our strongest novelty reservation ****and**** our highest from-scratch-validation risk. The **measured-before-apply auditable reconciler bundle** is novel by absence-of-counterexample, not by a positive gap ? a weaker form of evidence.
- ****Verdict where uncertain:**** the ****detection?decision substrate**** sits closest to flipping toward "reinventing" ? frozen-feature re-scoring is field-default, so **only** the persist-once/idempotent/versioned discipline is ours. Treat any "novel substrate" language as overclaim.
- ****Not yet searched / open research areas:**** (1) whether a recent badminton **full-match rally/interruption frame-level dataset** (BWF World Tour annotations surfaced in 2025 papers) exists that would relieve scarcity and serve as an external benchmark; (2) doubles-specific cue behavior (cadence/alternation priors break); (3) calibration methods specific to **count-valued** cues (net-crossings) vs probabilistic ones; (4) whether RL00CV's rebalancing interacts badly with grouped (by-match) folds at N=6 ? the cited paper studies record-wise LOOCV, not group-wise.
- ****Detector-domain-gap risk inherited downstream:**** WASB/TrackNet/MonoTrack detectors are trained on broadcast/specific rigs; per-frame shuttle accuracy degrades off-distribution (MonoTrack ~88.6%) and every cue (net-crossings, state, contact) inherits that error. **Quantify detector accuracy on our footage before trusting any cue column** ? currently unmeasured.

7. Recommended citation list (verified-real only)

- **§ TAL/TAS skeleton, refinement & post-processing (Theme: TAD-TAS)****
- Temporal Action Segmentation: An Analysis of Modern Techniques ? Ding, Sener, Yao, 2022 ? arXiv:2210.10352 (TPAMI)
 - MS-TCN: Multi-Stage Temporal Convolutional Network for Action Segmentation ? Abu Farha & Gall, 2019 ? arXiv:1903.01945 (CVPR 2019)
 - ASRF: Alleviating Over-segmentation Errors by Detecting Action Boundaries ? Ishikawa et al., 2021 ? arXiv:2007.06866 (WACV 2021)
 - Activity Grammars for Temporal Action Segmentation ? Gong et al., 2023 ? arXiv:2312.04266 (NeurIPS 2023)

- Post-Processing Temporal Action Detection (GAP) ? Nag et al., 2023 ? arXiv:2211.14924 (CVPR 2023)
- BMN: Boundary-Matching Network ? Lin et al., 2019 ? arXiv:1907.09702 (ICCV 2019)
- Action Spotting using Dense Detection Anchors Revisited (SoccerNet 2022) ? Soares, Shah, Biswas, 2022 ? arXiv:2206.07846
- *Read before citing:* Improving TAS via Constraint-Aware Decoding ? 2026 ? arXiv:2605.10149 (closest reconciler analog)

****§ Two-stage / cascade / decoupled detection (Theme: cascade-rescore)****

- S-CNN: Temporal Action Localization via Multi-stage CNNs ? Shou, Wang, Chang, 2016 ? arXiv:1601.02129 (CVPR 2016)
- Cascaded Boundary Regression (CBR) ? Gao, Yang, Nevatia, 2017 ? arXiv:1705.01180 (BMVC 2017)
- Decoupled Classification Refinement (DCR) ? Cheng, Wei, ? Shi, 2018 ? arXiv:1810.04002 (ECCV 2018)
- Rapid Object Detection using a Boosted Cascade (Viola-Jones) ? Viola & Jones, 2001 ? 10.1109/CVPR.2001.990517
- Decoupling Localization and Classification in SSAD ? 2019 ? arXiv:1904.07442

****§ Late/score-level fusion & stacked arbiter (Theme: late-fusion)****

- Early versus late fusion in semantic video analysis ? Snoek, Worring, Smeulders, 2005 ? 10.1145/1101149.1101236 (ACM MM)
- Stacked Generalization ? Wolpert, 1992 ? Neural Networks 5(2):241-259
- MoCaE: Mixture of Calibrated Experts ? Oksuz et al., 2023 ? arXiv:2309.14976 (TMLR)
- Modality Competition ? Huang et al., 2022 ? arXiv:2203.12221 (ICML 2022)
- DBF: Dynamic Belief Fusion ? Lee & Kwon ? WACV 2016 / arXiv:1511.03183 (extended arXiv:2204.02890)
- Highlight Detection with Pairwise Deep Ranking ? Yao, Mei, Rui, 2016 ? CVPR 2016 (IEEE 7780481)
- Less is More: Learning Highlight Detection from Video Duration ? Xiong et al., 2019 ? arXiv:1903.00859 (CVPR 2019)
- A framework for flexible summarization of racquet sports video ? **Liu, Huang, Jiang, Xing, Ye, Gao**, 2009 ? CVIU, S1077314208001355 *(fix author line)*

****§ "ML linter" ? assertions, repair, data validation (Theme: output-linting)****

- Model Assertions for Monitoring and Improving ML Models ? Kang, Raghavan, Bailis, Zaharia, 2020 ? arXiv:2003.01668 (MLSys)
- HoloClean: Holistic Data Repairs with Probabilistic Inference ? Rekatsinas et al., 2017 ? arXiv:1702.00820 (VLDB)
- Data Validation for Machine Learning (TFDV) ? Breck, Polyzotis, Roy, Whang, Zinkevich, 2019 ? MLSys 2019
- Snorkel: Rapid Training Data Creation with Weak Supervision ? Ratner et al., 2017 ? arXiv:1711.10160 (VLDB)
- Confident Learning ? Northcutt, Jiang, Chuang, 2021 ? arXiv:1911.00068 (JAIR)
- EventAnchor (racket-sports human-in-the-loop correction) ? 2021 ? arXiv:2101.04954 (CHI 2021)

****§ Scarce-label evaluation, point-supervision, CV pitfalls (Theme: weak-sup-eval)****

- SF-Net: Single-Frame Supervision for TAL ? Ma et al., 2020 ? arXiv:2003.06845 (ECCV 2020; eval on THUMOS14/GTEA/BEOID)
- Point-Level Temporal Action Localization ? Ju, Zhao et al., 2020 ? arXiv:2012.08236
- Distributional bias compromises leave-one-out cross-validation ? Austin, Pe'er, Korem, 2025 ? arXiv:2406.01652 / 10.1126/sciadv.adx6976 (Science Advances)
- WTAL via Explicit Subspaces (action-context confusion) ? Liu et al., 2021 ? arXiv:2103.16155 (AAAI 2021)
- SoccerNet (dataset + tolerance mAP@?) ? Giancola et al., 2018 ? arXiv:1804.04527

****§ Hierarchical / play-break / phase (Theme: multiscale-phase)****

- Generic play-break event detection for hierarchical sports video ? Tjondronegoro, Chen, Pham, 2003 ? IEEE ICME 2003
- GL-MSTCN (global-local phase recognition) ? Fang et al., 2022 ? 10.1186/s12938-022-01048-w
- PCL-Former (hierarchical context-aware TAL) ? 2025 ? arXiv:2507.06411 *(authorship unverified ? cite without asserting authors)*

****§ Domain ? badminton/racket-sport rally & cue prior art (Themes: cue-novelty + cross-cutting)****

- Towards Structured Analysis of Broadcast Badminton Videos (point segmentation) ? Ghosh, Singh,

Jawahar, 2018 ? arXiv:1712.08714 (WACV 2018)

- SmartTennisTV ? **Ghosh & Jawahar**, 2018 ? arXiv:1801.01430 *(fix author line)*
- MonoTrack ? Liu & Hodgins et al., 2022 ? arXiv:2204.01899 (CVPRW 2022)
- Enhancing Badminton Game Analysis: Shot Refinement via Fusion ? **Hsu, Yu, Cheng**, 2024 ? Sensors 24(13):4372, 10.3390/s24134372 *(fix author line)*
- Service and End of Rally Detection in Badminton Videos ? See et al., 2024 ? 10.1007/978-3-031-69073-0_15 (closest non-broadcast head-to-head baseline)
- A Comparative Analysis ? Rally Detection in Badminton ? See et al., 2025 ? 10.1007/s42979-025-03935-0
- Automated Service Height Fault Detection ? Goh et al., 2023 ? Sensors 23(24):9759
- Automatic rally detection on broadcast tennis videos ? 2013 ? 10.1080/19346182.2013.819007
- ShuttleSet ? Wang, Huang et al., 2023 ? arXiv:2306.04948 / 10.1145/3580305.3599906 (KDD 2023); ShuttleSet22 ? arXiv:2306.15664
- Unsupervised Temporal Feature Aggregation (unstructured table-tennis rally detection) ? Chaudhury et al., 2019 ? arXiv:2002.08097 (ISM 2019)

****Key files referenced (absolute paths):**** `C:\Users\avidu\Projects\badminton-highlight-indexer\docs\ANALYZERS\_AND\_RECONCILERS.md` (the reconciler/analyzer seam spec ? \$1 "signal not verdict" and the DETECTION/DECISION/RECONCILER diagram this memo maps to the field's vocabulary), `docs\MULTI\_SIGNAL\_FUSION\_PLAN.md` (`RallyCue` producers, `\_enrich\_candidate\_stats`, scale/translation-invariant features), `docs\EXPERIMENT\_HARNESS.md` (`Variant`/`compare`/`compare\_unlabeled`, the no-regression A/B + LOVO gate that C1/C3/C4 touch), `docs\HOW\_RALLY\_DETECTION\_WORKS.md` (2-layer mental model), `backend\eval\classifier.py` (the ADR-004 logistic-regression scorer = the "level-1 combiner" that C2/C7/C8/C10 modify), and `backend\eval\experiment.py:203` (`experiment.predict()` ? `net\_crossings`/`players\_in\_court` already treated as feature columns).